Bayesian T_2 Distribution Estimation Using Conditional Normalizing Flows: Technical Documentation

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Contents

1 Introduction **3**

2 Background and Motivation **4**
2.1 Optimization-Based T_2 Distribution Estimation 4
2.1.1 Nonnegative Least-Squares (NNLS) 5
2.1.2 Regularized NNLS 5
2.1.3 Generative Models for Explicit Sparsity 6

3 Methods **7**
3.1 Bayesian Posterior Sampling 7
3.2 Probabilistic Model and Posterior Factorization 8
3.2.1 Quantized Rician Likelihood 8
3.2.2 Signal Model Reparameterization 8
3.2.3 Posterior Factorization 9
3.2.4 Computing Laplace's Approximation 9
3.3 Distributions on the Simplex and the Aitchison Geometry 10
3.3.1 Distributions over ILR Coefficients 10
3.3.2 A Shallow, Interpretable Prior 11
3.3.3 Deep Parameterized Distributions on the Simplex 11
3.4 Normalizing Flows for Posterior Approximation 12
3.4.1 Conditional Normalizing Flows 13
3.4.2 Flow Architecture 14
3.4.3 Joint Flow for Posterior Approximation 16
3.5 Training 17
3.5.1 Variational Inference 17
3.5.2 Path Gradient 17
3.5.3 Self-Supervised Regularization 18
3.5.4 Encoder Regularization 18
3.5.5 Training 18
3.6 Inference 19
3.6.1 Flow Inference Implementation Details 20
3.7 Joint Prior 21

4 Results **22**
4.1 Data and Experimental Setup 22
4.2 Synthetic Data Evaluation 25
4.2.1 Posterior Calibration 25
4.2.2 Bimodal Synthetic Spectra 27
4.3 In Vivo Evaluation 27

4.3.1 Posterior Quality and MCMC Diagnostics . . . 29
4.4 Preconditioning Ablation . . . 30

5 Discussion **32**
5.1 Limitations . . . 34

A Additional Applications of the Learned Change of Variables **37**
A.1 ELBO-Based Flip-Angle Estimation . . . 37
A.1.1 Cosine Series Representation for Efficient EPG Basis Computation and Differentiation . . . 39
A.2 Spatially Regularized MAP Estimation . . . 41
A.2.1 Efficient Log-Determinant Evaluation by Spectral Moment Matching . 44
A.3 Inference at a Different T_2 Resolution . . . 45
A.3.1 Deterministic Lift by Interpolation . . . 45
A.3.2 Lifted Flow Construction . . . 46
A.3.3 Lifted Prior, Likelihood, and Importance Ratio . . . 46
A.3.4 Posterior Mode Recentring . . . 47
A.3.5 Experiments . . . 47

B Extended Phase Graph Algorithm **49**

C Quantized Rician Noise Model **50**

D Gaussian Initialization for Laplace's Approximation **51**

E Softmax Concentration and Coefficient Scaling under Low-Pass Filtering **52**
E.1 Softmax Concentration and Extremal Values . . . 52
E.2 Coefficient Scaling under Regridding . . . 53
E.3 Scaling Consistency with the Sampling Interval . . . 54
E.4 Scaling in Regions of Local Maxima with Multiple Peaks . . . 54

F Distributions on the Probability Simplex **55**
F.1 The Probability Simplex . . . 55
F.2 The Aitchison Geometry . . . 56
F.3 Distributions over Isometric Log-Ratio Coefficients . . . 56
F.3.1 Geometry of the ILR Transformation . . . 57
F.3.2 Canonical Distribution on ILR Coefficients . . . 57
F.3.3 Marginal Properties of the Canonical ILR Distribution . . . 58
F.3.4 Power Distribution on Δ_{n-1} . . . 59
F.3.5 Normal Distribution on Δ_{n-1} . . . 59

G Custom Normalizing Flows **59**
G.1 Simplex Flow . . . 59
G.2 Orthogonal Flow . . . 60
G.2.1 Optimization on $\mathcal{O}_n$ . . . 61
G.2.2 Accelerating the Retraction . . . 61

H The Symmetric Dirichlet Distribution on Δ_{n-1} **62**

I The Symmetric Gaussian Restricted to Δ_{n-1} **62**
I.1 Normalization Constant of the Symmetric Gaussian on Δ_{n-1} 63
I.1.1 Wide Width Limit ($\sigma \gg \sigma_0$) 63
I.1.2 Narrow Width Limit ($\sigma \ll \sigma_0$) 65
I.1.3 Moderate Width ($\sigma \approx \sigma_0$) 66

J Optimization on Embedded Riemannian Manifolds **68**

K Densities on Riemannian Manifolds with Application to the Standard Simplex **69**
K.1 Density Transformations on Manifolds 69
K.2 The Standard Simplex Manifold Δ_{n-1} 69
K.2.1 Jacobian Determinant of the ILR Transform 69
K.2.2 Jacobian Determinant of the Inverse ILR Transform 71
K.2.3 Changing to the Intrinsic Measure 73
K.3 The Softmax Manifold $\Delta_{n-1} \times \mathbb{R}$ 73
K.3.1 Forward Transformation 73
K.3.2 Inverse Transformation 74
K.3.3 Changing to the Intrinsic Measure on $\Delta^{n-1} \times \mathbb{R}$ 76

1 Introduction

Magnetic resonance imaging (MRI) is a non-invasive imaging technique used to visualize the structure and function of the human body. By employing strong magnetic fields and radiofrequency pulses to manipulate the magnetic moments of atomic nuclei within tissues, subsequent changes in bulk magnetization transverse to the main magnetic field can be detected and reconstructed into high-resolution images.

Myelin water imaging (MWI) is a specialized MRI technique designed to quantify myelin content in white matter tissue. Myelin, a lipid-rich substance forming an insulating sheath around axons, facilitates rapid and efficient electrical impulse conduction; its degradation, as seen in diseases such as multiple sclerosis, impairs neural function. MWI exploits the differing transverse relaxation times (T_2) of water in distinct tissue compartments. Specifically, the T_2 of water within the myelinated lipid bilayers is several times shorter than that of the surrounding tissue. This gives rise to a transverse magnetization which is composed of signal components decaying at different relaxation rates, measured for example using a multi-spin echo (MSE) scan with uniform echo spacing.

Quantitative analysis of MSE signals commonly involves multi-exponential decomposition, where the observed signal is modelled as a sum of exponentially decaying components, each characterized by a distinct T_2 relaxation time [1]. The myelin water fraction (MWF), defined as the fraction of signal attributed to rapidly decaying T_2 components, is a marker for myelin content. However, estimating T_2 distributions and their derived biomarkers from MSE data is challenging. Traditional methods, such as nonnegative least squares (NNLS), yield point estimates that are sensitive to noise and require regularization for stability; moreover, they provide no measure of uncertainty in the T_2 spectrum, leaving the reliability of derived metrics unassessed. A more robust approach considers the entire space of plausible T_2 distributions.

Bayesian posterior sampling, which explores the full posterior distribution of T_2 spectra consistent with the measured signal, is well-suited for this task. This approach not only provides estimates of the mean T_2 distribution but also enables the quantification of uncertainty for individual T_2 components and derived metrics, such as MWF. However, sampling from

the posterior distribution can be computationally demanding for high-dimensional problems. To address this, we propose a hybrid framework that combines traditional posterior sampling with a learned change-of-variables obtained through variational inference. We train a conditional normalizing flow to approximate the true posterior distribution and utilize the resulting learned bijection as a preconditioning transformation for traditional Bayesian sampling methods, such as Metropolis–Hastings or Langevin dynamics. While this proposed framework contains a learned component, the network serves primarily as a change of variables that preconditions the problem geometry for more efficient sampling, analogous to preconditioners in iterative solvers for linear inverse problems and to coordinate transformations in numerical integration. Additionally, we introduce a novel, interpretable, shallow prior over the space of T_2 distributions based on the Aitchison geometry, learned simultaneously with the approximate posterior. We demonstrate that our integrated approach enables efficient posterior sampling, leading to improved estimation of T_2 distributions and MWF in myelin water imaging.

Figure 1 illustrates representative in vivo estimates.

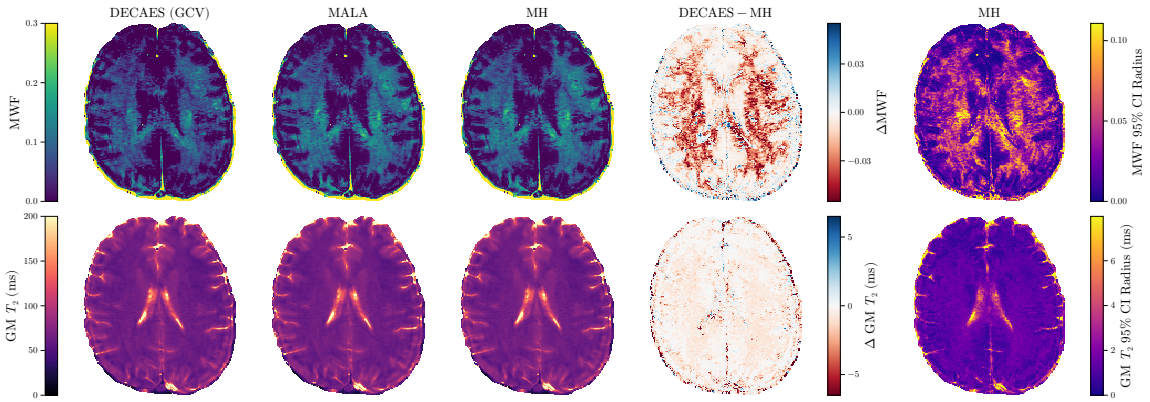


Figure 1: Representative in vivo brain biomarker maps for a single axial slice. Rows show MWF and geometric mean T_2. Columns show DECAES with GCV regularization, MALA posterior mean, MH posterior mean, DECAES − MH difference, and MH 95 % CI half-width.

2 Background and Motivation

We begin by considering the problem of estimating the T_2 distributions from multi-spin echo (MSE) signals. Our discussion proceeds in two parts: first, we describe traditional optimization-based approaches for computing point estimates; then, we describe Bayesian methods, which allow exploration of the full posterior distribution over T_2 distributions.

2.1 Optimization-Based T_2 Distribution Estimation

In the optimization-based framework, the goal is to determine the T_2 distribution coefficients $x \in \mathbb{R}_+^{n_{T_2}}$ given a measured MSE signal $b \in \mathbb{R}_+^{n_{\mathrm{TE}}}$ and a matrix of extended phase graph (EPG) basis signals $A \in \mathbb{R}_+^{n_{\mathrm{TE}} \times n_{T_2}}$. This can be formulated as the following minimization problem:

$$x^* = \arg\min_{x \geq 0} \mathcal{L}(Ax, b) + \mathcal{R}(x) \tag{1}$$

where the data-fidelity term $\mathcal{L}$ ensures that Ax closely approximates the measured signal b, and $\mathcal{R}$ is a regularization term that incorporates prior knowledge about the expected shape of the T_2 distribution.

The EPG basis matrix A encodes the MRI physics of the MSE sequence, including the effects of radiofrequency pulses, stimulated echoes, B_1-inhomogeneity, and transverse and longitudinal relaxation, as modelled by the EPG algorithm, which simulates the evolution of spin populations under repeated refocusing pulses. While the full details of the EPG algorithm are provided in appendix B, it is sufficient to note that each column of A corresponds to an exponentially decaying basis signal with transverse relaxation time $T_{2,j}$. The $T_{2,j}$ values are logarithmically spaced between $T_2^{\min}$ (first column) and $T_2^{\max}$ (last column). The coefficients x therefore quantify the relative contributions of these basis signals to the measured signal b, which is why they are referred to as the T_2 *distribution*. We construct A using an optimized implementation of the EPG algorithm provided by the DECAES open-source software package [1].

2.1.1 Nonnegative Least-Squares (NNLS)

Under the assumption of homoscedastic Gaussian noise on the measured signal, i.e. $b \sim \mathcal{N}(b \mid Ax, \sigma_b^2\mathrm{I})$, the *maximum likelihood estimate (MLE)* of $x$can be computed by solving the nonnegative least-squares (NNLS) problem:

$$x^* = \underset{x \geq 0}{\arg\min} \, \|Ax - b\|_2^2. \tag{2}$$

Equation (2) can be solved efficiently; however, it is highly ill-conditioned when A is a matrix of EPG basis signals. Precise theoretical understanding of this ill-conditioning is difficult due to the complexity of the EPG algorithm, but it can be partially understood in terms of the simpler multi-exponential inference problem, which has been explored extensively [2]. There, given a continuous-time signal modelled as $b(t) = \int_0^\infty x(\lambda) e^{-\lambda t} \, d\lambda$, the continuous spectrum $x(\lambda)$ is in principle given in closed-form by the inverse Laplace transform (ILT): $x(\lambda) = \frac{1}{2\pi i} \int_{\gamma - i\infty}^{\gamma + i\infty} b(t) e^{\lambda t} \, dt$, where $\gamma \in \mathbb{R}$ is fixed. The ILT, however, is a Fredholm integral equation of the first kind, and this class of equations is known to be ill-posed when the kernel is smooth and does not admit orthogonal bases over the real line [2]. This is the case with the ILT kernel $K(t, \lambda) = e^{\lambda t}$, unlike the complex exponential kernel $K(t, \omega) = e^{i\omega t}$ used in the inverse Fourier transform.

For perfect refocusing pulses with $\alpha = 180°$, EPG basis signals decay exactly exponentially. When $\alpha < 180°$, stimulated echoes may transiently increase the signal in early echoes, but exponential decay soon dominates, leaving small perturbations at late echoes. Since EPG basis signals approximate exponentially decaying curves – at least for $90° \ll \alpha \leq 180°$, typical of real data – the T_2-distribution estimation problem (2) becomes as ill-posed as the ILT as n_{T_2} increases. Thus, regularizing the solution x^* to (2) is essential.

2.1.2 Regularized NNLS

A common approach [3, 4] to stabilize the solution to (2) is Tikhonov regularization – also known as ℓ^2 regularization or ridge regression – which penalizes the solution norm:

$$x_\mu^* = \underset{x \geq 0}{\arg\min} \, \|Ax - b\|_2^2 + \mu^2 \|x\|_2^2. \tag{3}$$

This is equivalent to the *maximum a posteriori probability (MAP)* estimate of x under a homoscedastic half-normal prior $x \sim \mathcal{HN}(x \mid \sigma_x^2\mathrm{I})$, and a homoscedastic Gaussian likelihood $b \sim \mathcal{N}(b \mid Ax, \sigma_b^2\mathrm{I})$. The regularization parameter μ is related to the likelihood and prior

scales as $\mu = \sigma_b/\sigma_x$. Physically, this prior is reasonable: the smoothness imposed on x by the Tikhonov regularization reflects the expectation that biological tissue exhibits a smooth spectrum of relaxation rates among its constituent water pools, for example, due to spin population mixing processes such as motional narrowing.

Choosing the regularization parameter μ is challenging, and many methods have been proposed [5–8]. For example, the DECAES software package [1] implements three common approaches: (1) the χ^2 method, which implicitly selects μ such that the residual norm ratio satisfies $r^2(\mu)/r^2(0) = \chi^2$, where $r(\mu) = \|Ax^*_\mu - b\|_2$ and $\chi^2 > 1$ is a free parameter [9]; (2) the generalized cross-validation (GCV) method, which selects μ as the minimizer of the functional $G(\mu) = r^2(\mu)/t^2(\mu)$, where $t(\mu) = \mathrm{tr}\big(\mathrm{I} - A(A^TA + \mu^2\mathrm{I})^{-1}A^T\big)$ [5]; and (3) the L-curve method, which identifies the point of maximum curvature on the graph of $\log\|x^*_\mu\|_2$ vs. $\log r(\mu)$ [5, 8].

The regularization parameter has a large effect on the computed T_2 distribution; if it is too large, the distribution becomes over-smoothed, and if it is too small, the inversion becomes ill-conditioned. Another perspective is that regularization reduces the intrinsic dimensionality of the solution: as μ increases, the components $x^*_{\mu,j}$ of the Tikhonov-smoothed distribution become increasingly correlated. In this way, a sufficiently large μ constrains the degrees of freedom of x^*_μ, preventing it from overfitting the data b. Equivalently, x^*_μ becomes increasingly *sparse* in some domain [10, 11].

2.1.3 Generative Models for Explicit Sparsity

An alternative to regularization parameter tuning is to make the low intrinsic dimensionality of the solution explicit. Let the solution space of T_2 distributions be $\mathcal{X} = \mathrm{Image}(\mathcal{G}) \subset \mathbb{R}_+^{n_{T_2}}$, the image of a *deep generative model* $\mathcal{G} : \mathbb{R}^{n_z} \to \mathbb{R}_+^{n_{T_2}}$ over a latent space of dimension $n_z \ll n_{T_2}$. The inference problem then becomes

$$z^* = \arg\min_z \mathcal{L}(A\mathcal{G}(z), b), \quad x^* = \mathcal{G}(z^*), \tag{4}$$

where z^* is the solution in the n_z-dimensional latent space. Here, T_2 distributions $x = \mathcal{G}(z)$ are by definition sparse in the z domain.

This formulation introduces two optimization problems: (1) learning the generative model $\mathcal{G}$ from training data; and (2) solving (4) at inference time. This contrasts with single-shot learned inference methods, where x^* is directly predicted via a forward pass of the data through a learned model. While single-shot methods are computationally efficient, they are often less robust: the learned model must generalize across all possible inputs, which can be challenging due to the ill-conditioning of A and noise in b. Inference-time optimization, on the other hand, allows the solution to adapt to the specific data instance, thereby improving stability and accuracy.

Deep generative models of this form are widely used to represent data sparsity explicitly [12–14]. However, they require fixing the latent dimension n_z *a priori* for all candidate solutions. This is problematic because the intrinsic dimensionality of regularized solutions, such as x^*_μ in (3), is data-dependent. For instance, as the signal-to-noise ratio (SNR) of b decreases, the number of independently resolvable T_2 components also decreases [2]. To compensate, the regularization parameter μ must increase to reduce the effective dimensionality of x^*_μ and prevent overfitting. This adaptivity is why μ must be chosen carefully.

Moreover, a fundamental challenge in training generative models for T_2 distributions is the need for a "gold standard" dataset of T_2 distributions as training targets, which must be generated using existing methods. As a result, any biases or limitations in these methods, such as the heuristics used to select the regularization parameter in Tikhonov-regularized

NNLS, are inherited by the generative model. An alternative is to train the model by minimizing (4) directly over datasets of A and b, but this requires careful regularization to prevent mode collapse due to the problem's ill-posed nature. This highlights the need for a principled approach, motivating Bayesian methods that explicitly model the posterior distribution over T_2 distributions, balancing data fidelity against posterior entropy to mitigate over-concentration and account for uncertainty in the solution space.

Related studies using deep-learning methods for MRI inverse problems fall into three broad groups. First are those which learn mappings directly from measured data to target image biomarkers [15–17]. Second are those which unroll iterative schemes, such as for image reconstruction, using trainable sub-networks as learned regularizers or proximal operators; this preserves more of the physical model structure but still computes only point estimates [18–20]. Third are those using image-level variational or generative priors for variational inference and approximate posterior sampling in MR reconstruction [21–24]. In this work, we instead use deep learning methods to construct parameterized deep posterior distributions; this allows for the use of Bayesian methods for uncertainty quantification of the T_2 spectrum and derived biomarkers via Markov chain Monte Carlo (MCMC) posterior sampling methods.

3 Methods

In the previous section, we introduced traditional optimization-based methods for estimating the T_2 distribution and their limitations, including the challenges of ill-posedness, regularization, and model bias. We now turn to Bayesian methods, which address these limitations by exploring the full posterior distribution of T_2 distributions given the measured data. Unlike point estimates such as the MLE solution (2), the MAP solution (3), or generative latent-space solutions (4), Bayesian posterior sampling aims to characterize the uncertainty in the T_2 distribution estimates.

3.1 Bayesian Posterior Sampling

The goal of Bayesian posterior sampling is to generate samples x from the posterior distribution:

$$x \sim p(x \mid b) \propto p(b \mid x)p(x), \tag{5}$$

where $p(x)$ is a prior distribution reflecting prior knowledge about the T_2 distribution, and $p(b \mid x)$ is the likelihood of the measured signal b given x. In general, the likelihood $p(b \mid x)$ will also depend on the EPG basis matrix A and other non-random acquisition parameters, but we omit these dependencies for brevity.

By generating posterior samples $\{x^{(i)}\}$, one can compute both point estimates (e.g., the posterior mean) and uncertainty measures for derived metrics, such as the MWF. For a given summary statistic $f(x)$, its posterior expectation can be approximated as:

$$\mathbb{E}_{p(x|b)}[f(x)] \approx \frac{1}{N} \sum_{i=1}^{N} f(x^{(i)}), \tag{6}$$

where N is the number of posterior samples. This framework naturally accommodates the estimation of variances, quantiles, or other measures of uncertainty, enabling a robust characterization of T_2 distributions and their derived statistics.

3.2 Probabilistic Model and Posterior Factorization

Measured MSE signals b are often modelled as Gaussian:

$$b \sim \mathcal{N}(b \mid Ax, \sigma^2 \mathrm{I}), \tag{7}$$

where $\sigma > 0$ represents the (unknown) noise level. However, magnitude MRI data more accurately follows a Rician distribution [25–27]. In particular, as decaying signals approach zero, the expected measured magnitude approaches the Rician noise floor $\sigma\sqrt{\frac{\pi}{2}}$; this is misattributed to long-lived signal by a Gaussian likelihood model, introducing bias into T_2 distribution estimates and derived metrics such as MWF, which depend on accurately quantifying rapidly decaying signals.

3.2.1 Quantized Rician Likelihood

MRI data is commonly quantized to 12- to 16-bit integers. Given linearly rescaled data $b = \text{slope} \cdot k + \text{offset}$ corresponding to integer measurements k, the quantization bin width is $\delta = |\text{slope}|$. Quantization can be coarse; for instance, multi-echo (3+1)D images may contain only a few thousand unique values across millions of measurements.

As a modelling assumption, each reported amplitude b_i is interpreted as the lower edge of its quantization bin, since the stored counts do not record whether the scanner rounds or truncates. To account for both the Rician noise and quantization, we model b using a quantized Rician likelihood, defined as:

$$p_{\mathrm{QRice}}(b \mid \nu = Ax, \sigma, \delta) = \prod_{i=1}^{n_{\mathrm{TE}}} p_{\mathrm{QRice}}(b_i \mid \nu_i, \sigma, \delta) \tag{8}$$

$$p_{\mathrm{QRice}}(b_i \mid \nu_i, \sigma, \delta) = \int_0^{\delta} p_{\mathrm{Rice}}(b_i + \epsilon \mid \nu_i, \sigma)\, d\epsilon, \tag{9}$$

where ν is the modelled signal, σ is the noise level, and p_{Rice} is the Rician probability density function. Details for computing p_{QRice}, including efficient evaluation via Gauss–Legendre quadrature, are in appendix C.

3.2.2 Signal Model Reparameterization

We now introduce a parameterization that will allow a hybrid variational inference approach, using deep normalizing flows to approximate the complex part of the posterior and Laplace's approximation for the simpler part. The reparameterized signal model is:

$$\begin{aligned} b &\sim \mathrm{QRice}(b \mid \nu, \sigma, \delta) \\ \text{where} \quad \nu &= Ax = s\bar{\nu},\ \bar{\nu} = A\bar{x},\ x = s\bar{x}. \end{aligned} \tag{10}$$

Here, the scale $s > 0$ of the modelled signal is factored out, and $\bar{x}$ is the normalized T_2 distribution with $\|\bar{x}\|_1 = 1$. This factorization emphasizes the *shape* of the T_2 distribution, $\bar{x}$, which determines relative metrics like the MWF; the scale parameter s is tightly constrained by the data b and is easily estimated given $\bar{x}$. In the rest of this work, we refer to $\eta = (\log s, \log \sigma)$ as nuisance parameters, and we write the likelihood of the data under the noise model (10) as:

$$p(b \mid \bar{x}, \eta) = p_{\mathrm{QRice}}(b \mid \nu = s\bar{\nu}, \sigma, \delta). \tag{11}$$

3.2.3 Posterior Factorization

We now consider the factorized posterior distribution

$$p(\bar{x}, \eta \mid b) = p(\bar{x} \mid b)p(\eta \mid \bar{x}, b). \tag{12}$$

Due to the high dimensionality of $\bar{x}$, we approximate $p(\bar{x} \mid b)$ using a conditional normalizing flow $q_{\bar{x}}(\bar{x} \mid b)$, detailed in section 3.4. The conditional posterior $p(\eta \mid \bar{x}, b)$ is simpler: η is constrained by $n_{\mathrm{TE}} \approx 48 - -56$ measurements. Thus, by the Bernstein–von Mises theorem, we expect Laplace's approximation

$$p(\eta \mid \bar{x}, b) \approx q_\eta(\eta \mid \bar{x}, b) = \mathcal{N}(\eta \mid \mu_\eta, \Sigma_\eta) \tag{13}$$

to be accurate. This approximation is parameter-free and stabilizes training by mitigating loss spikes. Loss spikes can occur early in training when poor posterior approximations yield small σ values, causing the Rician log-likelihood, which scales approximately as $1/\sigma^2$ near the mode, to diverge. Laplace's approximation directly targets the posterior mode, producing σ with appropriate scale. Furthermore, the use of Laplace's approximation allows for direct generalization across scales s, so we can train on normalized data b.

3.2.4 Computing Laplace's Approximation

Given a normalized T_2 distribution $\bar{x}$, Laplace's approximation has mean μ_η at the conditional posterior mode and covariance Σ_η given by the inverse Hessian of its negative log density:

$$\begin{aligned}
\mu_\eta &= \arg\min_\eta -\log p(\eta \mid \bar{x}, b) && (14)\\
&= \arg\min_\eta -\log p(\eta, b \mid \bar{x}) && (15)\\
\Sigma_\eta^{-1} &= -\nabla_\eta^2 \left[\log p(\eta, b \mid \bar{x})\right]\big|_{\eta=\mu_\eta}, && (16)
\end{aligned}$$

where (15) follows from Bayes' rule and the fact that the normalizing constant does not depend on η. Here, the joint conditional density $p(\eta, b \mid \bar{x}) = p(b \mid \bar{x}, \eta)p(\eta)$ equals the likelihood (11) multiplied by a prior $p(\eta)$, which we take to be $\mathcal{N}(\eta \mid \mu_{\eta,0}, \Sigma_{\eta,0})$ with diagonal $\Sigma_{\eta,0}$. The mode (or MAP point) μ_η is computed using Newton's method, with the initial guess given by the mode of the corresponding problem with Gaussian instead of quantized Rician noise; details are provided in appendix D.

Since this will be part of a neural network architecture, we need to be able to differentiate through μ_η and Σ_η. We use the implicit function theorem (IFT) to compute derivatives of μ_η with respect to b, $\bar{\nu}$, and δ. Let $F(\eta, b, \bar{\nu}, \delta) = -\nabla_\eta \log p(\eta, b \mid \bar{x})$. Then, μ_η is defined implicitly by $F(\mu_\eta, b, \bar{\nu}, \delta) = 0$, and the IFT gives the derivatives

$$\begin{aligned}
\frac{\partial \mu_\eta}{\partial b_i} &= -\left(\frac{\partial F}{\partial \eta}\right)^{-1} \frac{\partial F}{\partial b_i} = \Sigma_\eta \frac{\partial^2 \log p_{\mathrm{QRice}}(b_i \mid s\bar{\nu}_i, \sigma, \delta)}{\partial \eta \partial b_i}\\
\frac{\partial \mu_\eta}{\partial \bar{\nu}_i} &= -\left(\frac{\partial F}{\partial \eta}\right)^{-1} \frac{\partial F}{\partial \bar{\nu}_i} = \Sigma_\eta \frac{\partial^2 \log p_{\mathrm{QRice}}(b_i \mid s\bar{\nu}_i, \sigma, \delta)}{\partial \eta \partial \bar{\nu}_i}\\
\frac{\partial \mu_\eta}{\partial \delta} &= -\left(\frac{\partial F}{\partial \eta}\right)^{-1} \frac{\partial F}{\partial \delta} = \Sigma_\eta \frac{\partial^2 \sum_i \log p_{\mathrm{QRice}}(b_i \mid s\bar{\nu}_i, \sigma, \delta)}{\partial \eta \partial \delta},
\end{aligned} \tag{17}$$

where the partial derivatives are computed at $(\mu_\eta, b, \bar{\nu}, \delta)$ using the explicit derivative expressions of the quantized Rician log-likelihood derived in appendix C.

3.3 Distributions on the Simplex and the Aitchison Geometry

We now consider the problem of constructing flexible parametric distributions over the space of normalized T_2 distributions. Herein, we refer to *the simplex* as the set Δ_{n-1} of n-dimensional vectors with positive components and unit sum, known also as the *standard simplex* or *probability simplex*. For example, $\Delta_2 \subset \mathbb{R}^3$ is an equilateral triangle with vertices on the coordinate axes. The simplex interior is an $(n-1)$-dimensional manifold embedded in $\mathbb{R}^n$; density estimation must account for its positivity and unit-sum constraints. Change-of-variables techniques have been developed for density estimation on other non-Euclidean spaces such as tori, spheres, and cylinders [28–31]. Density estimation on the positive orthant of the hypersphere [32] is notable, since the elementwise square root yields a bijection onto the simplex, allowing such methods to be easily adapted. Here, we instead leverage the *Aitchison geometry* [33, 34], which endows the simplex with a Euclidean vector space structure. Specifically, the *isometric log-ratio (ILR) transformation* provides a bijective map between the interior of Δ_{n-1} and $\mathbb{R}^{n-1}$, allowing us to work with unconstrained coordinates.

We favour the ILR transformation for its simplicity and interpretability. The ILR transformation and its inverse are given by [35]

$$u = \mathrm{ilr}(\bar{x}; \beta) = \frac{1}{\beta} \Psi^T \log \bar{x} \tag{18}$$

$$\bar{x} = \mathrm{ilr}^{-1}(u; \beta) = \mathrm{softmax}(\beta \Psi u), \tag{19}$$

where $\bar{x} \in \Delta_{n-1}$, $u \in \mathbb{R}^{n-1}$, $\beta > 0$, log is applied elementwise, $\mathrm{softmax}(z)_i = e^{z_i} / \sum_j e^{z_j}$, and $\Psi \in \mathbb{R}^{n \times (n-1)}$ is any orthonormal basis for the tangent space of the simplex (i.e. $\Psi^T \Psi = \mathrm{I}_{n-1}$ and $\Psi^T 1 = 0$). The ILR transformation allows us to convert distributions p_u over unconstrained ILR coefficients into distributions $p_{\bar{x}}$ over the simplex via the change-of-variables formula:

$$p_{\bar{x}}(\bar{x} \mid \beta) = p_u(u) / \left(\beta^{n-1} \sqrt{n} \prod_{i=1}^{n} \bar{x}_i \right). \tag{20}$$

See appendices F and G for further details.

The ILR transform naturally extends the elementwise logarithm, commonly used for transforming positive vectors to unconstrained vectors. For instance, given $x \in \mathbb{R}^n_+$, let $u_i = \log x_i$ with inverse $x_i = e^{u_i}$. Then, the change-of-variables formula is $p_x(x) = p_u(u) / \prod_i x_i$. These only differ from Equations (18)–(20) by up- and down-projection by Ψ and Ψ^T, scaling by β, and normalization via $\mathrm{softmax}(\cdot)$.

3.3.1 Distributions over ILR Coefficients

We consider two natural classes of distributions p_u over the ILR coefficients u:

- *ILR-based:* choose p_u directly in ILR space, e.g. standard normal $p_u = \mathcal{N}(0, \mathrm{I}_{n-1})$.
- *Simplex-based:* choose $p_{\bar{x}}$ native to the simplex and transform to p_u using equation (20).

The simplest simplex-based distribution is obtained by transforming uniform $p_{\bar{x}} = \mathcal{U}(\Delta_{n-1})$ via $u = \mathrm{ilr}(\bar{x}; \beta_0)$:

$$p_u(u) = \Gamma(n) \left(\beta_0^{n-1} \sqrt{n} \prod_{i=1}^{n} \bar{x}_i \right) \tag{21}$$

where $p_{\bar{x}}(\bar{x}) \equiv 1/|\Delta_{n-1}| = \Gamma(n) = (n-1)!$, and $\beta_0 = \pi/\sqrt{6}$ is chosen such that $\mathbb{E}_{p_u}[u] = 0$ and $\mathbb{E}_{p_u}[uu^T] = \mathrm{I}_{n-1}$. We refer to (21) as the *canonical distribution* on the ILR coefficients; see appendix F.3.2 for derivation of β_0.

3.3.2 A Shallow, Interpretable Prior

T_2 spectra in brain tissue are expected to vary smoothly, with broad peaks for intra-/extracellular water and smaller peaks for myelin water. Using the Aitchison geometry, we construct a prior based on low-pass filtered ILR coefficients that softly constrains plausible T_2 distributions to favour biologically plausible T_2 spectra.

The prior is constructed by choosing Ψ^T in (18) to be the orthonormalized type-II discrete cosine transform (DCT) matrix with the first row $\frac{1}{\sqrt{n}}1^T$ removed. The ILR transform (18) is then equivalent to the DCT-II transform of $\log \bar{x}$ (with the DC component fixed to zero) scaled by $1/\beta$. Low-pass filtering can then be performed by scaling the ILR coefficients to attenuate high-frequency components. This low-pass prior on $\bar{x}$ can be constructed as follows:

1. Choose a base distribution π_u over ILR coefficients.
2. Low-pass filter u into $u_\Lambda = \Lambda u$ where Λ is diagonal. Then $p_{u_\Lambda}(u_\Lambda) = \pi_u(u)/|\Lambda|$ where $u = \Lambda^{-1}u_\Lambda$.
3. Map u_Λ to the simplex, $\bar{x} = \mathrm{ilr}^{-1}(u_\Lambda; \beta)$. The low-pass prior over $\bar{x}$ is then given by

$$p_{\bar{x}}(\bar{x} \mid \beta) = \pi_u(u) / \left(\beta^{n-1}|\Lambda|\sqrt{n}\prod_{i=1}^{n}\bar{x}_i\right) \tag{22}$$

   where $u = \Lambda^{-1}u_\Lambda = \Lambda^{-1}\,\mathrm{ilr}(\bar{x}; \beta)$.

When the base distribution is ILR-based with π_u standard normal, the resulting distribution is equivalent to a zero-mean *normal-on-simplex* [36] distribution with a particular covariance structure. For motivation, the canonical distribution (21) corresponds to the uniform measure over the simplex, with $\beta = \beta_0$. Low-pass filtering then reweights this measure by frequency content, yielding a prior that uniformly spans the space of smooth T_2 distributions, resulting in the *low-pass prior*:

$$p_{\bar{x}}(\bar{x} \mid \Lambda) = \frac{\Gamma(n)}{|\Lambda|}\prod_{i=1}^{n}\left(\frac{\bar{x}_{0,i}}{\bar{x}_i}\right) \tag{23}$$

where $\bar{x}_0 = \mathrm{ilr}^{-1}\big(\Lambda^{-1}\,\mathrm{ilr}(\bar{x}; \beta_0); \beta_0\big)$. For the implemented model, we instead choose standard Gaussian coefficients $\pi_u = \mathcal{N}(0, \mathrm{I})$ and $\beta = \beta_0$ in (22).

3.3.3 Deep Parameterized Distributions on the Simplex

We can generalize the methodology used to construct the low-pass prior in order to create deep, flexible distributions on the simplex. Prepending the diagonal filter $\Lambda \in \mathbb{R}^{(n-1)\times(n-1)}$ from (22) with a smooth invertible function $\varphi_\xi : \mathbb{R}^{n-1} \to \mathbb{R}^{n-1}$ with a tractable Jacobian determinant gives

$$p_{\bar{x}}(\bar{x}) = \pi_u(u) \Big/ \left(\beta_0^{n-1}|\Lambda|\big|J_{\varphi_\xi}(u)\big|\sqrt{n}\prod_{i=1}^{n}\bar{x}_i\right), \tag{24}$$

where u is the base coordinate, $\xi = \varphi_\xi(u)$ is the pre-window ILR coefficient, $J_{\varphi_\xi} = \frac{\partial \varphi_\xi}{\partial u}$ is the Jacobian of φ_ξ at u, and $\bar{x} = \mathrm{ilr}^{-1}(\Lambda\xi; \beta_0)$. Sampling from $p_{\bar{x}}$ is via $\varphi_{\bar{x}} = \mathrm{ilr}^{-1}(\cdot; \beta_0) \circ \Lambda \circ \varphi_\xi$:

$$\bar{x} \sim p_{\bar{x}} \Leftrightarrow \bar{x} = \varphi_{\bar{x}}(u) = \mathrm{ilr}^{-1}(\Lambda\varphi_\xi(u); \beta_0) \text{ where } u \sim \pi_u. \tag{25}$$

Identity φ_ξ recovers the low-pass prior of (22), in which case $\xi \equiv u$. We will henceforth reserve the base coordinate label u for the deep flow transformation $\varphi_{\bar{x}}$ of equation (25), and

refer to both the base coordinate of the prior and the pre-window coordinate of the deep flow as ξ.

This framework allows for the construction of flexible distributions over the simplex with desirable properties: (1) expressiveness that scales with the complexity of φ_ξ, (2) closed-form densities via (24), and (3) efficient sampling via (25).

3.4 Normalizing Flows for Posterior Approximation

Posterior sampling is notoriously difficult in high dimensions [37, 38]. This difficulty largely stems from the curse of dimensionality: posterior sampling explores the entire solution space, but obtaining sufficient coverage of a high-dimensional space becomes exponentially harder as the dimension increases. While our problem is not extremely high-dimensional ($n_{T_2} \approx 40$ is typical), navigating the ambient $\bar{x}$-space through naive random walks, for example, is not feasible. Equally important is that performing random, uncorrelated Brownian jumps in $\bar{x}$-space – typical of *Markov chain Monte Carlo (MCMC)* methods – does not respect the smooth structure we expect of T_2 distributions. Posterior samples $\bar{x}_i$ should ideally exhibit smoothness and correlations between neighbouring T_2 components, reflecting the underlying physics.

These issues can be simultaneously addressed through a judicious choice of transformation of the sampling space. If we let $u = \varphi_{\bar{x}}^{-1}(\bar{x}; b)$ be a data-dependent transformation of $\bar{x}$ to an unconstrained vector u, where $\varphi_{\bar{x}}$ is a *diffeomorphism* – a differentiable, one-to-one function with a differentiable inverse – which also has a tractable Jacobian determinant, then the task becomes to find $\varphi_{\bar{x}}$ such that sampling in u-space is more efficient than sampling from $p(\bar{x} \mid b)$ directly.

Thus far, we have factored the joint posterior as $p(\bar{x}, \eta \mid b) = p(\bar{x} \mid b)p(\eta \mid \bar{x}, b)$, and have provided a nuisance posterior approximant $q_\eta(\eta \mid \bar{x}, b) \approx p(\eta \mid \bar{x}, b)$ using Laplace's approximation (Equations (13)–(16)). While approximating the marginal posterior $p(\bar{x} \mid b)$ is more challenging, by letting $\varphi_{\bar{x}} = \mathrm{ilr}^{-1}(\cdot; \beta_0) \circ \Lambda \circ \varphi_\xi(\cdot; b)$ (i.e. (25) with data-dependent φ_ξ), we can build a highly expressive deep distribution $q_{\bar{x}}(\bar{x} \mid b)$. The joint approximate posterior

$$q(\bar{x}, \eta \mid b) = q_{\bar{x}}(\bar{x} \mid b) q_\eta(\eta \mid \bar{x}, b) \tag{26}$$

can then be trained to approximate $p(\bar{x}, \eta \mid b)$ using *amortized variational inference*; see section 3.5.1. We train the flow $\varphi_{\bar{x}}$ primarily to improve posterior-sampling efficiency [30, 39–41]. Consider how the posterior $p(\bar{x} \mid b)$ transforms under $\varphi_{\bar{x}}^{-1}$:

$$p(\bar{x} \mid b) = p(u \mid b) / |J_{\varphi_{\bar{x}}}(u; b)|, \tag{27}$$

where $J_{\varphi_{\bar{x}}} = \frac{\partial \bar{x}}{\partial u}$ is the Jacobian of $\varphi_{\bar{x}}$ at u[1]. This transforms the sampling problem to

$$\begin{aligned} (u, \eta) \sim p(u, \eta \mid b) &= p(\bar{x}, \eta \mid b) |J_{\varphi_{\bar{x}}}(u; b)| \\ &\propto p(b \mid \bar{x}, \eta) p(\bar{x}, \eta) |J_{\varphi_{\bar{x}}}(u; b)|, \end{aligned} \tag{28}$$

where $\bar{x} = \varphi_{\bar{x}}(u; b)$. By training $\varphi_{\bar{x}}$ to transform $p(u, \eta \mid b)$ into a simpler distribution (e.g. standard normal), the posterior geometry becomes easier to navigate, effectively preconditioning the sampling problem; see figure 2 for a visualization of this preconditioning effect.

[1]The Jacobian $J_{\varphi_{\bar{x}}^{-1}}$ of $\varphi_{\bar{x}}^{-1}$ at $\bar{x}$ can also be used, as $\left|J_{\varphi_{\bar{x}}^{-1}}\right| = |J_{\varphi_{\bar{x}}}|^{-1}$.

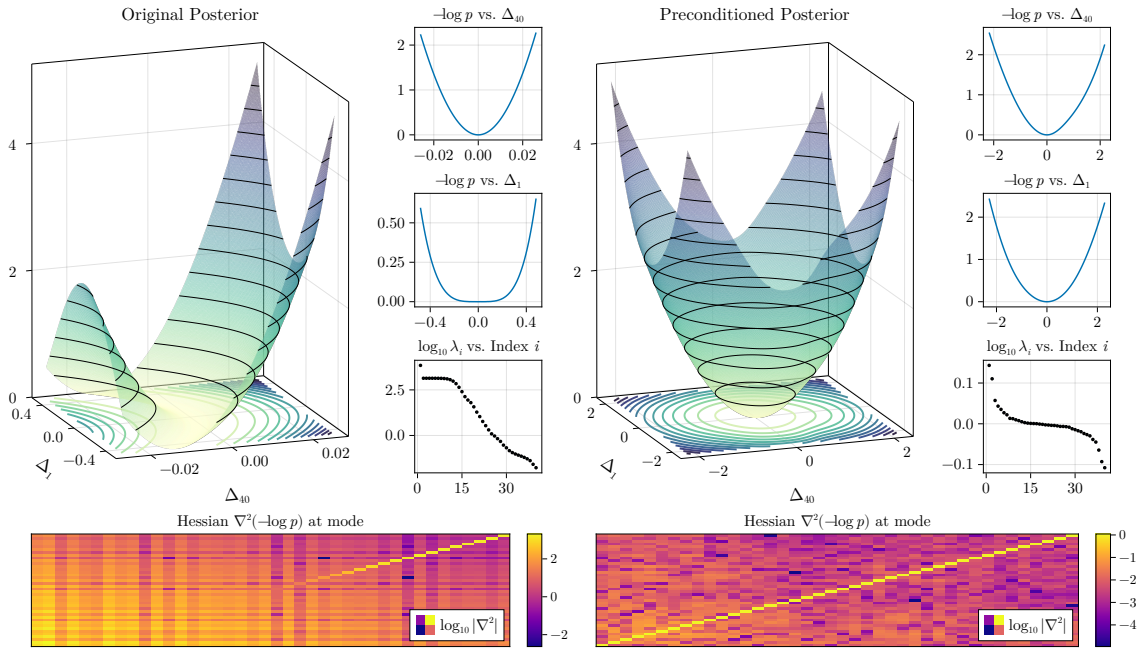


Figure 2: Effect of the learned conditional normalizing flow preconditioner on the negative log-posterior at the MAP solution for a representative in vivo voxel. The columns differ only in that the right column uses the learned change of variables from equation (25), whereas the left column replaces the learned backbone φ_ξ with the identity. Within each column, the surface plot shows the posterior surface, recentred to the mode $u^*_{\bar{x}}$, spanned by the Hessian eigenvectors corresponding to the largest and smallest eigenvalues, $u_{\bar{x}} = u^*_{\bar{x}} + \Delta_1 \hat{e}_1 + \Delta_{n_{T_2}-1}\hat{e}_{n_{T_2}-1}$, with nuisance dimensions $u_\eta = 0$ fixed at the mode; the three smaller panels show the negative log-posterior profiles along each axis through the origin, as well as the sorted eigenvalue spectrum $\log_{10}\lambda_i$; and the bottom panel shows the log-absolute Hessian values $\log_{10}|\nabla^2(-\log p)|$ at the mode. The trained flow produces a posterior geometry that is approximately an isotropic quadratic bowl.

3.4.1 Conditional Normalizing Flows

Let $q_{\bar{x}}$ be a conditional density defined by

$$\log q_{\bar{x}}(\bar{x} \mid b) = \log \pi_u(u) - \log|J_{\varphi_{\bar{x}}}(u; b)|, \tag{29}$$

$$\bar{x} \sim q_{\bar{x}}(\bar{x} \mid b) \Leftrightarrow \bar{x} = \varphi_{\bar{x}}(u; b) \text{ where } u \sim \pi_u. \tag{30}$$

We say that the *flow* $\varphi_{\bar{x}}$, a diffeomorphism, *pushes forward* samples u drawn from the *base distribution* π_u to samples $\bar{x} \sim q_{\bar{x}}$; we thus refer to $q_{\bar{x}} = \varphi_{\bar{x}} \circ \pi_u$ as the *pushforward* density of π_u by $\varphi_{\bar{x}}$. If we let $\pi_u = \mathcal{N}(0, \mathrm{I}_{n_{T_2}-1})^2$, we see from (27) and (29) that if $q_{\bar{x}}$ approximates the marginal posterior $p(\bar{x} \mid b)$, then the *inverse flow* $\varphi_{\bar{x}}^{-1}$ will approximately *normalize* the posterior:

$$p(\bar{x} \mid b) \approx q_{\bar{x}}(\bar{x} \mid b) \Rightarrow p(u \mid b) = \varphi_{\bar{x}}^{-1}(\cdot; b) \circ p(\bar{x} \mid b) \approx \pi_u(u),$$

hence the name (conditional) *normalizing* flow. Note that it is this property that makes $\varphi_{\bar{x}}^{-1}$ a powerful preconditioner for sampling; the complexities of $p(\bar{x} \mid b)$ are smoothed away by

[2]The dimension of the base distribution π_u must match the degrees of freedom of $\bar{x}$ (constrained by $\sum_i \bar{x}_i = 1$) for the flow to be invertible.

$\varphi_{\bar{x}}^{-1}$, resulting in a posterior surface with much simpler geometry: $-\log p(u \mid b) \approx \frac{1}{2}\|u\|^2 + C$ (see figure 2). Furthermore, the flow need not be perfectly trained (impossible in practice) to see benefits; any improvements in the posterior geometry will help downstream inference in u-space.

The base distribution π_u can be transformed into an arbitrarily complex density $q_{\bar{x}}$ by $\varphi_{\bar{x}}$, and so normalizing flows can be seen as a generalization of the *reparameterization trick*: the sampling procedure (30) can be differentiated by caching $u \sim \pi_u$ during the forward pass of the backpropagation algorithm and then differentiating through $\varphi_{\bar{x}}$ as usual. It is furthermore crucial in practice that flows are *composable*: given L diffeomorphisms $\{\varphi_{\bar{x},i}\}_{i=1}^{L}$, the function composition $\varphi_{\bar{x}} = \varphi_{\bar{x},L} \circ \cdots \circ \varphi_{\bar{x},1}$ is itself a diffeomorphism which transforms π_u into $q_{\bar{x}}$ such that

$$\log q_{\bar{x}}(\bar{x} \mid b) = \log \pi_u(u) - \sum_{i=1}^{L} \log\left|J_{\varphi_{\bar{x},i}}(u_{i-1}; b)\right|, \tag{31}$$

where $u_i = \varphi_{\bar{x},i}(u_{i-1}; b)$, $u_0 = u$, $u_L = \bar{x}$, and $J_{\varphi_{\bar{x},i}}$ is the Jacobian of $\varphi_{\bar{x},i}$ at u_{i-1}. Invertibility follows from $\varphi_{\bar{x}}^{-1} = \varphi_{\bar{x},1}^{-1} \circ \cdots \circ \varphi_{\bar{x},L}^{-1}$.

3.4.2 Flow Architecture

The conditional normalizing flow $\varphi_{\bar{x}}$ is composed of B flow blocks, the low-pass window Λ of section 3.3.2, and an inverse ILR transform:

$$\varphi_{\bar{x}} = \mathrm{ilr}^{-1}(\cdot; \beta_0) \circ \Lambda \circ \varphi_{\text{block}}^{(B)} \circ \cdots \circ \varphi_{\text{block}}^{(1)}. \tag{32}$$

Each block has the form:

$$\varphi_{\text{block}} = E \circ T^{-1} \circ C \circ T, \tag{33}$$

where T denotes an affine linear flow, C a symmetric affine coupling flow, and E an elementwise affine flow. The block structure derives from the real-NVP family of architectures [42–45], with a slight modification inspired by Fourier neural operators [46], where the learned affine flow T replaces the Fourier transform. Each block applies four transformations: (1) linearly transform ILR coefficients to a learned space, allowing global interactions between dimensions; (2) apply the nonlinear symmetric coupling flow C; (3) linearly transform back to ILR space via T^{-1}; and (4) apply an elementwise affine transformation E in ILR space. See figure 3 for the factored joint flow and figure 4(b) for the backbone block structure.

For simplicity, we assume the number of T_2 components n_{T_2} is odd, ensuring the spectrum-coordinate dimension $d = n_{T_2} - 1$ is even[3].

To condition the flow on the data, C and E also take an encoding vector $z \in \mathbb{R}^{n_z}$ as input, produced by a transformer-style encoder [47]. The measured signal b and its projection $A^T b$ are encoded alongside MRI-specific data: the quantization level δ, the echo spacing TE, the number of echoes n_{TE}, the EPG matrix A, and the parameters defining A (longitudinal T_1, transverse range $T_2^{\min}$ and $T_2^{\max}$, and flip angle α). The continuous parameters are formed into a vector θ, where

$$\theta_0 = (\delta, \log\delta, \cos\alpha, \sin\alpha, \tau_{2,1}, \tau_{2,2}, \log\tau_{2,1}, \log\tau_{2,2}, \tau_1, \log\tau_1), \quad \theta = \frac{\theta_0 - \mu_{\theta_0}}{\sigma_{\theta_0}}, \tag{34}$$

$\tau_{2,1} = T_2^{\min}/\text{TE}$, $\tau_{2,2} = T_2^{\max}/\text{TE}$, $\tau_1 = T_1/\text{TE}$, and θ equals θ_0 standardized to zero mean and unit variance using fixed (augmented-)training data statistics μ_{θ_0} and σ_{θ_0}.

[3]Matrix multiplications within φ_{block} are typically faster with even-dimensional vectors, too.

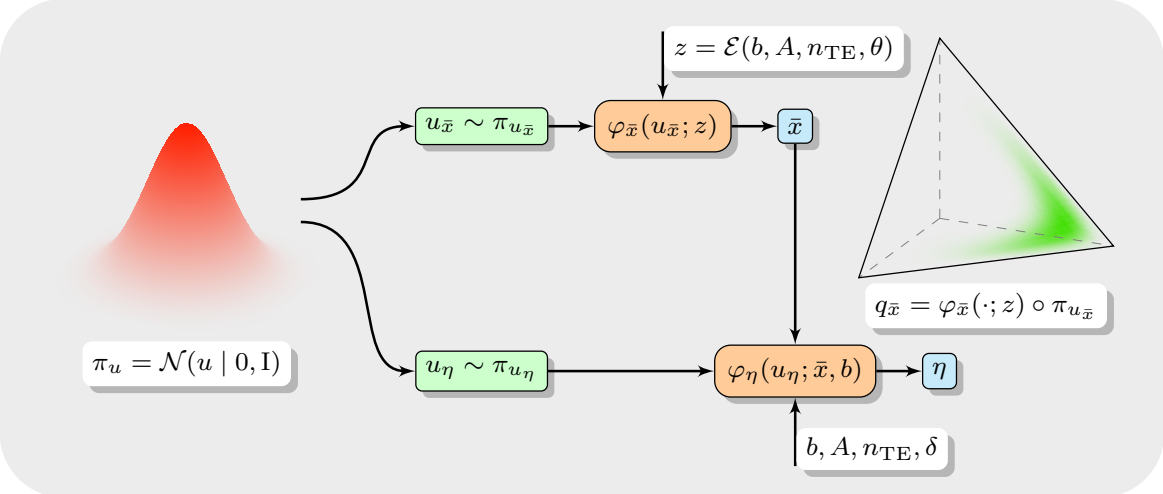


Figure 3: Factored conditional normalizing flow architecture. A deep flow $\varphi_{\bar{x}}$ is trained to approximate the marginal posterior $p(\bar{x} \mid b)$ over the normalized T_2 distribution, while the conditional posterior $p(\eta \mid \bar{x}, b)$ implements Laplace's approximation over the nuisance parameters $\eta = (\log s, \log \sigma)$ via the affine flow φ_η.

Figure 4(a) illustrates the encoder architecture. The number of echoes n_{TE} is treated as a discrete token and input to the encoder via an embedding layer. The token dimension is $n_{\mathrm{tok}} = 64$ and the MLP hidden dimension $4n_{\mathrm{tok}} = 256$. Sinusoidal positional encodings are added separately to the 4 query tokens and the $4 + n_{T_2}$ key/value tokens. A stack of $L = 2$ cross-attention transformer blocks with 8 heads is applied sequentially to the query tokens; the transformed queries are then mapped from $\mathbb{R}^{4\times n_{\mathrm{tok}}}$ to $z_0 \in \mathbb{R}^{n_z}$, with $n_z = 128$, and a final RMSNorm layer projects z_0 to the sphere of radius $r = \sqrt{n_z}$, giving $z \in S_r^{n_z-1}$. These encoding vectors are regularized during training using the Wasserstein-2 regularizer described in section 3.5.4.

The affine linear flow T is parameterized using a matrix exponential:

$$Tu = \exp(M)u + c, \qquad T^{-1}v = \exp(-M)(v - c), \tag{35}$$

where $M \in \mathbb{R}^{d\times d}$ and $c \in \mathbb{R}^d$, with $d = n_{T_2} - 1$. This parameterization is invertible for every finite M, with log-Jacobian determinant $\mathrm{tr}(M)$. The inverse T^{-1} shares parameters with T.

The elementwise affine flow E scales and shifts each ILR coefficient independently as:

$$u' = s \odot u + t, \ (a, t) = f_E(z), \quad s = h(a), \tag{36}$$

where $f_E : \mathbb{R}^{n_z} \to \mathbb{R}^d \times \mathbb{R}^d$ is an MLP. The positive scale activation is $h(a) = (a + \sqrt{a^2 + 4})/2$, applied elementwise in both E and C. The log-Jacobian determinant for (36) is

$$\log|J_E(u, z)| = \sum_i \log s_i. \tag{37}$$

The symmetric coupling flow C is composed of two coupling flows. The input u is split in half into u_1 and u_2, and u is transformed one half at a time:

$$\begin{aligned} u_1' &= s_1 \odot u_1 + t_1, \ (a_1, t_1) = f_{C_1}(z, u_2), \quad s_1 = h(a_1) \\ u_2' &= s_2 \odot u_2 + t_2, \ (a_2, t_2) = f_{C_2}(z, u_1'), \quad s_2 = h(a_2) \end{aligned} \tag{38}$$

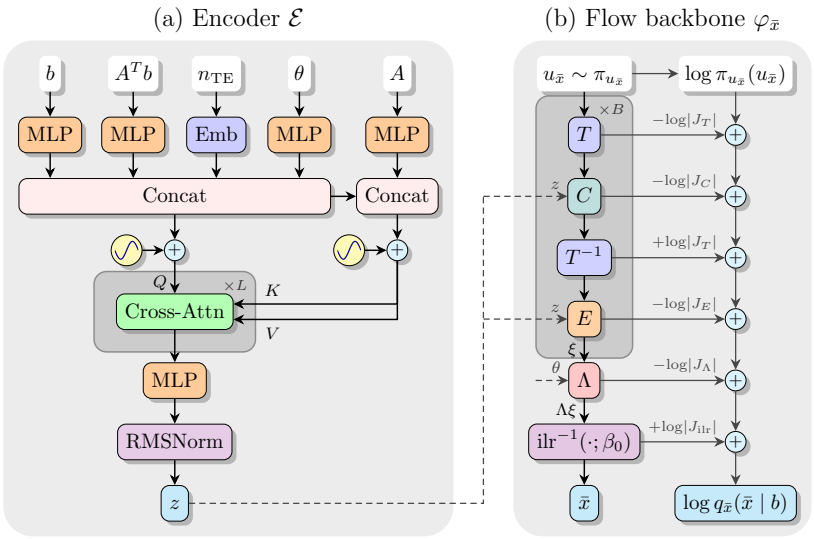


Figure 4: Network architecture. (a) The encoder architecture, mapping the tokenized signal and scan parameters to the encoding vector $z \in S_r^{n_z-1}$ via cross-attention. (b) The flow backbone architecture, pushing a base sample $u_{\bar{x}} \sim \mathcal{N}(0, \mathrm{I})$ through the B flow blocks, low-pass window Λ, and inverse ILR transform to produce a normalized T_2 distribution sample $\bar{x}$.

where $f_{C_1}, f_{C_2} : \mathbb{R}^{n_z} \times \mathbb{R}^{d/2} \to \mathbb{R}^{d/2} \times \mathbb{R}^{d/2}$ are MLPs. The log-Jacobian determinant for (38) is:

$$\log|J_C(u, z)| = \sum_i \log(s_1)_i + \sum_i \log(s_2)_i. \tag{39}$$

3.4.3 Joint Flow for Posterior Approximation

In section 3.2.4, we introduced Laplace's approximation q_η to $p(\eta \mid \bar{x}, b)$ for the nuisance parameters $\eta = (\log s, \log \sigma)$. Since q_η is Gaussian, it is equivalent to the pushforward of $\pi_{u_\eta} = \mathcal{N}(0, \mathrm{I}_2)$ by an *affine flow* φ_η conditional on $\bar{x}$ and b:

$$\begin{aligned} q_\eta(\eta \mid \bar{x}, b) &= \varphi_\eta(\cdot; \bar{x}, b) \circ \pi_{u_\eta} \\ \varphi_\eta(u_\eta; \bar{x}, b) &= \mu_\eta + \Sigma_\eta^{1/2} u_\eta \end{aligned} \tag{40}$$

where μ_η and Σ_η are the mean and covariance of $q_\eta(\eta \mid \bar{x}, b)$ computed via (14) and (16). Since Σ_η is positive definite at μ_η, φ_η is a diffeomorphism with Jacobian $\Sigma_\eta^{1/2}$.

The Laplace flow φ_η can be neatly integrated with the deep normalizing flow $\varphi_{\bar{x}}$. Let $\pi_u = \pi_{u_{\bar{x}}} \times \pi_{u_\eta} \cong \mathcal{N}(0, \mathrm{I}_{n_{T_2}+1})$ be the product of the base distributions $\pi_{u_{\bar{x}}} = \mathcal{N}(0, \mathrm{I}_{n_{T_2}-1})$ and $\pi_{u_\eta} = \mathcal{N}(0, \mathrm{I}_2)$. Then, $q(\bar{x}, \eta \mid b) = q_{\bar{x}}(\bar{x} \mid b) q_\eta(\eta \mid \bar{x}, b)$ is the pushforward density $q = \varphi(\cdot; b) \circ \pi_u$ with φ given by

$$\varphi(u; b) = (\bar{x}, \eta) \text{ where } \begin{cases} \bar{x} &= \varphi_{\bar{x}}(u_{\bar{x}}; b) \\ \eta &= \varphi_\eta(u_\eta; \bar{x}, b) \end{cases} \tag{41}$$

and $u = (u_{\bar{x}}, u_\eta)$. Since $\bar{x}$ does not depend on u_η, the Jacobian J_φ of φ with respect to u is block-triangular with $J_{\varphi_{\bar{x}}}$ and $J_{\varphi_\eta} = \Sigma_\eta^{1/2}$ on the diagonal. Therefore, the update rule for the pushforward $q = \varphi(\cdot; b) \circ \pi_u$ is given by:

$$\begin{aligned} \log q(\bar{x}, \eta \mid b) &= \log \pi_u(u) - \log|J_\varphi(u; b)|, \\ \log|J_\varphi(u; b)| &= \log|J_{\varphi_{\bar{x}}}(u_{\bar{x}}; b)| + \frac{1}{2} \log|\Sigma_\eta|. \end{aligned} \tag{42}$$

3.5 Training

3.5.1 Variational Inference

We train the joint flow φ in (41) using amortized variational inference [30, 48–50]. For the approximate posterior $q = \varphi(\cdot; b) \circ \pi_u$ from (42), the evidence lower bound (ELBO) is given by:

$$\begin{aligned} \log p(b) &\geq \mathbb{E}_q\big[\log p(b, \bar{x}, \eta) - \log q(\bar{x}, \eta \mid b)\big] \\ &= \mathbb{E}_q\big[\log p(b \mid \bar{x}, \eta)\big] - \mathrm{KL}\big[q(\bar{x}, \eta \mid b) \,\big\|\, p(\bar{x}, \eta)\big] \\ &= \mathbb{E}_q\big[\log p(b, \bar{x}, \eta)\big] + \mathcal{H}\big(q(\bar{x}, \eta \mid b)\big) \end{aligned} \tag{43}$$

where $\mathrm{KL}[q \,\|\, p] = \mathbb{E}_q[\log \frac{q}{p}]$ is the *Kullback–Leibler (KL) divergence* and $\mathcal{H}[q] = \mathbb{E}_q[-\log q]$ is the *differential entropy*. Maximizing the ELBO in (43) has three equivalent interpretations: (1) maximize the likelihood of the data while keeping q close to the prior; (2) maximize both the joint probability of the data and the entropy of q; and (3) minimize $\mathrm{KL}[q(\bar{x}, \eta \mid b) \,\|\, p(\bar{x}, \eta \mid b)]$ between q and the true posterior[4].

While the KL divergence and entropy have closed forms for Gaussian base distributions π_u, a fully Monte Carlo estimate of the ELBO has lower variance when q approaches the true posterior [51]. We therefore use the first form of the ELBO in (43) and define the *free energy* $\mathcal{F}[q]$ to be its negative. In terms of φ, the free energy is given by:

$$\begin{aligned} \mathcal{F}[q] &= \mathbb{E}_{\pi_u}\left[\log q(\bar{x}, \eta \mid b) - \log p(b, \bar{x}, \eta)\right] \\ &= \mathbb{E}_{\pi_u}\left[\log \pi_u(u) - \log|J_\varphi(u; b)| - \log p(b, \bar{x}, \eta)\right] \end{aligned} \tag{44}$$

where $(\bar{x}, \eta) = \varphi(u; b)$. Given a dataset $\{b\}$ with empirical distribution $\tilde{p}(b)$, the objective is to minimize the free energy:

$$\mathcal{L}_{\text{free}} = \mathbb{E}_{\tilde{p}}\left[\mathcal{F}[q]\right] \text{ where } q = \varphi(\cdot; b) \circ \pi_u. \tag{45}$$

Training optimizes the encoder, flow backbone, shared low-pass window, and nuisance-prior parameters; the nuisance Laplace approximation has no trainable parameters.

3.5.2 Path Gradient

To reduce the variance of mini-batch gradient estimates of the free energy, especially as q approaches the true posterior, we use the *path gradient* [51–53]:

$$\blacktriangledown_\phi f(T(u; \phi), \phi) \coloneqq \big(\nabla_\phi f(T(u; \phi), \phi')\big)_{\phi' = \phi} \tag{46}$$

where f is a function of the output of the flow T and its parameters ϕ; $\blacktriangledown_\phi f$ treats f as a constant function of ϕ, only accumulating gradients through the flow "path" $T(u; \phi)$. Let the flow φ in (44) have parameters ϕ, and let sg denote the stop-gradient which holds ϕ constant with respect to ∇_ϕ. Then, $\blacktriangledown_\phi \log \frac{q}{p} = \nabla_\phi \log \frac{q_{\text{sg}}}{p}$, and we have

$$\blacktriangledown_\phi \mathcal{F}[q] = \mathbb{E}_{\pi_u}\left[\nabla_\phi \left(\log q_{\text{sg}}(\bar{x}, \eta \mid b) - \log p(b, \bar{x}, \eta)\right)\right]. \tag{47}$$

The reduced finite-sample variance of (47) comes at the cost of evaluating both the forward flow for $(\bar{x}, \eta) = \varphi(u; b)$ and the inverse flow for $\log\left|J_{\varphi_{\text{sg}}^{-1}}\right|$. Note that (47) is an unbiased estimate of the total gradient, i.e. $\blacktriangledown_\phi \mathcal{F}[q] = \nabla_\phi \mathcal{F}[q]$, since the *score* satisfies $\mathbb{E}_q[\frac{\partial}{\partial \phi} \log q] = 0$.

[4] To see this, subtract $\log p(b)$ from both sides of (43) and use Bayes' rule.

3.5.3 Self-Supervised Regularization

We regularize the flow $q(\bar{x}, \eta \mid b)$ using synthetic data. First, labels $(\bar{x}, \eta) \sim p(\bar{x}, \eta)$ are sampled from the prior. Then, synthetic data $b \sim p(b \mid \bar{x}, \eta)$ is simulated using the quantized Rician likelihood model (10). The objective is to maximize the likelihood of the labels $(\bar{x}, \eta)$ under q conditioned on the simulated signal b:

$$\mathcal{L}_{\text{self}} = \mathbb{E}_{p(\bar{x},\eta)}\mathbb{E}_{p(b|\bar{x},\eta)}\big[-\log q(\bar{x}, \eta \mid b)\big]. \tag{48}$$

We use its marginal counterpart, supervising the spectrum posterior rather than the conditional nuisance approximation:

$$\mathcal{L}'_{\text{self}} = \mathbb{E}_{p(\bar{x},\eta)}\mathbb{E}_{p(b|\bar{x},\eta)}\big[-\log q_{\bar{x}}(\bar{x} \mid b)\big]. \tag{49}$$

The omitted nuisance term depends on the shared learned prior parameters, so the marginal objective is a deliberate choice rather than an equivalent joint objective. Neither the prior nor likelihood sampling processes are differentiated; the sampled labels and data are treated as fixed. Evaluating $\mathcal{L}'_{\text{self}}$ requires an additional encoding of b and pushing of $\bar{x}$ through the inverse marginal flow $\varphi_{\bar{x}}^{-1}$.

For a fixed prior and signal model, minimizing $\mathcal{L}'_{\text{self}}$ minimizes the expected forward KL divergence between the marginal spectrum posteriors, since

$$\begin{aligned}\mathcal{L}'_{\text{self}} &= \mathbb{E}_{p(b)}\, \mathcal{H}[p(\bar{x} \mid b), q_{\bar{x}}(\bar{x} \mid b)] \\ &= \mathbb{E}_{p(b)}\, \mathrm{KL}[p(\bar{x} \mid b) \,\|\, q_{\bar{x}}(\bar{x} \mid b)] + \mathbb{E}_{p(b)}\, \mathcal{H}[p(\bar{x} \mid b)]\,.\end{aligned}$$

The entropy term is independent of $q_{\bar{x}}$ under this fixed model. This marginal forward KL encourages $q_{\bar{x}}$ to cover spectrum-posterior modes, complementing the reverse-KL free-energy objective (44).

3.5.4 Encoder Regularization

Regularizing the encoder outputs $z \in \mathbb{R}^{n_z}$ is necessary to ensure stable representations and prevent overfitting. A common approach is ℓ^2-regularization, equivalent to an i.i.d. unit normal prior on each z_i. Inspired by VICReg [54], we find that decorrelating z components improves encodings. We use half the squared Gaussian Wasserstein-2 distance:

$$\mathcal{L}_{\text{enc}} = \tfrac{1}{2}\mathcal{W}_2^2(\mathcal{N}(\mu, \Sigma), \mathcal{N}(0, \mathrm{I})) = \sum_{i=1}^{n_z} \frac{\mu_i^2 + (\sqrt{\lambda_i} - 1)^2}{2} \tag{50}$$

$$\nabla_\mu \mathcal{L}_{\text{enc}} = \mu, \quad \nabla_\Sigma \mathcal{L}_{\text{enc}} = \frac{1}{2}(\mathrm{I} - \Sigma^{-1/2}) \tag{51}$$

where μ, Σ are the mean and covariance of the encodings $\{z^j\}$ of the current mini-batch of real and synthetic signals, and λ_i are the eigenvalues of Σ. Minimizing $\mathcal{L}_{\text{enc}}$ pushes $\mu \to 0$ and $\Sigma \to \mathrm{I}$, whitening the components of z.

We find further improvement by normalizing encodings to unit RMS norm, i.e. $\|z\|_{\text{RMS}}^2 = \frac{1}{n_z}\|z\|_2^2 = 1$, constraining z to the sphere $S_r^{n_z - 1}$ with radius $r = \sqrt{n_z}$. The loss $\mathcal{L}_{\text{enc}}$ then encourages zero mean and identity covariance on this sphere.

3.5.5 Training

Training consisted of 160 000 optimizer updates over approximately 44 h on one NVIDIA Tesla V100 GPU with 32 GB memory and six CPU threads on UBC ARC Sockeye [55]. The full training objective is

$$\mathcal{L} = \mathcal{L}_{\text{free}} + \lambda_{\text{self}}\mathcal{L}'_{\text{self}} + \lambda_{\text{enc}}\mathcal{L}_{\text{enc}}, \tag{52}$$

combining the variational free energy (45), the self-supervised forward-KL loss (49), and the Wasserstein-2 encoding regularization (50). The variational free energy is a reverse-KL objective over batches of measured data b, whereas the forward-KL loss requires batches of samples from the prior $p(\bar{x}, \eta)$ and forward simulated synthetic data $b \sim p(b \mid \bar{x}, \eta)$. We use equal weighting $\lambda_{\text{self}} = 1$ throughout, encouraging the model to learn posteriors over both synthetic signals and real signals equally well. The encoder weighting was also $\lambda_{\text{enc}} = 1$.

The flow and encoder were optimized using preconditioned stochastic gradient descent (PSGD) with Kronecker-factored triangular preconditioners [56, 57] and schedule-free momentum [58, 59]. We use preconditioner learning rate $\eta_Q = 0.1$, update RMS-clip of 3, schedule-free momentum with $\beta_1 = 0.99$, base learning rate 10^{-2}, a linear warm-up from zero over 500 optimizer steps, and decoupled weight decay $\lambda_{\text{wd}} = 10^{-3}$ on dense weight matrices.

Training used a batch size of 16 384 with online data augmentation: signals and their corresponding EPG rows were retained at a uniformly chosen ordered subset of n'_{TE} echoes, where $P(n'_{\text{TE}} = n) \propto e^{\lambda n}$ for $n \in \{16, \ldots, 56\}$ and $\lambda > 0$ sets $\mathbb{E}[n'_{\text{TE}}] = 48$, then zero-padded; quantized signals were uniformly dequantized and requantized using $\delta' = \delta Z$, where Z is log-normal with mean 1 and standard deviation 0.5; the DECAES flip-angle estimate α was perturbed by zero-mean Gaussian noise with standard deviation 1°; and T_1 and the T_2 range were randomized by drawing T_2^{min} uniformly from $[\text{TE} - 1\,\text{ms}, \text{TE} + 3\,\text{ms}]$ and T_2^{max} and T_1 uniformly from $[1\,\text{s}, 2\,\text{s}]$.

3.6 Inference

Given a trained flow φ and corresponding approximate posterior $q = \varphi(\cdot; b) \circ \pi_u$, we must choose a posterior sampling method. Even limiting ourselves to MCMC methods, there exists a dauntingly large literature of approaches to choose from [60]. For this reason, we consider an exhaustive comparison of MCMC methods beyond the scope of this work. Instead, we compare three established, simple methods and leave the investigation of more advanced techniques for future work.

(1) Direct and Importance-Weighted Sampling The simplest inference method samples directly from the approximate posterior $q(\bar{x}, \eta \mid b)$. This requires only a single forward pass through the flow per sample, but relies entirely on the model's generalization and may not fully capture the true posterior structure [61]. If only $\bar{x}$ is needed, one can directly sample the marginal posterior $q_{\bar{x}}(\bar{x} \mid b)$.

Importance-weighted sampling reweights proposal samples using the importance weights:

$$\begin{cases} (\bar{x}_t, \eta_t) \sim q(\bar{x}, \eta \mid b), \\ w_t = \frac{p(b \mid \bar{x}_t, \eta_t) p(\bar{x}_t, \eta_t)}{q(\bar{x}_t, \eta_t \mid b)} \propto \frac{p(\bar{x}_t, \eta_t \mid b)}{q(\bar{x}_t, \eta_t \mid b)}. \end{cases} \tag{53}$$

Posterior expectations are then estimated as weighted averages,

$$\mathbb{E}_{p(\bar{x}, \eta \mid b)}(f(\bar{x}, \eta)) \approx \sum_{t=1}^{T} \tilde{w}_t f(\bar{x}_t, \eta_t), \tag{54}$$

where $\tilde{w}_t = w_t / \sum_{t'} w_{t'}$. The accuracy of importance sampling depends critically on the proposal distribution; poorly matched proposals yield high-variance estimates [41, 62]. Neural networks have been used to improve proposals, though careful training is necessary to prevent mode-seeking behaviour [63].

(2) Metropolis–Hastings (MH) Algorithm Given an initial sample $u_0 \sim \pi_u(u)$ and a *proposal distribution* $Q(\tilde{u} \mid u)$, a sequence $(u_t)_{t\geq 1}$ of posterior samples is generated as follows [64, 65]:

$$\begin{cases} \tilde{u} & \sim Q(\tilde{u} \mid u_t) \\ A_t & = \min\left\{1, \frac{p(\tilde{u}|b)}{p(u_t|b)} \frac{Q(u_t|\tilde{u})}{Q(\tilde{u}|u_t)}\right\} \\ & = \min\left\{1, \frac{p(b|\tilde{\bar{x}},\tilde{\eta})}{p(b|\bar{x}_t,\eta_t)} \frac{p(\tilde{\bar{x}},\tilde{\eta})}{p(\bar{x}_t,\eta_t)} \frac{|\det(\tilde{J}_\varphi)|}{|\det(J_{\varphi,t})|} \frac{Q(u_t|\tilde{u})}{Q(\tilde{u}|u_t)}\right\} \\ u_{t+1} & = \tilde{u} \text{ with probability } A_t, \text{ else } u_t. \end{cases} \tag{55}$$

In the experiments below, MH uses the learned approximate posterior as an independent proposal distribution. Equivalently, in u-space we set $Q(\tilde{u} \mid u) = \mathcal{N}(0, \mathrm{I})$, draw each proposed $\tilde{u}$ from the base distribution, and map it through φ. If φ is well trained, then $q(\bar{x}, \eta \mid b) \approx p(\bar{x}, \eta \mid b)$ and the acceptance ratio is high. If the approximation is poor, states with large target-to-proposal density ratios can be retained through repeated rejections. Random-walk Metropolis (RWM) proposals such as $Q(\tilde{u} \mid u) = \mathcal{N}(u, \sigma^2_{\mathrm{RWM}}\mathrm{I})$ may also be used.

(3) Metropolis-Adjusted Langevin Algorithm (MALA) The unadjusted Langevin algorithm (ULA) [66] generates posterior samples by performing noisy gradient ascent on the log-posterior starting from $u_0 \sim \pi_u(u)$:

$$u_{t+1} = u_t + \tau_t \nabla_u \log p(u_t \mid b) + \sqrt{2\tau_t}\xi_t, \tag{56}$$

where τ_t is the step size, and $\xi_t \sim \mathcal{N}(0, \mathrm{I})$. Under suitable regularity and decreasing-step-size conditions, the distribution of u_t approaches the stationary distribution $p(u \mid b)$. By Bayes' theorem, the gradient of the log-posterior can be exactly computed as

$$\begin{aligned} \nabla_u \log p(u_t \mid b) &= \nabla_u(\log p(\bar{x}_t, \eta_t \mid b) + \log|J_{\varphi,t}|) \\ &= \nabla_u(\log p(b, \bar{x}_t, \eta_t) + \log|J_{\varphi,t}|). \end{aligned} \tag{57}$$

Langevin methods efficiently explore high-probability regions by leveraging gradient information, making them particularly effective in high dimensions compared to random walk Metropolis algorithms. However, fixed step sizes $\tau_t = \tau$ generally introduce discretization bias in the stationary distribution [67].

The *Metropolis-adjusted Langevin algorithm (MALA)* corrects this issue by incorporating a Metropolis–Hastings accept-or-reject step [68] using a proposal based on the ULA (57):

$$\begin{cases} Q(\tilde{u} \mid u_t) & = \mathcal{N}(u_t + \tau \nabla_u \log p(u_t \mid b), 2\tau\mathrm{I}) \\ u_{t+1} & = \text{Metropolis–Hastings step (55)}. \end{cases} \tag{58}$$

For sufficiently regular product target distributions in the high-dimensional limit, MALA achieves an asymptotically optimal mean acceptance rate near $A \approx 0.574$ [68].

3.6.1 Flow Inference Implementation Details

All sampling methods operate over the joint base-space coordinates $u \in \mathbb{R}^{n_u}$, with $n_u = n_{T_2} + 1$, using the learned flow $\varphi(\cdot; b)$ as a change of variables onto the joint parameter space over $(\bar{x}, \eta)$, with one chain initialized at $u_0 \sim \pi_u = \mathcal{N}(0, \mathrm{I})$ per voxel.

Because the trained flow approximately normalizes the base-space posterior in the sense that $p(u \mid b) \approx \mathcal{N}(0, \mathrm{I})$, the RWM and MALA proposal parameters may be set analytically using the constants derived for a standard normal target, rather than tuned: we use the RWM proposal variance $\sigma^2_{\mathrm{RWM}} = 2.382^2/(n_u - 1)$ [69] and the MALA step size $\tau \approx 1.650^2/(2\sqrt[3]{n_u})$,

equivalently a proposal standard deviation of $\sqrt{2\tau} \approx 1.650/\sqrt[6]{n_u}$ [68]. The numerical scale factors are asymptotically optimal for standard normal target distributions, achieving the more generally optimal acceptance rates $A^*_{\mathrm{RWM}} = 0.234$ and $A^*_{\mathrm{MALA}} = 0.574$.

We report posterior means and central 95 % credible intervals (CIs), summarized by their half-widths, for both $\bar{x}$ and scalar biomarkers of interest: the myelin (MWF), intra-/extracellular (IEWF), and cerebrospinal (CWF) water fractions, and the geometric mean T_2 over the full spectrum (GM T_2) and myelin-water pool (MW T_2). Data is batched across voxels, computing encoder and flow quantities in parallel on the GPU; the exception is the no-U-turn sampler, which, due to dynamic chain lengths, was run on the CPU (see section 4.4).

3.7 Joint Prior

The joint prior over the inference variables $(\bar{x}, \eta)$ is taken to be factorized as

$$p(\bar{x}, \eta) = p(\bar{x})p(\log s)p(\log \sigma), \tag{59}$$

reflecting the modelling assumption that the normalized T_2 spectrum of the underlying tissue is approximately independent of the per-voxel signal scale and noise level.

Low-Pass Prior on $\bar{x}$. The prior on $\bar{x}$ is given by the shallow low-pass distribution of equation (22), obtained by pushing a standard Gaussian $\pi_{u_{\bar{x}}} = \mathcal{N}(0, \mathrm{I})$ through a diagonal window filter $\Lambda = \mathrm{diag}\left(\lambda_1^+, \ldots, \lambda_{n_{T_2}-1}^+\right)$ in DCT frequency space and then applying the inverse ILR transform. We use a scaled Gaussian window

$$\lambda_i = \lambda_0 \exp\!\left(-(f_i/F_c)^2/2\right), \tag{60}$$

where $f_i = i/n_{T_2}$ indexes the non-zero DCT frequency bins, $F_c > 0$ is a learnable cutoff, and $\lambda_0 > 0$ is a learnable amplitude. Because the low-pass filtering is applied to DCT coefficients corresponding to $\log \bar{x}$ over the $\log T_2$ grid (appendix E), F_c is a cutoff *in log-T_2 frequency*, and consequently, the learned width is scale-invariant in T_2: altering $T_2^{\min}$ or $T_2^{\max}$ rescales the $\log T_2$ grid but does not change the relative smoothness of spectra favoured by the prior. The amplitude λ_0 scales the low-passed coefficients, sharpening the T_2 peaks by increasing the amplitude of the logits as λ_0 grows, concentrating the softmax distribution. A positive floor $\lambda_{\min} > 0$ is also added to λ_i in quadrature via $\lambda_i^+ = \sqrt{\lambda_i^2 + \lambda_{\min}^2}$ – equivalent in distribution to adding isotropic Gaussian noise of variance $\lambda_{\min}^2$ in ILR space – preventing complete attenuation of high-frequency components to zero. This same window Λ forms the terminal layer of the generator flow $\varphi_{\bar{x}}$ such that the prior and the flow naturally reside in the same space of smooth T_2 spectra.

Priors on $\eta = (\log s, \log \sigma)$. Both components of η are assigned independent Gaussian priors. The log-scale prior $p(\log s) = \mathcal{N}(\log s \mid \mu_{\log s,0}, \sigma^2_{\log s,0})$ is initialized such that $\mathbb{E}[s] = 1$ and the scale is $s \approx 1 \pm 0.05$. The log-noise prior $p(\log \sigma) = \mathcal{N}(\log \sigma \mid \mu_{\log \sigma,0}, \sigma^2_{\log \sigma,0})$ is initialized with $\mu_{\log \sigma,0} = -30\log(10)/20$ and $\sigma_{\log \sigma,0} = 10\log(10)/20$; for $\mathrm{SNR}_{\mathrm{dB}} = -20\log_{10}\sigma$, these values correspond to mean 30 dB and standard deviation 10 dB. The hyperparameters $(\mu_{\eta,0}, \Sigma_{\eta,0})$, window cutoff F_c, amplitude λ_0, and floor $\lambda_{\min}$ (initialized at 0.1) are learned jointly by approximate empirical Bayes during training, and the same nuisance priors are shared inside Laplace's approximation (14). At the final 160 000-update checkpoint, the learned window has $F_c = 0.2093$ for $n_{T_2} = 41$ over 8–1000 ms, $\lambda_0 = 8.041$, and $\lambda_{\min} = 0.02695$. The learned Gaussian priors on $\log s$ and $\log \sigma$ have mean/standard-deviation pairs $(-0.002845, 0.005426)$ and $(-5.6575, 0.6774)$, respectively. Using fixed floors between 0.025 and 0.4 changed the held-out free energy by approximately 0.01 nats per voxel.

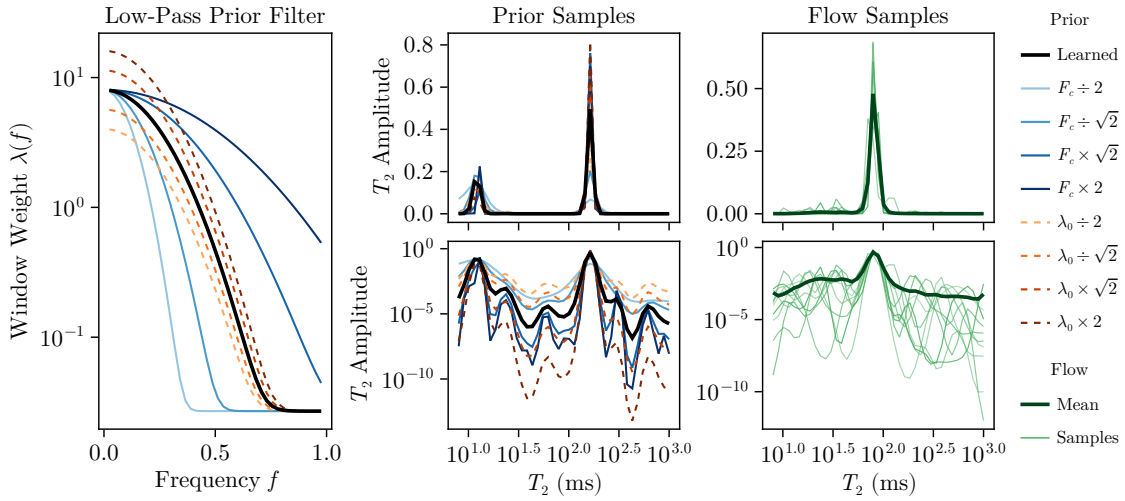


Figure 5: Learned low-pass window Λ and its effect on T_2 spectra. The left panel shows the learned window weights $\lambda(f)$ alongside similar weights given by scaling either the cutoff F_c or the amplitude λ_0 by factors $\frac{1}{2}$, $\frac{1}{\sqrt{2}}$, $\sqrt{2}$, and 2. The middle panels show the effect of the differing parameters on the T_2 spectrum by pushing a fixed sample u through each window. The cutoff determines the overall smoothness while the amplitude controls the peak sharpness. The right panels show ten posterior samples from the trained flow for a representative voxel, shaded by importance weight, alongside the mean over 1000 samples.

4 Results

4.1 Data and Experimental Setup

Datasets. All images were acquired with a 32-channel head coil on a Philips 3 T Ingenia Elition X scanner, subsequently upgraded to an MR 7700 (table 1). The studies were approved by the University of British Columbia Clinical Research Ethics Board. All participants provided written informed consent. Images were split by subject into training (376 images, 156 subjects; 298 acquired before and 78 after the upgrade), validation (25 images, 11 subjects), and test (94 images, 34 subjects) partitions, with no subject in more than one partition. The partitions were separated before voxel extraction, and the validation partition was reserved for training monitoring. Training and validation images, and the 21 in-distribution (ID) test images from 11 held-out subjects, were acquired using a 3D MSE sequence with 56 echoes at 7 ms uniform echo spacing. Acquired voxels are $1.5 \times 2 \times 3\,\mathrm{mm}^3$ (3 mm slices with 1.5 mm overlap), reconstructed at $1 \times 1 \times 1.5\,\mathrm{mm}^3$, except for 12 images (10 training, 1 validation, and 1 ID test) acquired at $2.5 \times 3 \times 3\,\mathrm{mm}^3$ and reconstructed at 1.5 mm isotropic. Quantitative in vivo evaluations used 1000 masked voxels sampled uniformly from the ID test images. For visualization, we also used an out-of-distribution (OOD) test image from a further held-out subject, acquired with 48 echoes at 8 ms spacing and 5 mm slices, reconstructed at $0.96 \times 0.96 \times 2.5\,\mathrm{mm}^3$, using a protocol excluded from training; a representative axial slice was used for the in vivo maps.

Brain masks were computed from the first echo with the Brain Extraction Tool (BET) [70] (fractional intensity threshold 0.6, robust centre estimation) using the image voxel sizes; for fields of view extending into the neck, the image was first reoriented to standard orientation and cropped with `robustfov`. The training partition retained all masked voxels except for those with zero first-echo magnitude. Signals are stored as integer echo amplitudes together

Table 1: Image counts by scanner and acquisition parameters for each partition.

	Training	Validation	Test ID	Test shifted
Total images	376	25	21	73
Scanner (Philips, 3 T)				
Ingenia Elition X	298	20	14	73
MR 7700	78	5	7	0
Echoes / spacing (ms)				
18 / 16	0	0	0	7
24 / 12	0	0	0	7
30 / 9.6	0	0	0	7
36 / 8	0	0	0	7
48 / 6	0	0	0	6
48 / 6.05	0	0	0	1
48 / 7.5	0	0	0	1
48 / 8	0	0	0	21
56 / 7	376	25	21	16
Repetition time (s)				
1.07–1.14	376	25	21	62
1.5	0	0	0	6
2.0	0	0	0	4
3.0	0	0	0	1
Slice thickness / spacing (mm)				
3 / 1.5	376	25	21	15
3 / 1.8	0	0	0	35
4 / 2	0	0	0	3
5 / 2.5	0	0	0	20
Reconstructed voxel size (mm^3)				
$0.936 \times 0.936 \times 1.5$	0	0	0	1
$0.938 \times 0.938 \times 1.5$	0	0	0	2
$0.938 \times 0.938 \times 2$	0	0	0	3
$0.958 \times 0.958 \times 2.5$	0	0	0	19
$1 \times 1 \times 1.5$	366	24	20	12
$1 \times 1 \times 2.5$	0	0	0	1
$1.5 \times 1.5 \times 1.5$	10	1	1	0
$1.797 \times 1.797 \times 1.8$	0	0	0	35

with the signal bin width δ from the scan parameters; signals b and bin widths δ were normalized by the DECAES estimate of the initial signal magnitude, and the quantized Rician likelihood uses the resulting bin width δ. Echo trains were zero-padded to $n_{\mathrm{TE}} = 56$ echoes if needed.

The EPG basis matrix A is constructed on a log-spaced T_2 grid using $n_{T_2} = 41$ bins over $[T_2^{\min}, T_2^{\max}]$. This gives even Aitchison dimension $n_{T_2} - 1 = 40$, convenient for the coupling layers in the flow backbone. At evaluation, $T_2^{\min} = \mathrm{TE}$ and $T_2^{\max} = T_1 = 1\,\mathrm{s}$; during training, $T_2^{\min}$, $T_2^{\max}$, T_1, and the flip angle are jittered by the augmentation pipeline of section 3.5.5.

Synthetic Evaluation Sets. We generated two synthetic test sets of 1000 spectra each. ID spectra were sampled from the learned prior $p(\bar{x})$; we refer to them as prior-sampled spectra. We also generated OOD synthetic T_2 spectra with two Gaussian lobes in $\log T_2$. The lobe centres were drawn uniformly from 10–25 ms and 50–100 ms, log-standard deviations from $[0.1, 0.2]$, and short-T_2 mixture weights from $[0, 0.25]$. For both sets, signals were simulated using the same EPG forward model as in training and acquisition settings from the held-out ID test cohort, including voxelwise DECAES flip-angle estimates and normalized bin widths. Ground-truth biomarkers were computed from the simulated T_2 spectra. Quantized Rician noise was added at $\mathrm{SNR} \in \{30\,\mathrm{dB}, 40\,\mathrm{dB}, 50\,\mathrm{dB}\}$, with signal scale $s = 1$ and $\sigma = 10^{-\mathrm{SNR}_{\mathrm{dB}}/20}$. The lowest SNR, 30 dB, is below typical in vivo levels. All Bayesian methods and DECAES baselines were evaluated on the same 1000 synthetic voxels and noise realizations for direct comparison.

Baselines, References, and Metrics. DECAES was run using default settings with three regularization parameter selection methods (L-curve, GCV, and χ^2) [1].

Posterior references use AdvancedHMC.jl [71], with 500 HMC chains per voxel (1000 warmup and 1000 retained draws) and 32 NUTS chains (5000 warmup and 5000 retained draws). Both use the learned coordinates of the transformed posterior density. Chains start from standard-normal draws and adapt their step size and diagonal mass to target acceptance 0.8; HMC uses 16 leapfrog steps. Identity-backbone HMC in section 4.4 instead investigates the unpreconditioned geometry.

NUTS evaluates densities, gradients, and sampler coordinates in double precision. HMC evaluates log densities and gradients in single precision, while its internal state (positions, momenta, diagonal mass matrices, step sizes, and acceptance ratios) uses double precision. The full NUTS reference covered 96 held-out ID voxels (15 360 000 retained draws). Its maximum rank-normalized split $\hat{R}$ was 1.0006, and minimum spectrum-bin bulk and tail effective sample sizes were 93 878 and 96 951. There were 46 retained divergences across 28 chains and 15 voxels, and 27 retained depth-limit transitions; mean acceptance was 0.874. The three displayed HMC references had maximum $\hat{R} = 1.0018$, minimum bulk/tail spectrum ESS 297 989/267 155, and 0 numerical-error iterations. Across the 12 displayed voxel/biomarker pairs, the maximum differences from the full NUTS means were 1.20 combined Monte Carlo standard errors for MALA and 1.65 for the displayed, thinned HMC estimator. Monte Carlo errors use autocorrelation-adjusted MALA errors, full-chain NUTS diagnostics, and the independent retained-chain means for the thinned HMC estimator.

Removing all NUTS chains with retained divergences shifted the four biomarker means by at most 0.77 full-reference Monte Carlo standard errors across the cohort, and 0.23 for the displayed voxels. This sensitivity analysis and the independent sampler comparisons do not exclude divergence-related bias.

Accuracy on synthetic voxels is reported as the mean absolute error (MAE) of MWF in % and geometric mean T_2 in ms, and as the Wasserstein-1 (W_1) distance between estimated and ground-truth normalized T_2 distributions over $\log T_2$. Bayesian methods additionally

Table 2: Reference mixing diagnostics before pooling or thinning, across the voxels listed. The $\hat{R}$ maximum is over base coordinates, and bulk/tail effective-sample minima are over spectrum bins. Errors count retained HMC iterations with any numerical error and divergent NUTS transitions, respectively.

Reference	Voxels	Chains	Max. $\hat{R}$	Min. bulk ESS	Min. tail ESS	Errors
HMC	3	500	1.0018	297989	267155	0
NUTS	96	32	1.0006	93878	96951	46

report 95 % credible-interval coverage of MWF and GM T_2, defined as the fraction of voxels for which the true value lies within the posterior 95 % CI.

4.2 Synthetic Data Evaluation

We first evaluated each method on the prior-sampled spectra described in section 4.1, comparing DECAES against four Bayesian estimators: direct sampling, importance-weighted (IW) sampling, Metropolis–Hastings (MH), and the Metropolis-adjusted Langevin algorithm (MALA). Direct, IW, and MH estimates consumed 1000 flow proposals per voxel, with MALA also requiring 1000 gradient evaluations. DECAES methods compute point estimates for comparison but report no analogous Bayesian coverage metrics.

Table 3 shows similar recovery errors across the Bayesian estimators, with lower mean errors than all DECAES methods on each reported recovery metric. Figure 6 illustrates this for a representative voxel, where the MALA posterior concentrates near the true spectrum, in contrast to the less accurate DECAES point estimates.

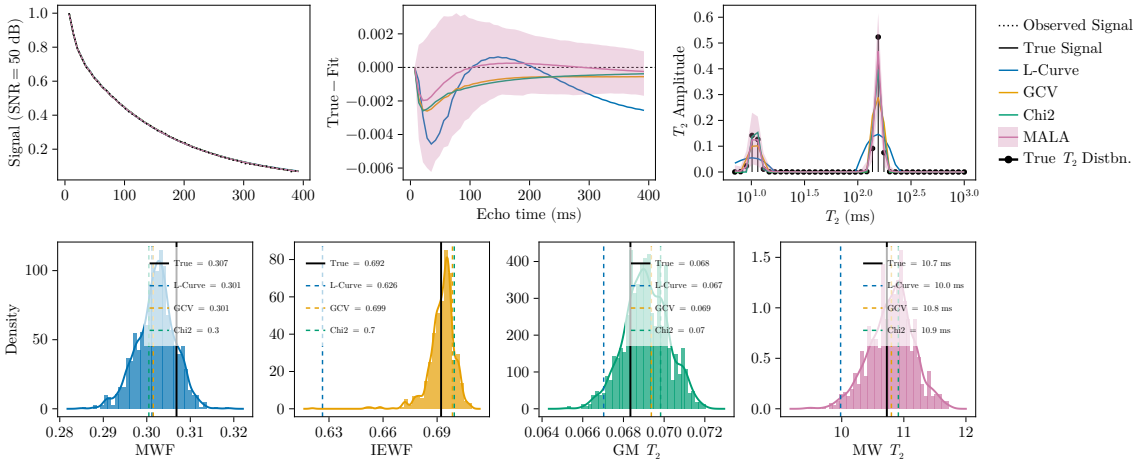


Figure 6: Inference comparison for the first prior-sampled spectrum at 50 dB. Top row: multi-spin echo signal decay, signal fit residual, and recovered T_2 spectra, comparing the DECAES estimates with MALA posterior means and 95 % credible bands. Bottom row: MALA posterior histograms for the MWF, IEWF, GM T_2, and MW T_2.

4.2.1 Posterior Calibration

A well-calibrated approximate posterior should produce observed coverage close to nominal coverage over a range of credible levels. Figure 7 compares the coverages for MALA, MH, and IW at SNR = 30 dB, 40 dB, and 50 dB. The three methods' calibration curves overlapped

Table 3: T_2 distribution recovery metrics over a batch of 1000 spectra drawn from the learned prior at SNR levels of 30 dB, 40 dB, and 50 dB. MWF MAE in %, geometric mean T_2 MAE, and Wasserstein-1 (W_1) distances between estimated and ground-truth T_2 distributions are reported as mean ± standard deviation. 95 % CI coverage of MWF and GM T_2 is reported for the Bayesian methods. Bold marks the lowest displayed mean error at each SNR.

	Method	Mean errors			95% coverage	
		MWF (%)	GM T_2 (ms)	W_1	MWF	GM T_2
SNR = 30 dB	L-Curve	9.5 ± 10.6	28.9 ± 47.7	0.44 ± 0.21	—	—
	GCV	6.9 ± 7.9	26.4 ± 52.9	0.34 ± 0.14	—	—
	Chi2	6.8 ± 7.9	26.5 ± 52.8	0.34 ± 0.14	—	—
	Direct	3.8 ± 5.8	6.1 ± 10.4	$\mathbf{0.18 \pm 0.11}$	93.6%	94.9%
	IW	$\mathbf{3.7 \pm 5.7}$	$\mathbf{5.8 \pm 9.6}$	$\mathbf{0.18 \pm 0.11}$	93.7%	93.9%
	MH	$\mathbf{3.7 \pm 5.6}$	$\mathbf{5.8 \pm 9.6}$	$\mathbf{0.18 \pm 0.11}$	93.1%	93.7%
	MALA	$\mathbf{3.7 \pm 5.6}$	$\mathbf{5.8 \pm 9.7}$	$\mathbf{0.18 \pm 0.11}$	93.6%	94.2%
SNR = 40 dB	L-Curve	4.9 ± 6.8	12.9 ± 22.3	0.27 ± 0.1	—	—
	GCV	3.7 ± 5.3	12.0 ± 22.2	0.21 ± 0.1	—	—
	Chi2	3.6 ± 5.0	12.1 ± 22.7	0.2 ± 0.1	—	—
	Direct	$\mathbf{2.1 \pm 3.5}$	$\mathbf{3.2 \pm 5.9}$	$\mathbf{0.11 \pm 0.08}$	95.1%	96.9%
	IW	$\mathbf{2.1 \pm 3.5}$	$\mathbf{3.2 \pm 5.8}$	$\mathbf{0.11 \pm 0.08}$	94.7%	96.7%
	MH	$\mathbf{2.1 \pm 3.5}$	$\mathbf{3.2 \pm 5.8}$	$\mathbf{0.11 \pm 0.08}$	94.9%	96.5%
	MALA	$\mathbf{2.1 \pm 3.5}$	$\mathbf{3.2 \pm 5.8}$	$\mathbf{0.11 \pm 0.08}$	94.7%	96.4%
SNR = 50 dB	L-Curve	3.2 ± 5.1	6.3 ± 11.8	0.18 ± 0.08	—	—
	GCV	2.5 ± 4.1	5.7 ± 11.2	0.14 ± 0.08	—	—
	Chi2	2.2 ± 3.7	5.7 ± 11.1	0.13 ± 0.08	—	—
	Direct	$\mathbf{1.4 \pm 2.6}$	2.2 ± 4.1	$\mathbf{0.08 \pm 0.06}$	95.0%	98.7%
	IW	$\mathbf{1.4 \pm 2.5}$	$\mathbf{2.1 \pm 4.0}$	$\mathbf{0.08 \pm 0.06}$	95.2%	98.0%
	MH	$\mathbf{1.4 \pm 2.5}$	$\mathbf{2.1 \pm 4.0}$	$\mathbf{0.08 \pm 0.06}$	95.0%	98.0%
	MALA	$\mathbf{1.4 \pm 2.5}$	$\mathbf{2.1 \pm 4.0}$	$\mathbf{0.08 \pm 0.06}$	94.4%	97.3%

closely. MWF coverage remained close to nominal; GM T_2 intervals were conservative at 40 and 50 dB. At 30, 40 and 50 dB, nominal 95 % MWF intervals covered 93.1–93.7 %, 94.7–95.1 %, and 94.4–95.2 % of true values across the four Bayesian estimators (table 3). GM T_2 coverage was 93.7–94.9 %, 96.4–96.9 %, and 97.3–98.7 %, respectively. MW T_2 coverage was 92.6–94.9 % across the tested SNRs.

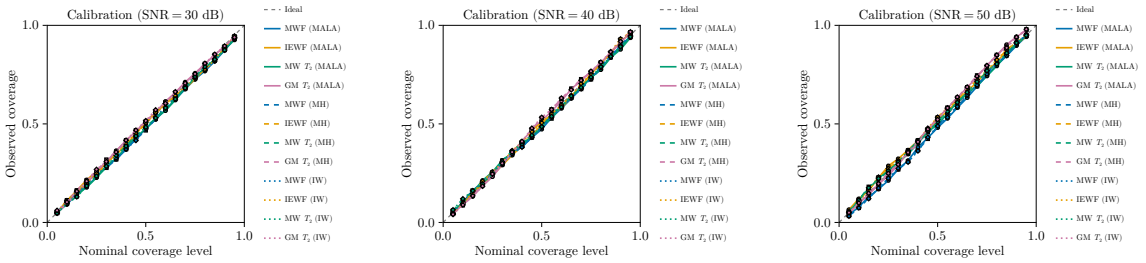


Figure 7: Calibration plots for MALA, MH, and IW on synthetic data with SNR = 30 dB (left), 40 dB (middle), and 50 dB (right). Observed posterior coverage is compared with nominal credible levels for each biomarker; perfect calibration lies along $y = x$.

4.2.2 Bimodal Synthetic Spectra

We repeated the recovery evaluation using the OOD bimodal spectra defined in section 4.1, retaining the acquisition settings, noise levels, and inference budgets (table 4 and figure 8). MWF coverage was 86.7–88.9 %, 91.7–92.4 %, and 98.2–98.5 % at 30, 40 and 50 dB, respectively, showing sensitivity to the spectrum distribution. The prior-sampled spectra span a broader range of compartment fractions and should not be interpreted as the empirical distribution of brain spectra. For the prior-sampled benchmark, median importance-weight ESS/S was 0.78, 0.85, and 0.76; no voxel had $\widehat{k} > 0.7$ at any SNR, with maximum $\widehat{k} = 0.69$. All voxels were retained.

4.3 In Vivo Evaluation

In vivo multi-spin echo brain data were processed using DECAES and two Bayesian sampling methods, MALA and MH. Figure 1 shows MWF and geometric mean T_2 maps for a single axial brain slice. The Bayesian posterior means produce MWF maps that are smoother and more spatially uniform than DECAES, with sharper white-matter and grey-matter contrast and boundaries. The DECAES − MH difference maps show coherent spatial structure rather than unstructured noise, suggesting nontrivial differences in the recovered spectra. The posterior CI half-width maps illustrate voxelwise estimated uncertainty, generally scaling in proportion to the magnitude of the estimated parameter. For the full volume (645 006 masked voxels), sampling required approximately 0.3 s/sample for IW/MH/RWM and 0.9 s/sample for MALA on one RTX 8000 GPU.

Figure 9 shows mean T_2 spectra for three voxel cluster partitions for a held-out test image slice, roughly corresponding to white matter, grey matter, and CSF, for each DECAES method as well as MH. Voxels were grouped into one of three clusters according to which water fraction – myelin, intra-/extracellular, or cerebrospinal-fluid – is most elevated relative to the slice average; the resulting clusters are shown in the rightmost panel. The MH spectra resolve more cleanly separated T_2 peaks than any DECAES method, whose intra-/extracellular peak is broader – plausibly an artifact of over-regularization in the NNLS solver – and whose myelin-water peak is smaller and more clustered toward the $T_2^{\min}$ boundary.

Table 4: T_2 recovery for 1000 synthetic bimodal T_2 spectra, using the acquisition settings and inference budgets of table 3. Errors are reported as mean ± standard deviation; 95 % CI coverage is reported for the Bayesian methods. Bold marks the lowest displayed mean error at each SNR.

	Method	Mean errors			95% coverage	
		MWF (%)	GM T_2 (ms)	W_1	MWF	GM T_2
SNR = 30 dB	L-Curve	10.2 ± 8.1	5.1 ± 4.0	0.36 ± 0.06	—	—
	GCV	7.4 ± 5.6	6.4 ± 5.4	0.33 ± 0.08	—	—
	Chi2	7.5 ± 5.6	6.7 ± 5.7	0.33 ± 0.09	—	—
	Direct	6.0 ± 4.7	2.3 ± 1.8	0.18 ± 0.08	86.7%	90.6%
	IW	$\mathbf{5.6 \pm 4.5}$	$\mathbf{2.1 \pm 1.6}$	$\mathbf{0.17 \pm 0.08}$	88.9%	91.1%
	MH	$\mathbf{5.6 \pm 4.5}$	$\mathbf{2.1 \pm 1.6}$	$\mathbf{0.17 \pm 0.08}$	88.4%	91.1%
	MALA	$\mathbf{5.6 \pm 4.5}$	$\mathbf{2.1 \pm 1.6}$	$\mathbf{0.17 \pm 0.08}$	88.0%	90.5%
SNR = 40 dB	L-Curve	5.1 ± 3.5	2.5 ± 2.2	0.2 ± 0.04	—	—
	GCV	4.6 ± 3.6	3.5 ± 3.0	0.17 ± 0.06	—	—
	Chi2	4.4 ± 3.6	3.6 ± 3.1	0.16 ± 0.06	—	—
	Direct	$\mathbf{3.0 \pm 2.8}$	$\mathbf{0.8 \pm 0.6}$	$\mathbf{0.09 \pm 0.03}$	92.3%	97.2%
	IW	$\mathbf{3.0 \pm 2.8}$	$\mathbf{0.8 \pm 0.7}$	$\mathbf{0.09 \pm 0.03}$	92.4%	96.9%
	MH	$\mathbf{3.0 \pm 2.8}$	$\mathbf{0.8 \pm 0.7}$	$\mathbf{0.09 \pm 0.03}$	92.4%	96.6%
	MALA	$\mathbf{3.0 \pm 2.8}$	$\mathbf{0.8 \pm 0.7}$	$\mathbf{0.09 \pm 0.03}$	91.7%	96.1%
SNR = 50 dB	L-Curve	3.2 ± 2.3	1.8 ± 1.5	0.12 ± 0.04	—	—
	GCV	2.4 ± 2.2	1.7 ± 1.6	0.08 ± 0.04	—	—
	Chi2	2.2 ± 2.0	1.7 ± 1.6	0.08 ± 0.04	—	—
	Direct	$\mathbf{1.3 \pm 1.2}$	$\mathbf{0.4 \pm 0.3}$	$\mathbf{0.06 \pm 0.02}$	98.3%	99.7%
	IW	$\mathbf{1.3 \pm 1.2}$	$\mathbf{0.4 \pm 0.3}$	$\mathbf{0.06 \pm 0.02}$	98.5%	99.7%
	MH	$\mathbf{1.3 \pm 1.2}$	$\mathbf{0.4 \pm 0.3}$	$\mathbf{0.06 \pm 0.02}$	98.3%	99.6%
	MALA	$\mathbf{1.3 \pm 1.2}$	$\mathbf{0.4 \pm 0.3}$	$\mathbf{0.06 \pm 0.02}$	98.2%	99.6%

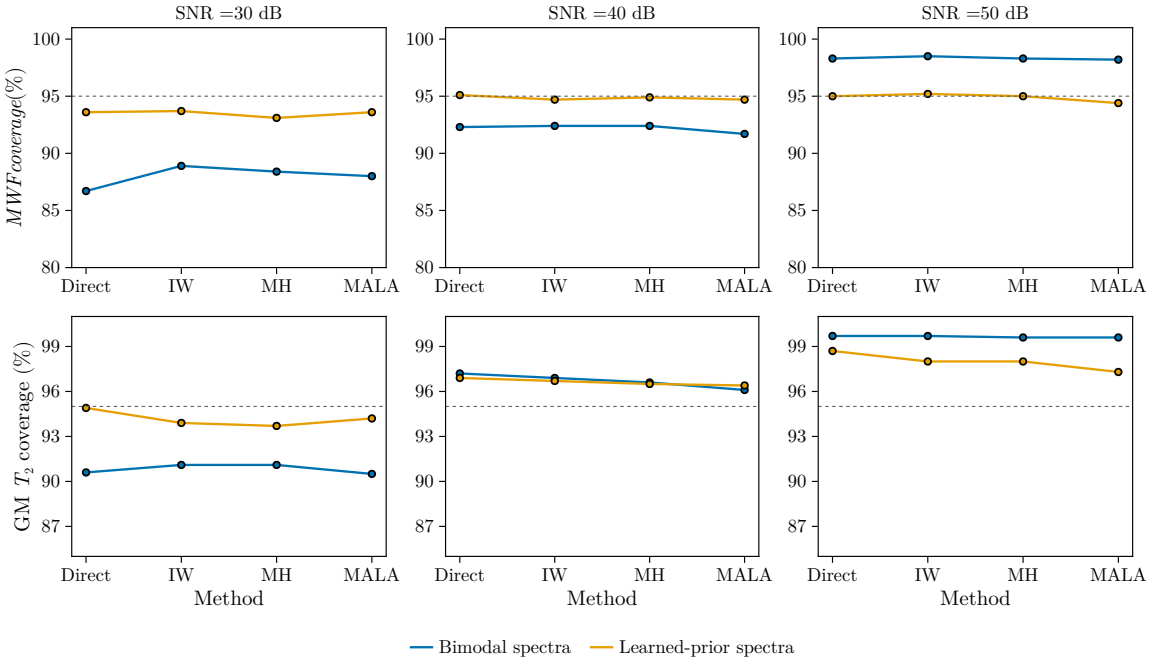


Figure 8: Observed 95 % interval coverage for synthetic bimodal and learned-prior spectra. Columns correspond to 30, 40 and 50 dB; rows show MWF and GM T_2. The dashed line marks nominal coverage.

Figure 10 compares learned-coordinate MALA with the pooled HMC and NUTS references for three in vivo voxels with high myelin, intra-/extracellular, or cerebrospinal-fluid fractions. The spectra and biomarker histograms agree closely, supporting the untuned MALA estimates and the consistency of the two references.

4.3.1 Posterior Quality and MCMC Diagnostics

The Pareto-smoothed importance sampling (PSIS) diagnostic $\hat{k}$ is used to flag unreliable posterior approximations by identifying voxels with heavy-tailed importance weights, characterized by $\hat{k} > 0.7$ [72, 73], which lead to high-variance estimators. Figure 11 shows that 948 of the 1000 ID test voxels satisfied $\hat{k} < 0.5$, and 3 exceeded 0.7. The mean normalized importance effective sample size was $\mathrm{ESS}/S = 0.826$ for $S = 1000$ proposals, with 939 voxels satisfying $\mathrm{ESS}/S > 0.7$.

Figure 12 shows sampling convergence diagnostics for MALA and MH on the ID test cohort. The MALA acceptance distribution concentrated around a mean of 0.547, close to the asymptotically optimal value 0.574 [68], while independent-proposal MH attained a mean acceptance of 0.785. For MALA, we report the integrated autocorrelation time τ^{int} of each biomarker sampling chain, as estimated by the self-consistent windowing method [74]. For independent-proposal MH, an autocorrelation time $\tau^{\mathrm{int}}_{\mathrm{Acc.}}$ may be estimated directly from the accept-reject sequence of the chain [75]. The median MWF autocorrelation time was 0.84 steps for MH and 3.22 steps for MALA; the median MH acceptance-sequence estimate was $\tau^{\mathrm{int}}_{\mathrm{Acc.}} = 0.82$ steps.

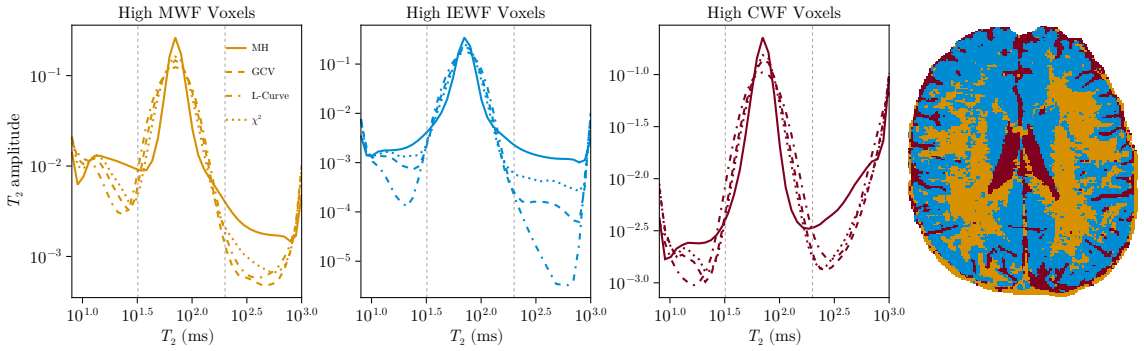


Figure 9: Cluster-mean T_2 spectra on a held-out test image slice, with voxels partitioned by dominant water fraction relative to the slice mean. First three panels: MH posterior mean overlaid on the DECAES L-curve, GCV, and χ^2 estimates. Rightmost panel: voxel cluster assignments.

4.4 Preconditioning Ablation

To investigate the benefit of preconditioning by the learned flow, we compared MALA in the flow-transformed base coordinates u against both MALA and HMC with an identity-flow backbone, replacing the deep conditional flow $\varphi_\xi(\cdot; b)$ with the identity map. This retains the low-pass window, inverse ILR, and conditional nuisance transformation, sampling the pre-window coordinates $\xi = \varphi_\xi(u_{\bar{x}}; b)$ directly. This ablation used the OOD bimodal spectra of section 4.1 at 50 dB. In figure 13, learned-backbone MALA used the theoretical step size without warmup; identity-backbone MALA adapted its step size per voxel over 4000 warmup iterations, targeting an acceptance ratio of 0.574; identity-backbone HMC used eight chains per voxel with 1000 warmup iterations and 16 leapfrog steps per draw. Each method retained 1000 draws per chain.

Figure 13 shows a roughly twentyfold reduction in median mean-coordinate autocorrelation time. Interpolating between the same approximate-posterior T_2 spectra, we see that extremely low-density regions in pre-window ξ coordinates must be traversed. This is in stark contrast to interpolation in learned u coordinates, wherein one remains in high posterior density regions throughout the entire interpolation path. We further see that the learned-backbone chain explores the displayed marginal posterior more broadly, with the identity-backbone chains instead becoming stuck in confined regions, unable to escape despite having access to posterior gradient information; even HMC, with an adapted diagonal metric, fails to explore the whole marginal space.

While figure 13 isolates the benefit for the MALA sampler, table 6 shows that the preconditioning benefits every sampler on synthetic data at SNR 50 dB. For fair comparison, the identity-backbone step sizes were tuned per voxel, whereas the flow-backbone samplers used the theory-derived values from section 3.6.1. The learned-backbone chains started without warmup; the simple identity-backbone chains used 4000 warmup iterations, and identity-backbone HMC used 1000. Each retained 1000 draws per chain; IW used 1000 proposal draws. With the learned backbone, all four samplers recovered the T_2 spectrum approximately equally accurately, with median absolute MWF errors near 1.0% and Wasserstein-1 distances near 0.06. Mixing efficiency varied: the MH and MALA samplers mixed much faster than RWM, as expected for an isotropic random walk in 42 dimensions. For each sampling method that was evaluated using both backbone variants, the identity backbone consistently gave larger median errors and lower median ESS/S; wide 95 % intervals on all quantities expose the high variability, with chains often failing to mix and to find high-

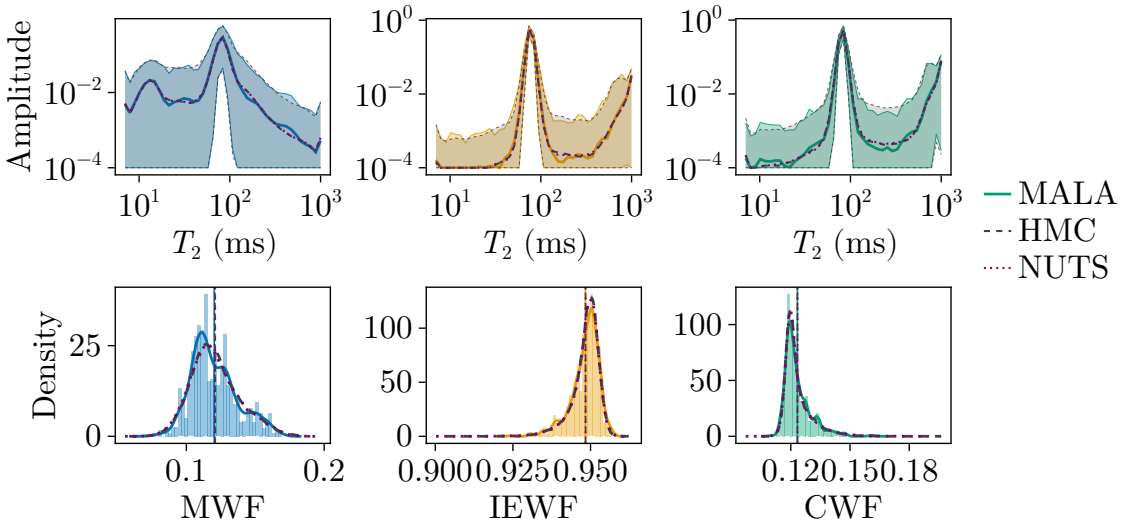


Figure 10: Posterior distributions for three representative in vivo voxels, comparing posterior samples using MALA (without warmup or step size tuning), HMC, and NUTS. Each column corresponds to a single voxel. Top row: T_2 spectra posterior mean and 95 % credible band. Bottom row: biomarker empirical posterior histograms with overlaid kernel density estimates and vertical lines indicating posterior means.

likelihood regions. The ESS/∇ column reports retained effective samples per gradient evaluation, equal to ESS/S for MALA and divided by the number of leapfrog steps for HMC. Identity-backbone HMC, with 16 leapfrog steps, and the separate 32-voxel identity-backbone NUTS investigation assess whether adaptation alone resolves the unpreconditioned sampling difficulty.

For 32 matched voxels with bimodal spectra, identity-backbone NUTS attained median within-chain biomarker ESS/S = 0.645 and ESS/∇ = 0.00122, with 12.6% of retained transitions reaching the tree-depth limit (table 5). No retained transitions were divergent. The long adaptive NUTS trajectories improved exploration over fixed-length HMC, with roughly twice the effective samples per gradient evaluation.

Table 5: Identity-backbone efficiency investigation on the first 32 voxels of the matched 50 dB bimodal cohort. Values are medians of within-chain biomarker diagnostics over voxels, with 1000 retained draws per voxel. MALA uses 4000 warmup iterations, and HMC and NUTS use 1000. HMC uses 16 leapfrog steps; NUTS adapts its trajectory length with maximum tree depth 10. These single-chain diagnostics do not establish between-chain convergence.

Identity sampler	Acceptance	ESS/S	ESS/∇
MALA	0.0010	0.00489	0.004893
HMC	0.9284	0.00860	0.000538
NUTS	0.9521	0.64502	0.001218

The purpose of these comparisons is to evaluate the benefit of the learned change of variables, rather than ranking the sampling algorithms on a common geometry. Comparing MWF recovery from the first and last 500 retained draws, learned-backbone MH/MALA retained steady mean absolute error between 1.34% and 1.35%.

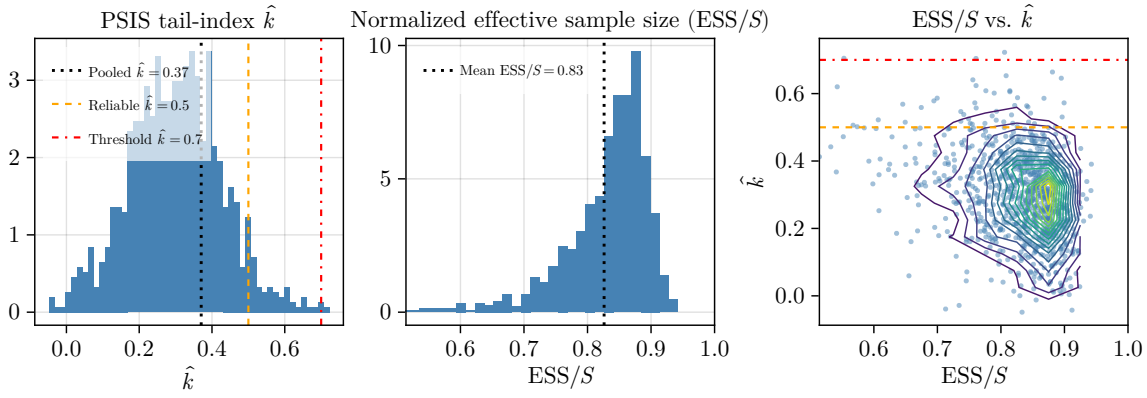


Figure 11: PSIS diagnostic $\hat{k}$ and normalized effective sample size ESS/S for the variational posterior. Both quantities are computed from self-normalized importance weights. The panels show the $\hat{k}$ and ESS/S marginal and joint distributions over the 1000 ID test voxels.

5 Discussion

In this work, we presented a Bayesian framework for T_2 distribution estimation in which a conditional normalizing flow was trained to approximate the marginal posterior over the normalized T_2 spectrum, coupled with a Laplace approximation to the conditional posterior over the nuisance parameters governing the signal scale and noise level. Unlike point-estimation methods, the learned flow simultaneously provides an approximate posterior that can be sampled directly and a data-dependent coordinate system, independently useful for running established sampling methods in the transformed space. On prior-sampled T_2 spectra at SNRs of 30, 40 and 50 dB, direct sampling, importance-weighted sampling, independent-proposal Metropolis–Hastings, and the Metropolis-adjusted Langevin algorithm produced similar recovery errors, with all four outperforming every regularized-NNLS baseline from DECAES (table 3). For example, at 50 dB the Bayesian methods recovered MWF with mean absolute errors of 1.4 % and $\bar{x}$ with Wasserstein-1 distances of approximately 0.08, compared with the best DECAES values of 2.2 % and 0.13, respectively (table 3). MWF coverage remained close to nominal on prior-sampled spectra, whereas GM T_2 intervals were conservative at 40 and 50 dB (figure 7).

An alternative view of the flow is geometric: it decorrelates the posterior over simplex-constrained spectra into approximately whitened coordinates (figures 2 and 13). When the learned flow backbone is instead replaced with the identity map, sampling operates over the curved pre-window ILR coordinates; using MALA, the median mean-coordinate autocorrelation time increased 20.2-fold (figure 13). On the ID in vivo cohort, the learned-backbone MALA acceptance rate of 0.547 (figure 12) was close to the asymptotically optimal rate of 0.574 [68]; on bimodal spectra with the identity backbone, MALA acceptance collapsed toward zero and the fixed-budget comparisons exhibited unresolved mixing (table 6). Identity-backbone HMC and NUTS assess how adaptation handles the unpreconditioned geometry (tables 5 and 6); the resulting comparisons support the learned transformation rather than establish sampler superiority. In this bimodal T_2 spectrum comparison, flow-preconditioned MALA attained over 200 times the ESS per gradient evaluation of identity-backbone HMC with an adapted diagonal metric (table 6). It is thus plausible that it is the preconditioned geometry, rather than gradient information alone, that makes sampling efficient.

The Aitchison geometry is a convenient and flexible framework for constructing the learned change of variables on the simplex. The ILR transform provides a differentiable

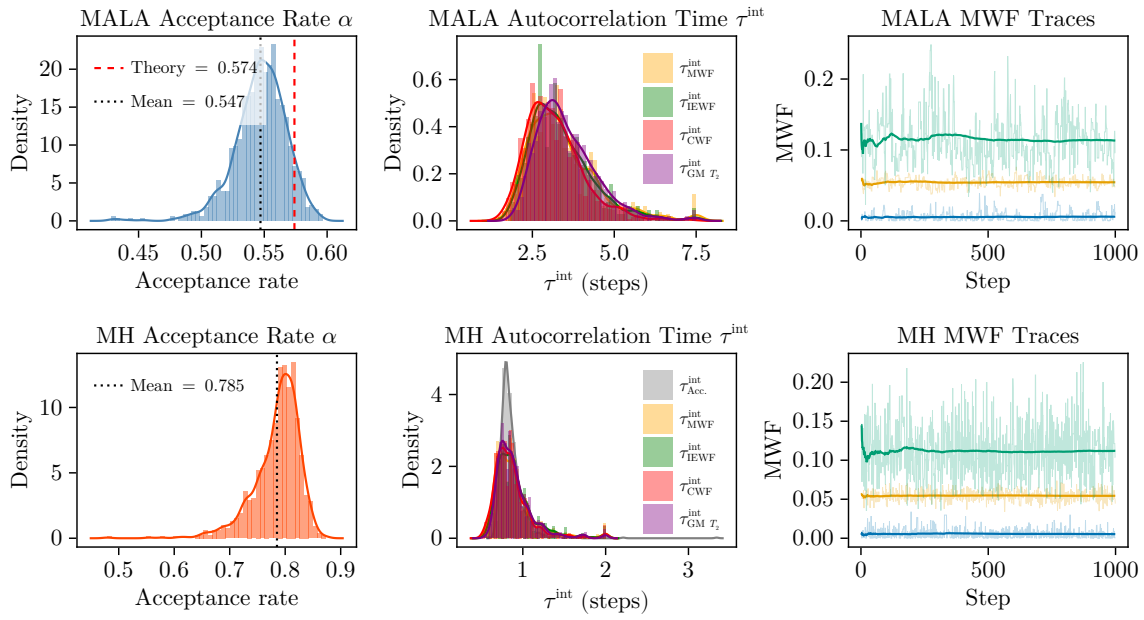


Figure 12: MCMC convergence diagnostics over the ID test voxels. Rows show MALA (top) and independent-proposal MH (bottom). The first column shows the acceptance rate; for MALA, the mean is close to the asymptotically optimal rate 0.574. The second column shows integrated autocorrelation times τ^{int}; for MALA, τ^{int} is estimated only from the biomarker chains, but for MH, it can additionally be estimated from the acceptance sequence. The third column shows MWF chain traces with overlaid cumulative means.

bijective map between the simplex interior and unconstrained Euclidean coordinates, allowing the deep flow to be constructed from standard flow components from the literature without explicit concern for positivity and unit-sum constraints. Over these coordinates, the choice of the DCT basis additionally lends the ILR coefficients the interpretation of DCT-transformed logits $\log \bar{x}$ over the $\log T_2$ grid. This interpretable representation was used to define a learnable low-pass prior whose cutoff controls the $\log T_2$ spectrum smoothness and whose amplitude controls the T_2 peak sharpness (figure 5). Using the same window Λ as the terminal flow layer places the prior and approximate posterior in the same filtered coefficient geometry at initialization.

Without ground truth, posterior quality in vivo can be evaluated from the importance weights: on the ID test cohort, 948 of 1000 PSIS diagnostic estimates satisfied $\hat{k} < 0.5$, and 3 exceeded 0.7 (figure 11). The mean normalized importance-sampling effective sample size was 0.826, supporting IW estimation for much of this ID cohort. We recommend IW for posterior expectations and credible intervals when the PSIS diagnostic is satisfactory, as for 997 of 1000 held-out test voxels (section 4.3.1). Both IW and independent-proposal MH are gradient-free; IW reweights proposals, whereas MH yields unweighted, correlated draws. MALA mixed more slowly than MH on the bimodal spectra (table 6), but its gradient-informed moves do not require the proposal to cover the posterior. On bimodal spectra (table 6), median ESS/S was 0.846 for IW, 0.534 for MH, and 0.134 for MALA, with similar recovery errors. Note that weight-based and autocorrelation-based ESS are distinct diagnostics, and these values do not establish wall-time speedups.

On the OOD in vivo slice, the Bayesian posterior means produced visually smoother MWF maps than DECAES, with sharper grey–white matter contrast, despite both methods operating voxelwise independently (figure 1). Cluster-averaged T_2 spectra also contained

Table 6: Preconditioning ablation with and without the learned flow backbone on 1000 synthetic bimodal T_2 spectra at SNR 50 dB. IW, independent-proposal MH, RWM, and MALA are each run on the learned-flow and identity backbones, the latter using per-voxel tuned step sizes and the former using the fixed step size from theory. HMC is included as a baseline comparison on the unpreconditioned geometry. Acceptance ratio, normalized effective sample size ESS/S, effective sample size per gradient evaluation ESS/∇, worst-biomarker integrated autocorrelation time τ^{int}, absolute MWF error $|\Delta\text{MWF}|$ in %, and the Wasserstein-1 distance $W_1(\bar{x}, \bar{x}_{\text{true}})$ between the estimated and ground-truth normalized spectra are reported with median values and the 2.5th and 97.5th empirical percentiles across voxels as sub- and superscript. ESS/S is computed from self-normalized importance weights (ESS^{iw}) for IW, and from the integrated autocorrelation time $\text{ESS}^{\text{int}} = S/(2\tau^{\text{int}})$ for the chain-based samplers.

Method	Backbone	α	ESS/S ↑	ESS/∇ ↑	τ^{int} ↓	$\lvert\Delta\text{MWF}\rvert$ (%) ↓	$W_1(\bar{x}, \bar{x}_{\text{true}})$ ↓
IW	Learned	—	$\mathbf{0.846^{0.9}_{0.7}}$	—	—	$0.983^{4.6}_{0.026}$	$\mathbf{0.0598^{0.1}_{0.031}}$
IW	Identity	—	$0.001^{0.002}_{0.001}$	—	—	$4.85^{17.0}_{0.23}$	$0.152^{0.35}_{0.051}$
MH	Learned	$0.786^{0.83}_{0.73}$	$0.534^{0.71}_{0.29}$	—	$\mathbf{0.936^{1.7}_{0.71}}$	$\mathbf{0.97^{4.5}_{0.031}}$	$0.0602^{0.1}_{0.031}$
MH	Identity	$0.0^{0.002}_{0.0}$	$0.001^{0.021}_{0.001}$	—	$500.0^{500.0}_{24.0}$	$3.47^{13.0}_{0.16}$	$0.124^{0.27}_{0.042}$
RWM	Learned	$0.232^{0.25}_{0.21}$	$0.0109^{0.022}_{0.0061}$	—	$45.8^{82.0}_{23.0}$	$1.1^{4.9}_{0.053}$	$0.0629^{0.11}_{0.031}$
RWM	Identity	$0.0223^{0.28}_{8.9e-5}$	$0.00645^{0.018}_{0.001}$	—	$77.6^{500.0}_{28.0}$	$2.11^{12.0}_{0.1}$	$0.0973^{0.31}_{0.033}$
MALA	Learned	$0.546^{0.58}_{0.52}$	$0.134^{0.19}_{0.078}$	$\mathbf{0.134^{0.19}_{0.078}}$	$3.72^{6.4}_{2.7}$	$0.987^{4.5}_{0.043}$	$0.0601^{0.1}_{0.031}$
MALA	Identity	$4.31e-5^{0.29}_{0.0}$	$0.001^{0.014}_{0.001}$	$0.001^{0.014}_{0.001}$	$500.0^{500.0}_{36.0}$	$8.69^{59.0}_{0.21}$	$0.351^{1.7}_{0.05}$
HMC	Identity	$0.932^{0.97}_{0.75}$	$0.00972^{0.037}_{0.0053}$	$0.000607^{0.0023}_{0.00033}$	$51.4^{94.0}_{13.0}$	$1.39^{9.3}_{0.049}$	$0.072^{0.18}_{0.027}$

more clearly delineated water peaks than the three DECAES estimates (figure 9). For three representative ID test voxels, samples generated via flow-preconditioned MALA, pooled HMC, and pooled NUTS agreed closely in both the full T_2 spectra posteriors and the derived biomarker empirical posteriors (figure 10).

The trained change of variables also supports Bayesian flip-angle estimation, spatially regularized MAP estimation, and sampling on different $\log T_2$ grids without retraining (appendices A.1 to A.3).

Related deep-learning methods reviewed in section 2 produce point estimates or image-level posteriors. To the best of our knowledge, the domain-specific combination used here – an ILR flow on the simplex, a quantized Rician likelihood, a Laplace approximation to the nuisance parameters, and learned preconditioning for full-T_2-spectrum posterior sampling – has not previously been used for T_2 relaxometry. Image-level score-based priors for MR reconstruction [22] address a complementary problem, which targets distributions over entire $(3 + 1)$-dimensional MRI signal data rather than intra-voxel inference. The voxel-wise uncertainty estimates presented here may additionally be clinically useful; for example, credible-interval maps could help distinguish genuinely low-MWF tissue from underestimates due to over-aggressive regularization of the NNLS problem, or could be used to differentiate between physiological changes and measurement uncertainty in longitudinal studies of the WM. However, calibration was only assessed on synthetic data (sections 4.2.1 and 4.2.2), and no clinical validation was attempted in this work.

5.1 Limitations

The primary recovery and calibration benchmark sampled spectra from the learned prior (table 3 and figure 7), while the bimodal experiments examined spectrum-distribution shift (section 4.2.2 and table 6). Both used the assumed EPG forward model, so forward-model mismatch remains untested. Training used 56-echo, 7 ms acquisitions with echo-train augmentation. In vivo maps are shown for one representative slice of a held-out acquisition with

a different protocol (figure 1); these suggest transfer, but cohort-wide posterior-sampling diagnostics used only 1000 voxels from the ID test cohort (figures 11 and 12). Coverage in figures 7 and 8 was evaluated at fixed signal scale and noise levels, rather than by drawing the full joint prior; calibration on these simulations does not establish calibration on in vivo data.

The likelihood model makes standard, but still simplifying, assumptions: the magnitude-data noise floor and quantization are accounted for by the quantized Rician model, but echoes are treated as conditionally independent, with each voxel assigned a single noise scale shared across echoes. Moreover, this noise level is more commonly assumed to be global per-image – that is, shared spatially as well as temporally – rather than varying per-voxel; this would simplify the Laplace approximation to a univariate approximation over the scale parameter, but would require separate tracking and estimation of per-image noise levels during training and inference. However, parallel imaging reconstruction is known to amplify noise non-uniformly across the image [76–78], an effect characterized by a spatially varying geometry factor g, and so a global noise level is not necessarily the more accurate description. Laplace's approximation to the nuisance parameters is not learned and plausibly does not always generate high-quality proposals; but since the joint change of variables remains invertible, asymptotic correctness remains guaranteed when using the IW or MH correction.

The conditional flow was trained using a reverse-KL objective, which has a mode-seeking tendency. This was supplemented by a self-supervised forward-KL loss, which is mode-covering but only for synthetically generated data, so posterior undercoverage remains possible. The PSIS results indicate predominantly well-behaved importance weights on the ID test cohort (figure 11), but importance-weight diagnostics cannot detect posterior modes that the flow misses entirely.

Inference in the main experiments was assumed voxelwise independent. The spatial regularization procedure of appendix A.2 produces a post-hoc spatially regularized MAP estimate, not to be confused with sampling from a truly spatially coupled posterior. The assumptions of the EPG model are also inherited, with B_1 correction performed by flip-angle scaling and fixed T_1 for each basis, without modelling exchange or diffusion. Flip-angle estimation is also assumed to be provided and accurate; we used the estimate from DECAES, which we found to be in close agreement with Bayesian flip-angle estimates (appendix A.1), albeit under the same EPG forward model. Training augmentation over α, T_1, and the T_2 grid provides some robustness to these assumptions (section 3.5.5). Posterior credible intervals are conditional on these modelling assumptions.

In summary, the learned flow makes full Bayesian T_2 distribution inference practical: it preconditions a difficult voxelwise posterior into geometrically decoupled coordinates wherein sampling becomes simple and efficient, and its tractable Jacobian determinant further enables importance weighting and MCMC acceptance corrections.

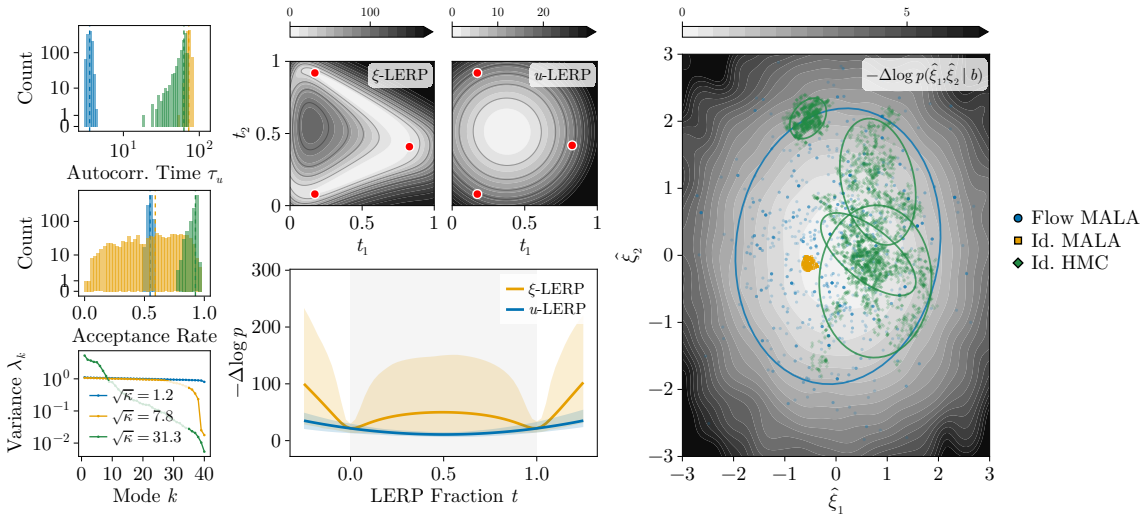


Figure 13: Illustration of the effect of the learned preconditioning on the posterior geometry and on sampling. Three sampling procedures are shown: MALA under the learned flow, and both MALA and HMC where the learned flow backbone φ_ξ is replaced with the identity transform. The left column shows the across-voxel diagnostics for batches of sample sequences $\{u_t\}$: histograms of the mean autocorrelation time τ_u and the acceptance rates, with their medians marked; and the eigenvalue spectrum λ_k of the importance-weighted proposal covariances in the learned and pre-window coordinates and the pooled HMC sample covariance in pre-window coordinates, with covariance aspect ratio $\sqrt{\kappa}$ annotated. The middle column shows negative-log-posterior slices through three i.i.d. standard-normal u points and their pre-window ILR images ξ (top), and straight-chord profiles with weight t, averaged over 100 independent standard-normal u-pairs with matched ξ endpoints (bottom). Each top triangle has its longest edge vertical and its third vertex to the right. Orthogonal coordinates (t_1, t_2) share one scale and centre the triangle bounding box at $(0.5, 0.5)$, with an 8 % margin inside $[0, 1]^2$; they are normalized positions, not interpolation weights. No endpoint-linear detrending is applied; each coordinate system subtracts its energy at the common physical point corresponding to $u = 0$. The right column shows representative sampler chains projected onto the top two principal directions of the importance-weighted pre-window proposal covariance. The last 1000 chain samples are drawn as translucent scatter points and outlined by the corresponding 2σ covariance ellipse. The single MALA flow chain spans the displayed marginal in one broad ellipse, whereas the identity MALA chain stalls in the curved, ill-conditioned ξ geometry. Each HMC chain improves on the MALA chain, but any individual chain still fails to cover the full marginal.

A Additional Applications of the Learned Change of Variables

The experiments from the main text use the trained encoder and flow as a learned, nonlinear change of variables in order to precondition the voxelwise posterior sampling problem. The same change of variables can be used to accelerate related optimization and sampling problems without retraining. This section illustrates three example use cases.

A.1 ELBO-Based Flip-Angle Estimation

The EPG basis matrix depends on the sequence of excitation and flip angles $\alpha_{\text{ex}}, \alpha_1, \ldots, \alpha_{n_{\text{TE}}}$. As detailed in appendix B, the ideal CPMG sequence of flip angles $\pi/2, \pi, \ldots$ is perturbed by B_1 inhomogeneity. This is modelled by a voxelwise effective flip angle α where $\alpha_{\text{ex}} = \alpha/2$ and $\alpha_j = \alpha$. In the main experiments, α is fixed per voxel, estimated by DECAES preprocessing, which finds the α which minimizes the NNLS residual norm $\hat{\alpha}_{\text{DECAES}} = \arg\min_\alpha \min_{x \geq 0} \|A(\alpha)x - b\|_2$. Instead, using the trained encoder, flow, and prior, we can optimize α directly against a variational objective. In our application, α varies over the voxels of a 3D brain volume; we investigate two representations: as an array of independent voxelwise values α_{ijk}, or as a low-order polynomial field $\alpha(x, y, z) = \sum_{0 \leq a+b+c \leq K} w_{abc} x^a y^b z^c$.

Each estimate of α affects $A(\alpha)$, which is used both in the likelihood and implicitly through the flow $\varphi(\cdot; z(\alpha))$ via the encoder conditioning vector $z(\alpha)$,

$$z(\alpha) = \mathcal{E}(b, A(\alpha), \theta(\alpha)), \tag{61}$$

where $\theta(\alpha)$ is the scalar feature vector from (34). We aim to avoid differentiating through $z(\alpha)$, since backpropagating through the encoder at every iteration would dominate per-step costs; instead, we hold $z(\alpha)$ fixed at the DECAES estimate $z(\hat{\alpha}_{\text{DECAES}})$ in the objectives below, refreshing intermittently outside of gradient computations only if needed. We consider three optimization objectives of increasing computational cost and statistical robustness, as well as a Bayesian sampling method.

The simplest estimate, and closest in spirit to DECAES, is a maximum a posteriori (MAP) estimate. A full MAP estimate jointly minimizes the negative log posterior over (u, α),

$$\begin{aligned} \mathcal{M}(u, \alpha) = &- \log p(b \mid \varphi(u; z(\alpha)), A(\alpha)) \\ &- \log p(\varphi(u; z(\alpha))) \\ &- \log \left|\det J_\varphi(u; z(\alpha))\right|. \end{aligned} \tag{62}$$

The trained flow acts only as a nonlinear change of variables, making the problem easy to solve from the initial guesses $u = 0$ and $\alpha = \hat{\alpha}_{\text{DECAES}}$.

An even cheaper estimate fixes the *pseudo-mode* $u = 0$ – which a well-trained φ maps close to the true posterior mode for any given $z(\alpha)$ – and minimizes the pseudo-MAP objective

$$\begin{aligned} \mathcal{L}(\alpha) = &- \log p(b \mid \bar{x}_0, \eta_0, A(\alpha)) \\ &- \log p(\bar{x}_0, \eta_0), \\ \text{where} \quad &\bar{x}_0, \eta_0 = \varphi(0; z(\alpha)), \end{aligned} \tag{63}$$

where $\bar{x}_0$ is the marginal pseudo-mode of the T_2 distribution, and $\eta_0 = (\log s, \log \sigma)$ are the nuisance parameters. With $z(\alpha)$ fixed, $\bar{x}_0$ does not depend on α and can be precomputed. The nuisance mode $\eta_0(\alpha)$ must be recomputed at each step, since its value is the Laplace mode (14) of the conditional posterior given the forward signal $A(\alpha)\bar{x}_0$; this is comparatively

cheap, as it reuses the Newton solver of section 3.2.4 and the implicit-function gradients (17). The pseudo-MAP relies more heavily on the quality of the variational posterior since it assumes that the marginal pseudo-mode $\bar{x}_0$ is a good enough approximation to the true MAP estimate, but it is much cheaper to compute and, as we will see, produces visually indistinguishable flip angles.

Both of these point estimates ignore the spread of the posterior over u. To account for this, α may instead be chosen to minimize the free energy

$$\mathcal{F}(\alpha) = \mathbb{E}_{u\sim\pi_u}\Big[-\log p(b \mid \varphi(u; z(\alpha)), A(\alpha)) \\ - \log p(\varphi(u; z(\alpha))) + \log q(\varphi(u; z(\alpha)) \mid b, A(\alpha))\Big], \tag{64}$$

with the trained flow held fixed. Since $\mathcal{F}(\alpha)$ is the negative evidence lower bound, minimizing it maximizes a variational lower bound on the marginal log-evidence $\log p(b \mid \alpha)$; the encoder is refreshed as α changes. The free energy averages over the approximate posterior rather than using a single mode, and is correspondingly more expensive. All point estimates are initialized from the DECAES estimate $\hat{\alpha}_{ijk}$, and the polynomial field from the linear least-squares fit $\arg\min_\alpha \sum_{ijk}(\alpha(x_i, y_j, z_k) - \hat{\alpha}_{ijk})^2$. The deterministic MAP objectives and the stochastic free-energy objective are minimized by PSGD with Newton-type preconditioners fitted to Hessian-vector products [56]. Note that the polynomial field is spatially coupled, so each coefficient is informed by many voxels and a single Monte Carlo sample of u gives a stable descent direction, whereas with one flip angle per voxel the gradients have more variance and several samples must be averaged per iteration to suppress the sampling noise.

Minimizing $\mathcal{F}(\alpha)$ still yields only a variational evidence point estimate of α. To quantify the uncertainty in the α estimates, we instead sample from the marginal posterior $p(\alpha \mid b) \propto p(b \mid \alpha)p(\alpha)$. There remains the usual complication that the marginal evidence $p(b \mid \alpha)$ is intractable, and we can only bound it from below; however, holding $z(\alpha_{\mathrm{DECAES}})$ fixed, the trained variational posterior allows for an efficient and unbiased estimator

$$\hat{p}(b \mid \alpha) = \frac{1}{S}\sum_{s=1}^{S} \frac{p(b \mid \varphi(u_s; z), A(\alpha))}{q(\varphi(u_s; z) \mid b, A(\alpha))} p(\varphi(u_s; z)), \tag{65}$$

satisfying $\mathbb{E}[\hat{p}(b \mid \alpha)] = p(b \mid \alpha)$ where $u_s \sim \pi_u$. Using this stochastic but unbiased estimate in a sampling routine gives rise to *pseudo-marginal* MCMC (PM-MCMC) methods which, remarkably, preserve the exact marginal posterior $p(\alpha \mid b)$ [79]. That is, given a symmetric proposal distribution, a move from α to α' is accepted with the usual Metropolis–Hastings probability

$$\min\left\{1,\ \frac{\hat{p}(b \mid \alpha')p(\alpha')}{\hat{p}(b \mid \alpha)p(\alpha)}\right\}, \tag{66}$$

with $\{u_s\}_{s=1}^S$ redrawn for each proposed α'. Empirically, most flip angles are concentrated near the nominal refocusing angle π. In order to respect this empirical finding as well as the 2π-periodic and even symmetries of the EPG basis (67), we place a von Mises prior $p(\alpha) \propto \exp(\kappa\cos(\alpha - \pi))$ on the flip angle, with mean $\mu = \pi$ fixed and concentration κ fit by maximum likelihood to the DECAES estimates $\{\hat{\alpha}_{ijk}\}$.

Apart from the polynomial field, all flip-angle estimates agree closely. Figure 14 compares the DECAES flip-angle map with three representative α estimates: the voxelwise PM-MCMC posterior median, the voxelwise pseudo-MAP, and the degree $K = 6$ polynomial free-energy minimizer, alongside difference maps of each against DECAES, and the voxelwise posterior standard deviation from the sampler. Table 7 reports the median pairwise voxelwise absolute differences alongside 2.5th and 97.5th percentile values. The voxelwise PM-MCMC and

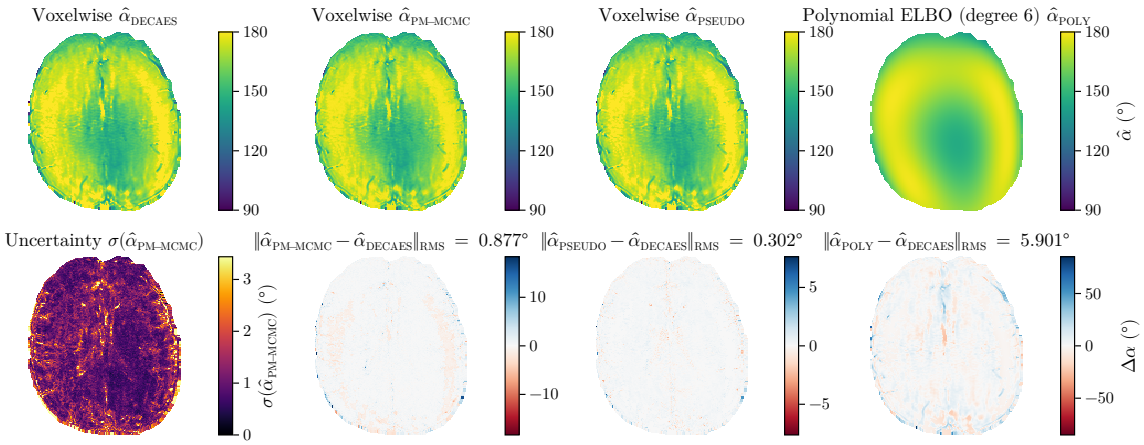


Figure 14: Flip-angle estimation using the pre-trained flow, optimizing α through its effect on the EPG basis matrix. Top row: the DECAES flip-angle map, the voxelwise PM-MCMC posterior median (65), the voxelwise pseudo-MAP estimate (63), and a degree-6 polynomial free-energy estimate (64). Bottom row: the voxelwise standard deviation of the PM-MCMC posterior samples, folded onto $[0, \pi]$, and the pointwise difference between the top-row estimate and the DECAES estimate.

Method	**DECAES**	**PM-MCMC**	**Pseudo-mode**
PM-MCMC	$0.32^{2.0}_{0.015}$	—	—
Pseudo-mode	$0.13^{0.75}_{0.0}$	$0.22^{1.9}_{0.0095}$	—
Polynomial	$2.6^{16.0}_{0.13}$	$2.4^{16.0}_{0.11}$	$2.6^{16.0}_{0.12}$

Table 7: Pairwise median absolute differences $|\hat{\alpha}_i - \hat{\alpha}_j|$ between DECAES, PM-MCMC, pseudo-MAP, and polynomial flip-angle estimates, shown with 2.5 and 97.5 percentile values as sub- and superscript.

pseudo-mode estimates have median absolute differences from DECAES of 0.32° and 0.13°, respectively, compared with 2.6° for the spatially smooth polynomial field. The close voxelwise agreement indicates that the least-squares DECAES estimate is already robust; the pseudo-MAP provides a cheap refinement, while the PM-MCMC sampler additionally yields voxelwise posterior uncertainty estimates. In figure 14 we see that the flip-angle uncertainty is largest near $\alpha = \pi$; this is consistent with the EPG basis being constant to first order in α in this region due to symmetry of the algorithm about $\alpha = \pi$, and therefore variations in α have comparatively small effect on the EPG basis. Because B_1 inhomogeneity is expected to vary smoothly and not to follow brain structure, the smooth polynomial field is physically motivated and easy to compute. Without an independent ground truth we cannot determine whether the smooth or the voxelwise estimate is the more accurate, though in figure 15 we can compare their downstream effect, illustrating the myelin water fraction, geometric-mean T_2, and intra-/extracellular water fraction maps computed by independent-proposal MH sampling under the voxelwise and polynomial free-energy flip-angle estimates.

A.1.1 Cosine Series Representation for Efficient EPG Basis Computation and Differentiation

As part of minimizing equations (63) and (64), we must evaluate the EPG basis matrix $A(\alpha)$ and its derivative $\partial A/\partial\alpha$ for many values of α at every step. Re-running the EPG recur-

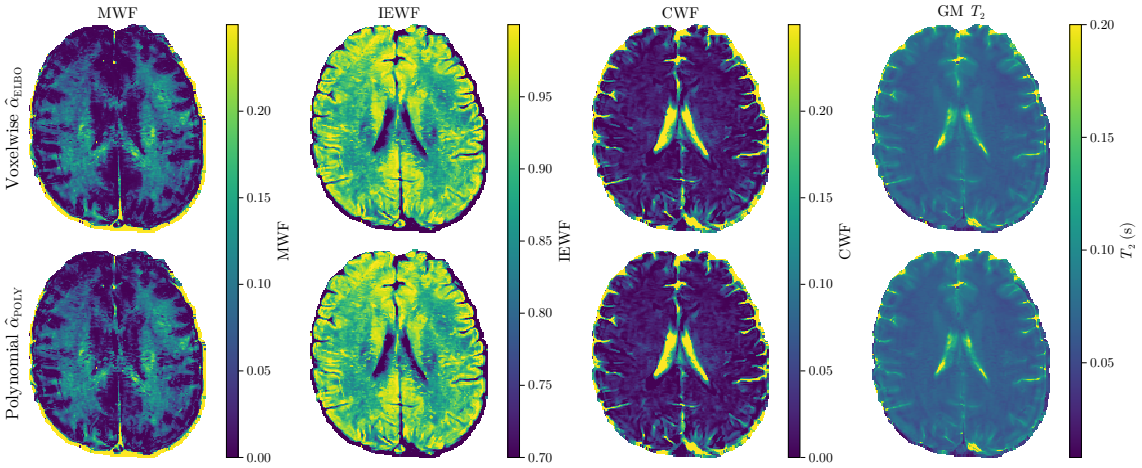


Figure 15: Biomarker maps using free-energy minimizing flip-angle estimates. Columns show the myelin water fraction, intra-/extracellular water fraction, cerebrospinal-fluid water fraction, and geometric-mean T_2, each computed via independent-proposal Metropolis–Hastings sampling with the pre-trained flow model. Top row: voxelwise flip-angle estimate (64). Bottom row: degree-6 polynomial estimate. The two flip-angle maps differ by a few degrees for typical voxels; the corresponding derived biomarkers agree closely, with one exception: the myelin water fraction appears spuriously underestimated using the polynomial flip-angle map.

sion (86) would be expensive, and the recursion currently has a fast implementation only on the CPU. This can, however, be avoided by exploiting the algebraic structure of the recursion under a constant refocusing flip angle. When all refocusing pulses are set to α, the rotation matrix (87) is affine in $\cos\alpha$ and $\sin\alpha$ via the half-angle identities $\cos^2(\alpha/2) = (1+\cos\alpha)/2$ and $\sin^2(\alpha/2) = (1-\cos\alpha)/2$. Each echo applies one such rotation with interwoven linear α-independent relaxation and transition operators. The transverse magnetization $(\mathbf{M}^{i,j})_1^+$ of (85) is therefore, having factored out the excitation amplitude $\sin(\alpha/2)$, a real trigonometric polynomial of degree at most n_{TE} in α. This polynomial is additionally an even function of α: the only α-dependence that is odd enters via $\sin\alpha$ factors in the off-diagonal entries of R_j (87) which couple the transverse and longitudinal magnetization; the remaining entries are even in α, and neither the relaxation nor transition matrices mix transverse with longitudinal signal. Since the EPG algorithm is initialized with purely transverse magnetization, the transfers to and from the longitudinal magnetization occur in pairs, each acquiring a factor of $\sin\alpha$; every transverse readout echo is therefore even in α, and the EPG basis permits a finite cosine series representation:

$$A_{ij}(\alpha) = \left|\sin(\alpha/2)\sum_{k=0}^{n_{\mathrm{TE}}} \hat{A}_{ijk}c_k(\alpha)\right|, \qquad (67)$$
$$c(\alpha) = (1, \cos\alpha, \cos 2\alpha, \ldots, \cos n_{\mathrm{TE}}\alpha),$$

where $c(\alpha)$ are the cosine features and $\hat{A} \in \mathbb{R}^{n_{\mathrm{TE}}\times n_{T_2}\times(n_{\mathrm{TE}}+1)}$ is a fixed tensor of cosine series coefficients depending only on T_1, the T_2 range, and TE. The coefficients for each $\hat{A}_{ij:}$ can be precomputed via discrete cosine transforms of the vectors $[\bar{A}_{ij}(\alpha_k)]_{k=0}^{n_{\mathrm{TE}}}$, where α_k are uniformly spaced in $[0,\pi]$ and $\bar{A}(\alpha)$ are the EPG bases $A(\alpha)$ before multiplying by the excitation amplitude $\sin(\alpha/2)$ and taking the absolute value. Evaluating $A(\alpha)$ and $\partial A/\partial\alpha$ then reduces to forming $c(\alpha)$ and performing a single matrix-vector multiplication followed by elementwise operations, all of which are trivially differentiable.

A.2 Spatially Regularized MAP Estimation

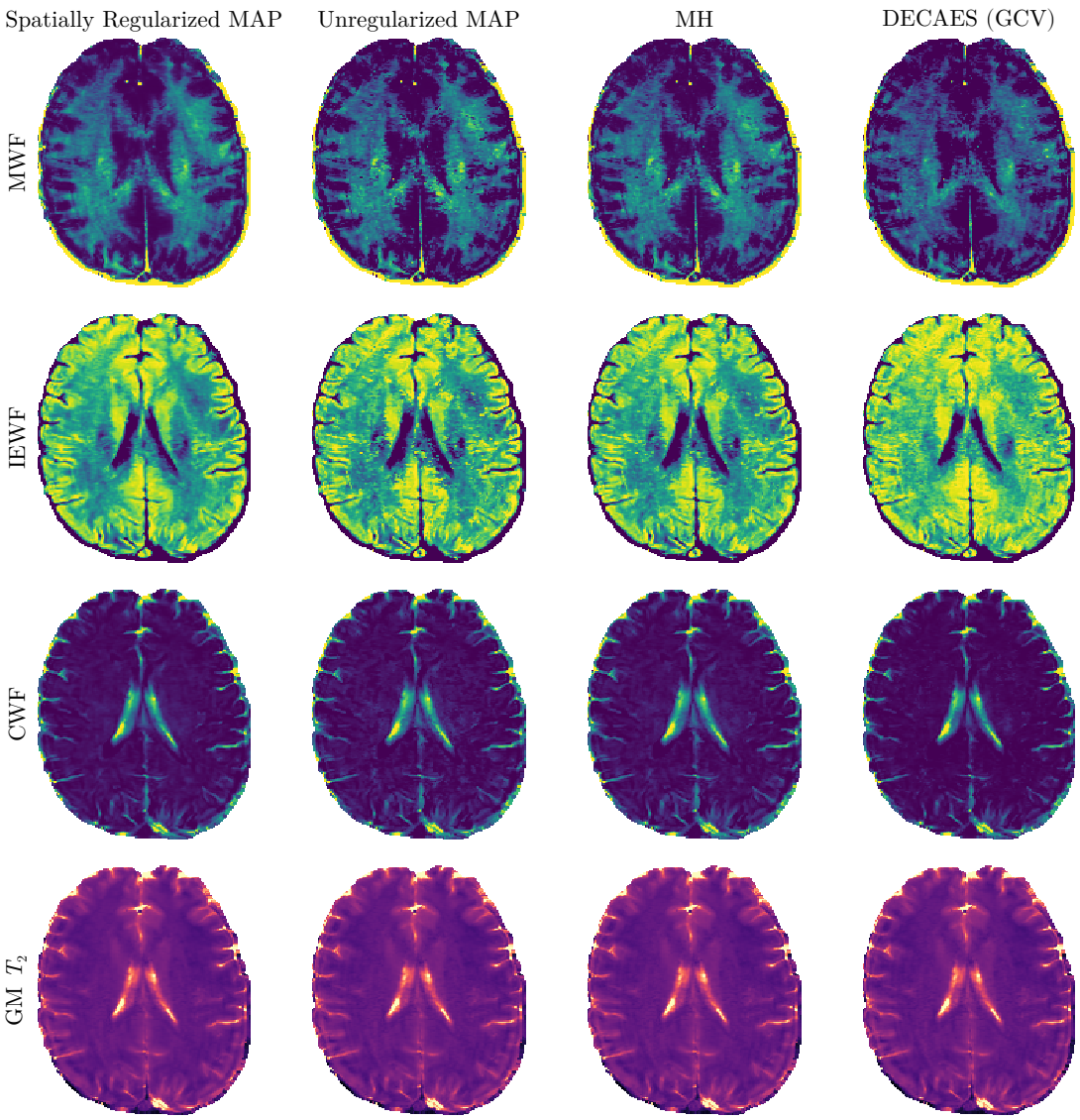


Figure 16: Spatially regularized two-step estimates. Biomarker maps derived from the spatially regularized MAP T_2 distribution are compared against the unregularized voxelwise MAP solution, the independent-proposal MH posterior mean, and DECAES with GCV regularization. The spatial prior smooths the estimates while preserving tissue boundaries.

Thus far we have considered only inference methods that are voxelwise-independent, with no voxel being influenced by its neighbours. For virtually all inference problems in MRI, some degree of spatial regularity is expected *a priori*, since the underlying tissue is organized into large, smoothly varying structures. In myelin water imaging of the brain, T_2 spectra are expected to vary smoothly within the grey and white matter tissue compartments, since it is atypical for the underlying tissue to have sharp discontinuities. This motivates the introduction of a spatial smoothing prior to couple the voxelwise estimates [80, 81]. We can impose this property *after* training the voxelwise model components by modifying the learned prior over T_2 spectra such that continuity across voxels is encouraged.

The natural coordinates in which to impose smoothness are the *whitened* ILR coordi-

nates $\xi_i = \Lambda^{-1} \operatorname{ilr}(\bar{x}_i; \beta_0) \in \mathbb{R}^{n_{T_2}-1}$, where i is a voxel index and Λ is the learned low-pass window of equation (60). These are the base coordinates of the low-pass prior (22), equal to the DCT-II coefficients of $\log \bar{x}_i$ before multiplication by Λ. Smoothing ξ across voxels encourages spatial correlations in the corresponding T_2 distributions, with derived biomarkers smoothed implicitly. We choose the simplest multivariate normal prior that imposes spatial correlations, a Gaussian Markov random field (GMRF) [82].

Let $\mathcal{A}$ be the weighted adjacency matrix of the masked grid of voxels and let $D = \operatorname{diag}(d)$ be the diagonal matrix of voxel *degrees*; that is, $d_i = \sum_j \mathcal{A}_{ij}$ is the total weight of the edges incident to the ith voxel. The corresponding symmetric normalized adjacency and graph Laplacian matrices are defined as

$$W = D^{-1/2} \mathcal{A} D^{-1/2}, \quad L = \mathrm{I} - W. \tag{68}$$

The standard-normal prior $p(\xi) = \mathcal{N}(\xi \mid 0, \mathrm{I})$ on ξ has an identity precision matrix. Here, we replace it with a block-diagonal GMRF prior with

$$Q = \alpha \mathrm{I} + \beta L = \alpha(\mathrm{I} + \kappa L), \tag{69}$$

repeated $n_{T_2} - 1$ times on the diagonal; $\alpha > 0$ controls the isotropic shrinkage, and $\beta \geq 0$ is a roughness penalty. Because the learned prior over ξ has unit marginal variance, we fix $\alpha = 1$ such that $Q = \mathrm{I} + \kappa L$ depends only on the smoothing strength $\kappa \geq 0$. Note that W has eigenvalues contained in $[-1, 1]$, and assuming the brain mask is connected,[5] $d^{1/2}$ is the unique eigenvector with unit eigenvalue, following from $\mathcal{A}1 = d$. Therefore L is positive semidefinite with a one-dimensional null space and eigenvalues in $[0, 2]$, and Q is positive definite for all $\kappa \geq 0$.

The global objective is a MAP optimization problem with respect to the joint base-coordinate measure of (28), in which the standard-normal prior on ξ has been replaced by the GMRF prior:

$$E(u) = \sum_{i=1}^{N} -\log p(u_i, b_i) + \frac{\kappa}{2} \sum_{j=1}^{n_{T_2}-1} \xi^{(j)}(u)^T L \xi^{(j)}(u). \tag{70}$$

The first term is the usual voxelwise MAP objective in u coordinates, summing the independent negative log joint densities; the second is the roughness penalty – the difference between the spatial prior and the i.i.d. prior – summed across coordinates, coupling neighbouring voxels. There are now two tasks: (1) to estimate the smoothing strength κ, and (2) to minimize the energy objective (70). The former is accomplished by an empirical Bayes approach, and the latter by a gradient-based optimization routine.

To estimate κ, we maximize an evidence lower bound on the marginal evidence $p(b \mid \kappa)$, similar to the flip-angle estimation in appendix A.1. Because the block-diagonal prior factorizes over coordinates, it suffices to consider a single coordinate. Let the trained amortized posterior $q(\xi \mid b)$ be the proposal distribution, where we drop the coordinate superscript on ξ for brevity. Then, the ELBO is given by the expected joint log density $\mathbb{E}_q[\log p(b, \xi \mid \kappa)]$ plus the entropy of q; since the likelihood $p(b \mid \xi)$ and entropy $\mathbb{E}_q[-\log q(\xi \mid b)]$ are independent of κ, maximizing the ELBO over κ reduces to maximizing the expected log prior under q:

$$\mathcal{L}(\kappa) = \mathbb{E}_q[\log p(\xi \mid \kappa)] = \frac{1}{2} \log|Q| - \frac{1}{2} \mathbb{E}_q\left[\xi^T Q \xi\right] + \text{const}, \tag{71}$$

[5] If the mask is not connected, there is one such eigenvector for each connected component. Given a partition of unity $1 = \sum_R 1_R$ over connected regions R, the corresponding eigenvectors are $d^{1/2} \odot 1_R$.

where we have substituted the Gaussian log-density $\log p(\xi \mid \kappa) = \frac{1}{2}\log|Q| - \frac{1}{2}\xi^T Q\xi + \text{const}$. By expanding the quadratic form $\xi^T Q\xi = \xi^T\xi + \kappa\xi^T L\xi$, we see that $\mathcal{L}$ depends on q only via the sufficient statistics

$$\begin{aligned} E_I &= \mathbb{E}_q\left[\xi^T\xi\right], \\ E_W &= \mathbb{E}_q\left[\xi^T W\xi\right], \\ E_L &= E_I - E_W = \mathbb{E}_q\left[\xi^T L\xi\right], \end{aligned} \tag{72}$$

thus simplifying the ELBO objective and its derivative to

$$\begin{aligned} \mathcal{L}(\kappa) &= \frac{1}{2}f(\kappa) - \frac{1}{2}(E_I + \kappa E_L) + \text{const}, \\ \mathcal{L}'(\kappa) &= \frac{1}{2}\left(f'(\kappa) - E_L\right), \end{aligned} \tag{73}$$

where $f(\kappa) = \log|\mathrm{I} + \kappa L|$ and $f'(\kappa) = \mathrm{tr}\big((\mathrm{I} + \kappa L)^{-1}L\big) = \sum_i \frac{\lambda_i}{1+\kappa\lambda_i}$, with $\lambda_i \in [0, 2]$ the eigenvalues of L. Since f' is strictly decreasing in κ from $f'(0) = \mathrm{tr}(L)$ to $f'(\infty) = 0$, a positive root of $f'(\kappa) = E_L$ exists and is unique whenever $0 < E_L < \mathrm{tr}(L)$; for $E_L \geq \mathrm{tr}(L)$, the maximizer is the boundary value $\kappa = 0$, indicating no smoothing penalty. Since q factorizes over voxels and L has unit diagonal, $E_L = \bar{\xi}^T L\bar{\xi} + \sum_i \mathrm{Var}_q(\xi_i)$ with $\bar{\xi} = \mathbb{E}_q[\xi]$. The amortized posterior is trained under the i.i.d. prior, so in coordinates that a single voxel's data barely inform, $\mathrm{Var}_q(\xi_i)$ stays close to the prior variance and the variance term alone drives E_L toward $\mathrm{tr}(L)$ and κ toward zero. We therefore evaluate the statistics at the posterior mean, $E_I = \bar{\xi}^T\bar{\xi}$ and $E_L = \bar{\xi}^T L\bar{\xi}$, a plug-in empirical Bayes estimate computed without finite-sample bias from posterior samples by U-statistics. Then $E_L = 0$ only if $\bar{\xi}$ is proportional to the null eigenvector $d^{1/2}$, which would imply $\kappa \to \infty$. We have two obvious choices to extend the single-coordinate analysis to the multi-coordinate case: independent κ_j can be fit to the respective coordinate statistics $E_I^{(j)}$ and $E_L^{(j)}$, or a single global κ shared across coordinates can be fit to the average statistics $\bar{E}_I = \frac{1}{n_{T_2}-1}\sum_j E_I^{(j)}$ and $\bar{E}_L = \frac{1}{n_{T_2}-1}\sum_j E_L^{(j)}$. We choose to use a shared global κ for simplicity.

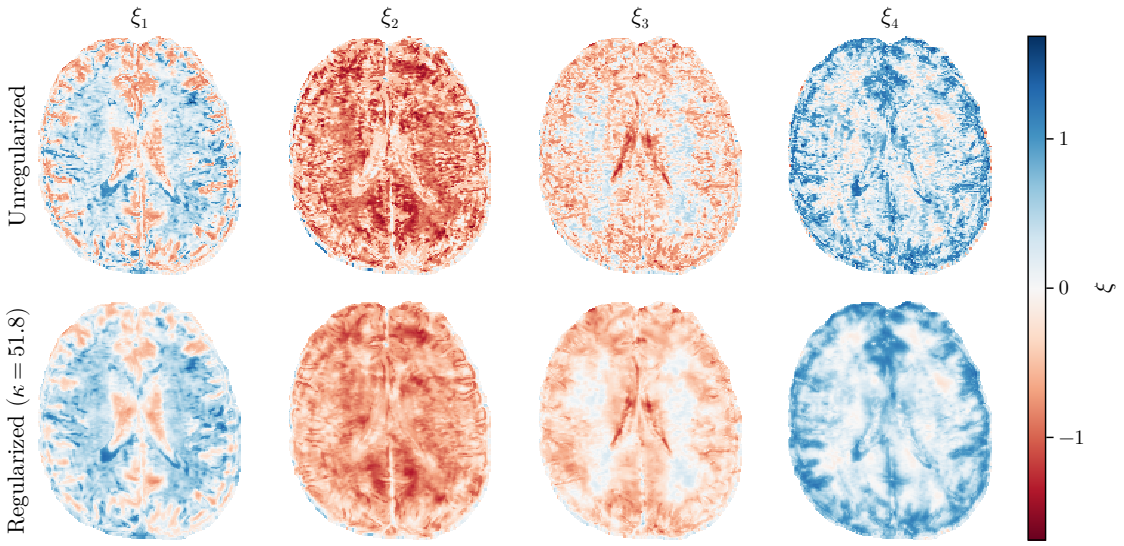


Figure 17: Effect of incorporating the spatial smoothing prior into the MAP estimate on the first four whitened ILR coordinates ξ.

After estimating κ, the spatially regularized solution u^* is found by minimizing equation (70) over u, initialized at the pseudo-mode $u = 0$. The nuisance coordinates $u_\eta = 0$ remain fixed at the conditional Laplace mode. Optimization uses PSGD with a low-rank Newton-type preconditioner fitted to Hessian-vector products [56, 57]. This results in smooth and spatially coherent images: figure 16 shows more spatially consistent biomarker maps than the unregularized MAP solution and the MH and DECAES baselines, and figure 17 illustrates the corresponding spatial smoothing of the ξ coordinates under the spatial prior.

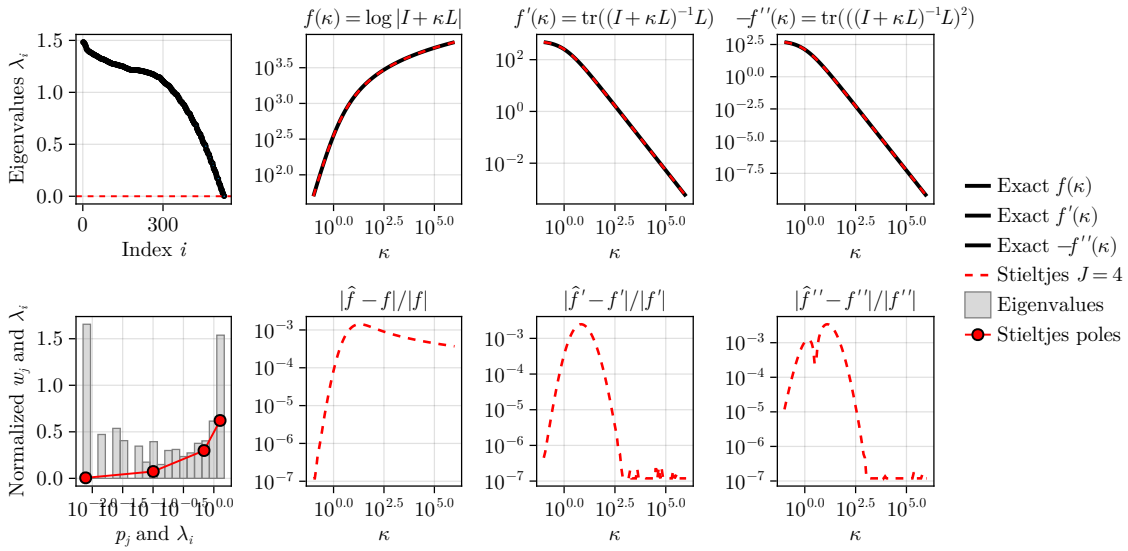


Figure 18: Comparison of the Stieltjes approximant ($J = 4$ poles) against closed-form expressions for the operator L on a central 23×23 crop of the selected slice. Top row: sorted eigenvalue spectrum of L, followed by exact and approximated values of $f(\kappa)$, $f'(\kappa)$, and $f''(\kappa)$. Bottom row: fitted poles and weights alongside the positive-eigenvalue histogram, followed by the relative approximation errors for $f(\kappa)$, $f'(\kappa)$, and $f''(\kappa)$.

A.2.1 Efficient Log-Determinant Evaluation by Spectral Moment Matching

Estimating the log-determinant $f(\kappa) = \log|\mathrm{I} + \kappa L|$ and its derivative for a graph of $N \sim 10^6$ voxels is nontrivial, since full eigendecomposition of L is impractically slow even if the graph is sparse. The derivative is, however, a *Stieltjes function* [83] of the spectral measure of L,

$$f(\kappa) = \log|\mathrm{I} + \kappa L| = \sum_{i=1}^{N} \log(1 + \kappa\lambda_i), \tag{74}$$

$$f'(\kappa) = \mathrm{tr}\big((\mathrm{I} + \kappa L)^{-1} L\big) = \sum_{i=1}^{N} \frac{\lambda_i}{1 + \kappa\lambda_i},$$

where λ_i are the eigenvalues of L. We approximate f' by a J-pole rational function [84]

$$f'(\kappa) \approx \sum_{j=1}^{J} \frac{w_j p_j}{1 + \kappa p_j} \tag{75}$$

whose poles $p_j > 0$ and weights $w_j > 0$ are chosen such that the approximant matches the asymptotic behaviour of $f'(\kappa)$ and its $J - 1$ derivatives as $\kappa \to 0^+$ and $\kappa \to \infty$, respectively. This requires estimating $\mathrm{tr}\big(L^k\big) = \sum_i \lambda_i^k$ for $1 \le k \le J$ and $\mathrm{tr}\big((L^\dagger)^k\big) =$

$\sum_{\lambda_i>0} \lambda_i^{-k}$ for $0 \le k \le J-1$, where $L^\dagger$ is the Moore–Penrose pseudo-inverse of L. The log-determinant approximation is then given by integrating the rational approximant term by term, $f(\kappa) \approx \sum_j w_j \log(1+\kappa p_j)$. We will also evaluate the accuracy of the rational approximation to $f''(\kappa) = -\operatorname{tr}\big(((\mathrm{I}+\kappa L)^{-1}L)^2\big)$, given by similar term-by-term differentiation as $f''(\kappa) \approx -\sum_j w_j p_j^2/(1+\kappa p_j)^2$. This moment-matching approach is equivalent to J-point Gaussian quadrature under the spectral measure of L, where the poles p_j are quadrature nodes and w_j are weights [84]. The forward moments $\operatorname{tr}\big(L^k\big)$ can be computed from $\operatorname{tr}\big(W^k\big)$, which, for small k, can be computed reasonably fast from summing over all closed walks of length k on the adjacency graph [85]. The pseudo-inverse moments $\operatorname{tr}\big((L^\dagger)^k\big)$ are estimated stochastically using Hutchinson's trace estimator [86] after deflating away the known null space of L. Figure 18 compares the $J = 4$-pole approximants with the closed-form expressions in equation (74), computed using exact eigenvalues of L for a two-dimensional grid.

A.3 Inference at a Different T_2 Resolution

While the learned change of variables acts on a fixed bin count n_{T_2}, each simplex sample $\bar{x}$ can be interpreted as a discretization of an underlying continuous T_2 density arising from a continuum of isochromats that each relax with their own T_2. The same trained components can therefore be reused at a different bin count $n'_{T_2} \neq n_{T_2}$, holding the physical range $(T_2^{\min}, T_2^{\max})$ fixed. This section compares two distinct ways to *lift* T_2 inference to a higher resolution. In the deterministic, or *pseudo*, lift, the native posterior is sampled and each output distribution $\bar{x}$ is then upsampled onto the target grid. The *product* lift instead samples the full n'_{T_2}-dimensional latent space, appending independent standard-normal high-frequency modes – the directions the trained flow never modelled – and evaluating the likelihood against the target-resolution basis $A_{n'}$.

Write $\ell = \log T_2$ and let $\Delta\ell = \log(T_2^{\max}/T_2^{\min})/(n_{T_2}-1)$, with $\Delta\ell'$ defined analogously at n'_{T_2}. At any resolution, a simplex sample $\bar{x}$ approximates the same continuous density via

$$\bar{x}_i \approx p(\ell_i)\Delta\ell, \quad p(\ell) \propto e^{g(\ell)}, \tag{76}$$

where g is a centred continuous logit field defined up to an additive constant. Cross-resolution inference amounts to the claim that the same g admits a faithful discretization on any grid spanning this physical range. The softmax normalization in $\mathrm{ilr}^{-1}(\cdot;\beta_0)$ (19) supplies the bin-width factor in (76). Appendix E shows that the corresponding DCT coefficients scale by $\sqrt{n'_{T_2}/n_{T_2}}$ between grids, while the bandwidth F_c transfers unchanged as a physical log-T_2 frequency.

A.3.1 Deterministic Lift by Interpolation

The trained generator maps the latent base vector $u \in \mathbb{R}^{n_{T_2}-1}$, with $u \sim \pi_u = \mathcal{N}(0, \mathrm{I}_{n_{T_2}-1})$, to a native-resolution distribution

$$\varphi_{\bar{x}}(u;b) = \mathrm{ilr}^{-1}(\Lambda\varphi(u;b);\beta_0), \tag{77}$$

where $\varphi(\cdot;b)$ is the trained Aitchison-space backbone and Λ is the trained low-pass window of (60). The deterministic, or *pseudo*, lift performs inference in this trained dimension and maps each posterior draw onto the target grid. Let $L \in \mathbb{R}^{n'_{T_2} \times n_{T_2}}$ be a nonnegative column-stochastic matrix that redistributes native bin masses onto the target grid – for instance the piecewise-linear interpolation matrix – so that $\bar{x}' = L\bar{x}$ is a valid target-resolution distribution

for any $\bar{x} \in \Delta_{n_{T_2}-1}$. Applying L to a trained posterior draw then defines the deterministic lift

$$\bar{x}'(u) = L\varphi_{\bar{x}}(u; b), \quad u \sim q_u(u \mid b). \tag{78}$$

The trained encoder and the native sampler operate in dimension n_{T_2} and cannot be applied to the target-resolution operator $A_{n'}$ directly. We therefore condition both on the composite *sketch basis* $B = A_{n'}L$, which compresses $A_{n'}$ back to n_{T_2} columns through L. Column-stochasticity gives the associativity $B\bar{x} = A_{n'}(L\bar{x}) = A_{n'}\bar{x}'$, so fitting $\bar{x}$ against B is identical to fitting the lifted distribution $\bar{x}'$ against the full target-resolution forward model. The sampled $\bar{x}$ is therefore not itself a T_2 distribution; it is a trained-dimension coordinate, likely under the trained prior, whose interpretable target-resolution image is $\bar{x}' = L\bar{x}$. Sampling remains n_{T_2}-dimensional, but replacing A by B changes both the likelihood and the encoder input, so the importance ratio need not be preserved. In the experiments below, the effective sample size decreases only slightly with the pseudo-lift, but note also that the lift merely interpolates the trained representation, and thus cannot recover target-resolution structure outside the column span of L.

A.3.2 Lifted Flow Construction

The product lift splices the trained backbone φ of (77) with an identity flow on the added coordinates. It operates on the padded base vector $u' = (u, v) \in \mathbb{R}^{n'_{T_2}-1}$ with $v \sim \mathcal{N}(0, \mathrm{I}_{n'_{T_2}-n_{T_2}})$ and is given explicitly by

$$\varphi_{\bar{x}'}(u'; b) = \mathrm{ilr}^{-1}\big(\Lambda'(\varphi \oplus \mathrm{I})(u'; b); \beta_0\big), \tag{79}$$

where $\varphi \oplus \mathrm{I}$ passes the first $n_{T_2} - 1$ coordinates through the trained backbone φ and the remaining $n'_{T_2} - n_{T_2}$ coordinates through the identity, $\mathrm{ilr}^{-1}(\cdot; \beta_0)$ uses the target-resolution basis Ψ' from (97), and Λ' is the lifted window with diagonal entries

$$\lambda'_i = \sqrt{\frac{n'_{T_2}}{n_{T_2}}}\lambda_i^+(f'_i), \quad f'_i = \frac{i}{n_{T_2}}, \tag{80}$$

where λ_i^+ is the noise-floored window of (60). The prefactor implements the coefficient scaling in (107). The normalized frequency f'_i is divided by the trained bin count rather than the target bin count, preserving the physical bandwidth F_c from (60).

A.3.3 Lifted Prior, Likelihood, and Importance Ratio

The lifted prior on $\bar{x}'$ mirrors (79) with φ replaced by the identity,

$$\bar{x}' = \mathrm{ilr}^{-1}\big(\Lambda' u'; \beta_0\big), \quad u' \sim \mathcal{N}(0, \mathrm{I}_{n'_{T_2}-1}), \tag{81}$$

yielding a low-pass windowed distribution on $\Delta_{n'_{T_2}-1}$. The quantized Rician likelihood (11) is evaluated against the lifted $\bar{x}'$ and the target-resolution EPG basis $A_{n'}$, built from the same physical acquisition parameters as A_n. Laplace's approximation (13) for the nuisance parameters $\log s$ and $\log \sigma$ is likewise reconditioned on $\bar{x}'$ and $A_{n'}$, so that they are estimated against the same forward operator the likelihood uses. The encoder, by contrast, still requires the trained dimension and is conditioned on the same sketch basis $B = A_{n'}L$ as the deterministic lift.

Since the trained backbone φ acts only on the first $n_{T_2} - 1$ base coordinates and v passes through the identity in *both* the lifted generator (79) and the lifted prior (81), the base-coordinate densities factor identically as

$$\pi_{u'}(u, v) = \pi_u(u) \cdot \mathcal{N}(v \mid 0, \mathrm{I}), \tag{82}$$

$$q_{u'}(u, v \mid b) = q_u(u \mid b) \cdot \mathcal{N}(v \mid 0, \mathrm{I}). \tag{83}$$

The forward Jacobians of the lifted flow therefore cancel in the independent-Metropolis importance ratio, as do the $\mathcal{N}(v \mid 0, \mathrm{I})$ base-density factors, leaving the trained-resolution ratio multiplied only by the likelihood contribution in v. If the likelihood were independent of v, the lifted importance ratio would reduce exactly to the trained one and Monte Carlo efficiency would be independent of $n'_{T_2} - n_{T_2}$; in practice, v enters the likelihood through $\mathrm{ilr}^{-1}(\cdot; \beta_0)$ and $A_{n'}$, so a residual high-mode mismatch produces finite importance-ratio variance. The experiments below compare the deterministic lift, the full product proposal, and a recentred product proposal that first shifts the lifted latent coordinates to their posterior mode.

A.3.4 Posterior Mode Recentring

The product lift in (79) uses the prior as the proposal for the added high modes. If the posterior mode in the lifted coordinates is displaced from $u' = 0$, then independent proposals waste samples near the wrong centre even when the local approximate posterior shape is reasonable. This motivates a shifted product proposal. Let u^* minimize the lifted negative log posterior using the same free energy as the sampler, with the encoder conditioned on $A_{n'}L$ and the likelihood evaluated using $A_{n'}$. The shifted generator is

$$\varphi^{\mathrm{MAP}}_{\bar{x}'}(u'; b) = \varphi_{\bar{x}'}(u' + u^*; b). \tag{84}$$

This shift has unit Jacobian and therefore changes only the proposal centre.

A.3.5 Experiments

Method	n_{T_2}	α_{MH}	α_{RWM}	α_{MALA}	$\mathrm{ESS}_{\mathrm{IS}}/S$	τ_{MH}	τ_{RWM}	τ_{MALA}
Native	41	$0.793^{0.81}_{0.77}$	$0.234^{0.24}_{0.23}$	$0.548^{0.56}_{0.54}$	$0.847^{0.87}_{0.8}$	$1.68^{2.0}_{1.5}$	$58.2^{80.0}_{43.0}$	$6.32^{7.7}_{5.3}$
Native + MAP	41	$0.771^{0.8}_{0.74}$	$0.234^{0.24}_{0.23}$	$0.55^{0.56}_{0.54}$	$0.82^{0.85}_{0.76}$	$1.72^{2.1}_{1.5}$	$57.4^{80.0}_{41.0}$	$6.34^{7.5}_{5.2}$
Product	81	$0.384^{0.52}_{0.26}$	$0.223^{0.23}_{0.21}$	$0.372^{0.46}_{0.28}$	$0.181^{0.35}_{0.075}$	$6.42^{13.0}_{3.8}$	$82.6^{110.0}_{60.0}$	$13.1^{20.0}_{9.6}$
Product + MAP	81	$0.511^{0.6}_{0.41}$	$0.223^{0.23}_{0.21}$	$0.374^{0.46}_{0.28}$	$0.303^{0.48}_{0.14}$	$4.03^{6.7}_{2.8}$	$82.2^{110.0}_{57.0}$	$13.5^{21.0}_{9.4}$
Pseudo-lift	81	$0.756^{0.79}_{0.7}$	$0.234^{0.24}_{0.23}$	$0.549^{0.56}_{0.54}$	$0.798^{0.85}_{0.71}$	$1.81^{2.3}_{1.6}$	$59.2^{79.0}_{42.0}$	$6.32^{7.6}_{5.2}$
Pseudo-lift + MAP	81	$0.766^{0.79}_{0.73}$	$0.234^{0.24}_{0.23}$	$0.548^{0.56}_{0.53}$	$0.809^{0.85}_{0.74}$	$1.76^{2.1}_{1.5}$	$56.2^{79.0}_{40.0}$	$6.23^{7.6}_{5.2}$
Product	121	$0.329^{0.47}_{0.2}$	$0.225^{0.23}_{0.21}$	$0.394^{0.48}_{0.31}$	$0.133^{0.29}_{0.051}$	$7.95^{17.0}_{4.5}$	$89.4^{120.0}_{62.0}$	$14.8^{22.0}_{11.0}$
Product + MAP	121	$0.506^{0.6}_{0.4}$	$0.226^{0.23}_{0.22}$	$0.397^{0.48}_{0.31}$	$0.307^{0.48}_{0.16}$	$4.18^{7.4}_{2.8}$	$93.0^{120.0}_{67.0}$	$14.4^{21.0}_{10.0}$
Pseudo-lift	121	$0.746^{0.78}_{0.68}$	$0.234^{0.24}_{0.23}$	$0.548^{0.56}_{0.53}$	$0.786^{0.84}_{0.69}$	$1.89^{2.5}_{1.6}$	$57.8^{82.0}_{42.0}$	$6.41^{7.7}_{5.3}$
Pseudo-lift + MAP	121	$0.762^{0.79}_{0.72}$	$0.235^{0.24}_{0.23}$	$0.548^{0.56}_{0.53}$	$0.804^{0.84}_{0.73}$	$1.8^{2.2}_{1.6}$	$56.3^{79.0}_{43.0}$	$6.36^{7.7}_{5.3}$

Table 8: Cross-resolution sampling diagnostics. Entries report median values with interquartile bounds across voxels. The native row is the trained-resolution reference; pseudo-lift denotes deterministic interpolation onto the target grid by (78); product denotes full n'_{T_2}-dimensional sampling; and MAP denotes recentring by (84).

To assess cross-resolution inference, we compare the pseudo-lift (78), the product lift (79), and the MAP-shifted product lift (84) at target resolutions $n'_{T_2} \in \{41, 81, 121\}$, holding n_{T_2} fixed at the trained resolution. The pseudo-lift samples the trained latent dimension against

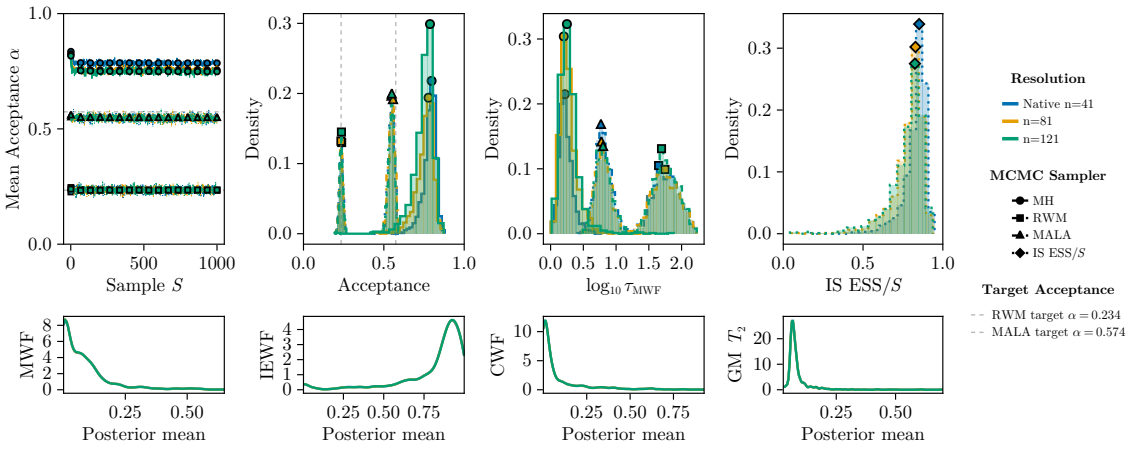


Figure 19: Cohort-level cross-resolution diagnostics for the pseudo-lift with MAP recentring. Acceptance trajectories, per-voxel acceptance distributions, MWF autocorrelation times, importance-sampling ESS/S, and posterior-mean biomarker distributions are shown across target resolutions.

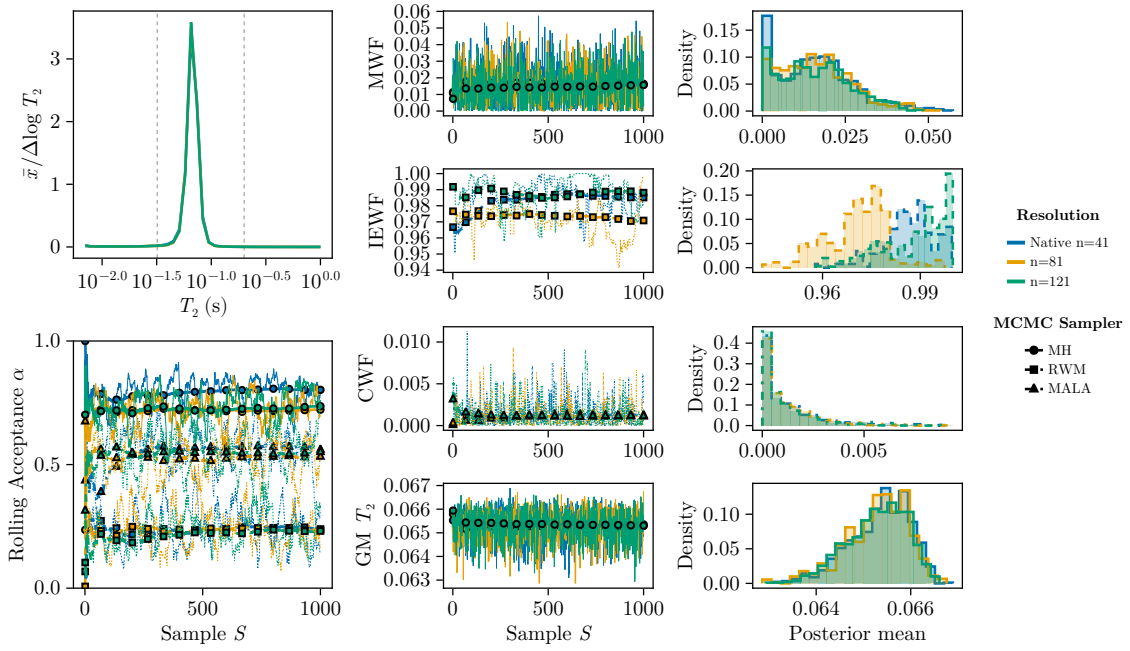


Figure 20: Voxel-level cross-resolution diagnostics for the pseudo-lift with MAP recentring. The top-left panel compares posterior T_2 densities, the bottom-left panel shows rolling independent-MH acceptance, and the remaining panels show biomarker traces and posterior biomarker distributions for one representative voxel.

$A_{n'}L$ and then interpolates each posterior draw to the target grid. The product methods sample the full lifted latent dimension against $A_{n'}$.

On the same ID test cohort, we ran independent MH, RWM, and MALA, using the step sizes of section 3.6.1, and importance sampling with the corresponding independent proposals. Table 8 and figures 19 and 20 report acceptance, autocorrelation, effective sample size, and biomarker diagnostics.

The pseudo-lift changed little with resolution, since it interpolates the native posterior without sampling added modes. The unshifted product lift degraded as n'_{T_2} increased; MAP recentring recovered part of the degradation.

B Extended Phase Graph Algorithm

The extended phase graph (EPG) basis matrix $A \in \mathbb{R}_+^{n_{\mathrm{TE}} \times n_{T_2}}$ is constructed using the EPG algorithm, which accounts for T_1 and T_2 relaxation, stimulated echoes, and B_1-inhomogeneity [9, 87]. Each column of A is a vector containing the magnitudes of n_{TE} spin echoes, forming a simulated multi-spin echo (MSE) signal. This signal is generated by evolving an initial magnetization state for n_{TE} periods of uniform echo spacing TE. The longitudinal relaxation time T_1 is the same for all columns, while the transverse relaxation time $T_{2,j}$ varies logarithmically from $T_2^{\min}$ (first column) to $T_2^{\max}$ (last column). The $(i,j)^{\mathrm{th}}$ entry of A is given by

$$A_{i,j} = |(\mathbf{M}^{i,j})_1^+|, \tag{85}$$

where $\mathbf{M}^{i,j} \in \mathbb{C}^{3n_{\mathrm{TE}}}$ is a *magnetization phase state vector* (MPSV) of n_{TE} spin population phase states. Each phase state $(\mathbf{M}^{i,j})_k$ is represented by a triplet $((\mathbf{M}^{i,j})_k^+, (\mathbf{M}^{i,j})_k^-, (\mathbf{M}^{i,j})_k^z)$ of Fourier coefficients, where $(\mathbf{M}^{i,j})_k^+$ and $(\mathbf{M}^{i,j})_k^-$ are the positive- and negative-order coefficients of the simulated complex transverse magnetization $M_{xy} = M_x + iM_y$, and $(\mathbf{M}^{i,j})_k^z$ is the corresponding coefficient of the simulated real longitudinal magnetization M_z.

The MPSV $\mathbf{M}^{i,j}$ is computed via the propagation equations

$$\begin{cases} \mathbf{M}^{i,j} & = E_j S R_j E_j \mathbf{M}^{i-1,j} \quad (1 \le i \le n_{\mathrm{TE}}), \\ (\mathbf{M}^{0,j})_1 & = (\sin\alpha_{\mathrm{ex}}, 0, 0), \\ (\mathbf{M}^{0,j})_k & = (0,0,0) \quad (2 \le k \le n_{\mathrm{TE}}) \end{cases} \tag{86}$$

where α_{ex} is the excitation pulse flip angle. The relaxation, rotation, and transition matrices E_j, R_j, and S are defined by their actions on an MPSV $\mathbf{M}$ as follows:

$$(E_j\mathbf{M})_k = \begin{bmatrix} E_{2,j} & 0 & 0 \\ 0 & E_{2,j} & 0 \\ 0 & 0 & E_1 \end{bmatrix} (\mathbf{M})_k, \tag{87}$$

$$(R_j\mathbf{M})_k = \begin{bmatrix} \cos^2(\alpha_j/2) & \sin^2(\alpha_j/2) & -i\sin(\alpha_j) \\ \sin^2(\alpha_j/2) & \cos^2(\alpha_j/2) & i\sin(\alpha_j) \\ -\frac{i}{2}\sin(\alpha_j) & \frac{i}{2}\sin(\alpha_j) & \cos(\alpha_j) \end{bmatrix} (\mathbf{M})_k, \tag{88}$$

$$S\mathbf{M} : \begin{cases} (S\mathbf{M})_1^+ = (\mathbf{M})_1^-, \\ (S\mathbf{M})_k^+ = (\mathbf{M})_{k-1}^+ \quad (2 \le k \le n_{\mathrm{TE}}), \\ (S\mathbf{M})_k^- = (\mathbf{M})_{k+1}^- \quad (1 \le k \le n_{\mathrm{TE}} - 1), \\ (S\mathbf{M})_{n_{\mathrm{TE}}}^- = 0, \\ (S\mathbf{M})_k^z = (\mathbf{M})_k^z. \end{cases} \tag{89}$$

Here, $(\mathbf{M}^{i,j})_k^+$ denotes the transverse component of the kth triplet of $\mathbf{M}^{i,j}$, and similarly $(\mathbf{M}^{i,j})_k^-$ and $(\mathbf{M}^{i,j})_k^z$ denote the anti-transverse and longitudinal components, respectively; $E_{2,j} = \exp(-\frac{\text{TE}/2}{T_{2,j}})$ and $E_1 = \exp(-\frac{\text{TE}/2}{T_1})$ are attenuation factors; $\alpha_{\text{ex}} = \bar{\alpha}\frac{\pi}{2}$ and $\alpha_j \equiv \bar{\alpha}\pi$ are excitation and refocusing flip angles; and $\bar{\alpha}(x, y, z) \in [0, 1]$ is a spatially varying map of B_1-inhomogeneity correction factors. This B_1 map can be measured separately or estimated per voxel [1, 9]. For convenience we refer to $\alpha = \bar{\alpha}\pi$ as *the* flip angle of a voxel, noting that α determines the excitation flip angle $\alpha_{\text{ex}} = \frac{\alpha}{2}$.

The EPG basis matrix A is constructed efficiently using the DECAES open-source software package [1].

C Quantized Rician Noise Model

Magnitude MRI data is more accurately modelled with Rician rather than Gaussian noise [25–27]. Given a true signal ν and noise level σ, the Rician probability density function (PDF) for the measured signal magnitude x is:

$$p_{\text{Rice}}(x \mid \nu, \sigma) = \frac{x}{\sigma^2} \exp\left(-\frac{x^2 + \nu^2}{2\sigma^2}\right) I_0\left(\frac{x\nu}{\sigma^2}\right), \tag{90}$$

where I_0 is the modified Bessel function of the first kind with order zero.

When the measurement x is quantized to a resolution of δ, the probability of observing x given ν and σ is the integral of p_{Rice} over the bin $[x, x + \delta)$:

$$p_{\text{QRice}}(x \mid \nu, \sigma, \delta) = \int_0^\delta p_{\text{Rice}}(x + \epsilon \mid \nu, \sigma)\, d\epsilon. \tag{91}$$

We approximate p_{QRice} using Gauss–Legendre quadrature:

$$p_{\text{QRice}}(x \mid \nu, \sigma, \delta) \approx \frac{\delta}{2} \sum_{k=1}^{N} w_k p_{\text{Rice}}\left(x_k \mid \nu, \sigma\right), \tag{92}$$
$$x_k = x + \frac{\delta}{2}(1 + \xi_k),$$

where $\xi_k \in (-1, 1)$ and $w_k > 0$ are the N Gauss–Legendre nodes and weights. We choose this method because p_{Rice} is analytic, and Gauss–Legendre quadrature is nearly optimal among quadrature rules for analytic functions [88, 89]. The implementation uses $N = 8$ nodes and weights generated at compile time for likelihood evaluation and the final nuisance Hessian. Its automatic derivatives use the same quadrature rule.

Note that the accuracy of this non-adaptive quadrature depends on the integrand and the integration interval. Pathological cases, such as $\delta \gg \sigma$ (a sharply peaked distribution relative to the integration domain) or $x \gg \nu$ (integration in the tail where p_{Rice} decays rapidly), can lead to errors. While adaptive methods could address these issues, they introduce additional complexity and computational overhead. In our posterior sampling application, the variational training optimizes for ν and σ that are likely given x, encouraging $[x, x + \delta)$ to contain the Rician PDF's mode. We also expect $\sigma \gtrsim \delta$, since σ is estimated from the data, and noise levels smaller than δ cannot be resolved due to quantization. Thus, we integrate over small intervals near the mode, where p_{Rice} is slowly varying and well-suited for quadrature.

Our implementation of the Rician and quantized Rician probability density functions, including numerical integration and derivative calculations for automatic differentiation, is open-sourced at `https://github.com/jondeuce/FastRicianLikelihoods.jl`. This implementation uses numerically stable expressions for the Rician log-density and its derivatives, including the explicit derivatives needed to compute Laplace's approximation in section 3.2.4.

D Gaussian Initialization for Laplace's Approximation

As described in section 3.2.4, we use Newton's method to compute the mean μ_η for Laplace's approximation. To obtain a suitable initial guess for this method, we approximate the posterior mode by substituting a Gaussian likelihood for the quantized Rician likelihood and solving for the mode of the resulting posterior. We assume that $\bar{x}$ is given and fixed (see Equation (12)).

We approximate the quantized Rician likelihood model with a Gaussian likelihood model for the mean dequantized signal $\bar{b} = b + \delta/2$, where δ denotes the quantization bin width. This approximation is motivated by considering the simplest case where the quadrature in (92) is computed with only $N = 1$ node. In this case, the quantized Rician model is exactly equivalent to a Rician model evaluated at the mean dequantized signal $\bar{b}$. For sufficiently large ν/σ, this Rician model is itself well-approximated by a Gaussian likelihood. Thus, to compute the initial guess, we minimize the negative log-joint posterior

$$-\log p(\eta, \bar{b} \mid \bar{x}) = -\log p(\bar{b} \mid \bar{x}, \eta) - \log p(\eta \mid \bar{x}) \tag{93}$$

$$\text{where} \quad -\log p(\bar{b} \mid \bar{x}, \eta) = \frac{n_{\mathrm{TE}}}{2} \log(2\pi\sigma^2) + \frac{\|\bar{b} - s\bar{\nu}\|^2}{2\sigma^2} \tag{94}$$

is the Gaussian likelihood, $\bar{\nu} = A\bar{x}$, and the prior $p(\eta \mid \bar{x}) \equiv p(\eta)$ on the nuisance parameters $\eta = (\log s, \log \sigma)$ is taken to be independent of $\bar{x}$.

We begin by computing the maximum likelihood estimate (MLE), i.e., the mode of the likelihood (94). Setting the gradient of (94) with respect to η to zero yields:

$$0 = \nabla_\eta \left[-\log p(\bar{b} \mid \bar{x}, \eta) \right] = \begin{pmatrix} -\frac{s}{\sigma^2} \bar{\nu} \cdot (\bar{b} - s\bar{\nu}) \\ n_{\mathrm{TE}} - \frac{1}{\sigma^2} \|\bar{b} - s\bar{\nu}\|_2^2 \end{pmatrix}.$$

Solving for s and σ^2 gives the maximum likelihood estimators:

$$s_{\mathrm{MLE}} = \frac{\bar{\nu} \cdot \bar{b}}{\|\bar{\nu}\|^2} \quad \sigma^2_{\mathrm{MLE}} = \frac{\|\bar{b} - s_{\mathrm{MLE}}\bar{\nu}\|^2}{n_{\mathrm{TE}}}. \tag{95}$$

Equivalently, s_{MLE} minimizes $\|\bar{b} - s\bar{\nu}\|^2$, and σ^2_{MLE} is the mean square of the residuals $r = \bar{b} - s_{\mathrm{MLE}}\bar{\nu}$. We will also need the Hessian of (94) at the MLE:

$$\begin{aligned} H(\eta) &= \nabla_\eta^2 \left[-\log p(\bar{b} \mid \bar{x}, \eta) \right] \\ &= \begin{bmatrix} -\frac{s}{\sigma^2} \bar{\nu} \cdot (\bar{b} - 2s\bar{\nu}) & \frac{2s}{\sigma^2} \bar{\nu} \cdot (\bar{b} - s\bar{\nu}) \\ \frac{2s}{\sigma^2} \bar{\nu} \cdot (\bar{b} - s\bar{\nu}) & \frac{2}{\sigma^2} \|\bar{b} - s\bar{\nu}\|^2 \end{bmatrix} \\ \Rightarrow H_{\mathrm{MLE}} = H(\mu_{\eta,\mathrm{MLE}}) &= \begin{bmatrix} \frac{s^2_{\mathrm{MLE}}}{\sigma^2_{\mathrm{MLE}}} \|\bar{\nu}\|^2 & 0 \\ 0 & 2n_{\mathrm{TE}} \end{bmatrix} \end{aligned}$$

where $\mu_{\eta,\mathrm{MLE}} = (\log s_{\mathrm{MLE}}, \log \sigma_{\mathrm{MLE}})$.

Now, we incorporate a Gaussian prior on the nuisance parameters, $p(\eta) = \mathcal{N}(\eta \mid \mu_{\eta,0}, \Sigma_{\eta,0})$, where $\Sigma_{\eta,0}$ is a diagonal covariance matrix. The prior parameters $\mu_{\eta,0}$ and $\Sigma_{\eta,0}$ are learned during training. We cannot compute the minimizer of (93) exactly with this prior, but since we are only looking for an initial guess for the quantized Rician problem, an approximate solution is sufficient. Taylor expanding the likelihood around the MLE gives

$$\begin{aligned} -\log p(\eta, \bar{b} \mid \bar{x}) \approx & \frac{1}{2}(\eta - \mu_{\eta,\mathrm{MLE}})^T \Sigma^{-1}_{\mathrm{MLE}} (\eta - \mu_{\eta,\mathrm{MLE}}) \\ & + \frac{1}{2}(\eta - \mu_{\eta,0})^T \Sigma^{-1}_{\eta,0} (\eta - \mu_{\eta,0}) + \text{const.} \end{aligned}$$

where $\Sigma_{\mathrm{MLE}}^{-1} = H_{\mathrm{MLE}}$ and the constant term is independent of η. If we assume that the prior is weak enough such that the new mode is not too far from the MLE, we can set the gradient of this quadratic approximation to zero and solve for the approximate posterior mode:

$$\begin{aligned} 0 &= \Sigma_{\mathrm{MLE}}^{-1}(\eta - \mu_{\eta,\mathrm{MLE}}) + \Sigma_{\eta,0}^{-1}(\eta - \mu_{\eta,0}) \\ \Rightarrow \mu_{\eta,\mathrm{MAP}} &\approx (\Sigma_{\mathrm{MLE}}^{-1} + \Sigma_{\eta,0}^{-1})^{-1}(\Sigma_{\mathrm{MLE}}^{-1}\mu_{\eta,\mathrm{MLE}} + \Sigma_{\eta,0}^{-1}\mu_{\eta,0}). \end{aligned} \tag{96}$$

Finally, we have our initial guess $\mu_{\eta,\mathrm{MAP}}$ for computing μ_η using Newton's method on the original problem with the quantized Rician likelihood. This approximation is very good in practice; typically only 3–4 iterations of Newton's method are needed to converge to the true mode in 32-bit arithmetic.

The required gradients and Hessians of the quantized Rician log-likelihood are evaluated using the explicit derivative formulas developed in section 3.2.4. The Newton step over $(\log s, \log \sigma)$ is computed using a spectral trust-region update, which limits the step norm when the local Hessian is poorly conditioned. From the Gaussian initialization, the solver takes one Gaussian Newton step and one midpoint-rule quantized-Rician step before switching to the eight-node Gauss–Legendre rule used for the final Hessian. Since $\eta \in \mathbb{R}^2$, the covariance square root $\Sigma_\eta^{1/2}$ used in the affine Laplace flow equation (40) is obtained from a 2×2 eigendecomposition of Σ_η.

E Softmax Concentration and Coefficient Scaling under Low-Pass Filtering

This section derives two scaling properties underlying the low-pass Aitchison prior and the cross-T_2 resolution experiments. First, we show how the low-pass filtering of ILR coefficients generated from a standard normal prior controls the concentration and dynamic range of the induced simplex distribution. Second, we derive how the amplitudes of the DCT-basis ILR coefficients of a fixed continuous logit field scale as a function of the number of T_2 grid points.

Let n_{T_2} denote the number of T_2 bins and let $n_u = n_{T_2} - 1$ denote the corresponding Aitchison dimension. Parameterize the log-T_2 range by $x \in [0,1]$ via $\ell(x) = \log T_2^{\min} + x \log(T_2^{\max}/T_2^{\min})$, and let $x_j = (j - 1/2)/n_{T_2}$ denote the midpoints of the n_{T_2} equispaced subintervals of $[0,1]$. The orthonormal DCT-II basis $\Psi \in \mathbb{R}^{n_{T_2} \times (n_{T_2}-1)}$ is given explicitly by

$$(\Psi)_{jk} = \sqrt{\frac{2}{n_{T_2}}} \cos(\pi k x_j) \quad \text{where} \quad x_j = \frac{j - 1/2}{n_{T_2}}. \tag{97}$$

The columns have unit norm and per-entry amplitude bounded by $\sqrt{2/n_{T_2}}$.

Let $u \sim \mathcal{N}(0, \mathrm{I}_{n_u})$, let $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_{n_u})$ be a diagonal low-pass filter, and let $\beta > 0$ be the ILR scale, typically with the canonical value $\beta = \beta_0 = \pi/\sqrt{6}$ from appendix F.3.2. The filtered logits and corresponding simplex sample are

$$z = \beta \Psi \Lambda u, \quad \bar{x} = \mathrm{softmax}(z), \tag{98}$$

where $\bar{x} \in \Delta_{n_{T_2}-1}$. Assume a high-frequency cutoff F_c measured in cycles per sampled log-T_2 interval, e.g. $\lambda_i = \exp(-(f_i/F_c)^2/2)$ where $f_i = i/n_{T_2}$ as in equation (60).

E.1 Softmax Concentration and Extremal Values

Given the effective cutoff frequency F_c of Λ, the correlation length L_c of z along the log-T_2 grid scales as

$$L_c \sim \frac{1}{F_c}. \tag{99}$$

Consequently, a local maximum of z typically extends over a cluster of m adjacent bins where

$$m = \frac{\alpha}{F_c} \sim \alpha L_c, \tag{100}$$

with $\alpha > 0$ a dimensionless constant. These m largest entries are the dominant contributors to the log-sum-exp denominator of softmax(z), allowing for the approximation

$$\mathrm{LSE}(z) = \log \sum_i e^{z_i} \approx \log\left(m e^{z_{\max}}\right) = z_{\max} + \log m \tag{101}$$

which can be used to estimate the largest simplex component value:

$$\bar{x}_{\max} = e^{z_{\max} - \mathrm{LSE}(z)} \approx \frac{1}{m} \approx \frac{F_c}{\alpha}. \tag{102}$$

Thus, F_c controls both the peak width and height of a typical $\bar{x}$ sample drawn from the prior.

The minimum component is governed by the dynamic range of z. For the low-pass Gaussian model in equation (98), the average marginal logit variance σ_z^2 of $z \sim \mathcal{N}(0, \beta^2 \Psi \Lambda^2 \Psi^T)$ is

$$\sigma_z^2 = \frac{\beta^2}{n_{T_2}} \sum_{i=1}^{n_u} \lambda_i^2 \tag{103}$$

where $\mathrm{tr}\big(\Psi \Lambda^2 \Psi^T\big) = \mathrm{tr}\big(\Lambda^2\big) = \sum_{i=1}^{n_u} \lambda_i^2$ follows from the cyclic property of the trace and the semi-orthogonality of Ψ. Define the typical number of independent clusters of z by

$$n_{\mathrm{eff}} \sim \frac{n_{T_2}}{m} = \frac{n_{T_2} F_c}{\alpha}. \tag{104}$$

Then, extreme-value theory for Gaussian random variables suggests [90]

$$z_{\max} \approx \sigma_z \sqrt{2 \log n_{\mathrm{eff}}}, \tag{105}$$

and symmetry under $z \mapsto -z$ suggests the typical relation $z_{\min} \approx -z_{\max}$, giving the approximations

$$\bar{x}_{\min} \approx \frac{1}{m} e^{-2 z_{\max}}, \quad \frac{\bar{x}_{\max}}{\bar{x}_{\min}} \approx e^{2 z_{\max}}. \tag{106}$$

These heuristics imply that the dynamic range grows exponentially with $\sigma_z \propto \beta \bar{\lambda}$ where $\bar{\lambda}^2 = \frac{1}{n_{T_2}} \sum_{i=1}^{n_u} \lambda_i^2$. For a hard cutoff F_c with $\lambda_i \propto 1_{f_i \le F_c}$, it follows that $\bar{\lambda}^2 \propto F_c$ and thus $\sigma_z \propto \beta \sqrt{F_c}$.

E.2 Coefficient Scaling under Regridding

Let $g(\ell(x))$ be a centred continuous logit field of bounded variation representing a true underlying smooth signal, and sample logits $z_j = g(\ell(x_j))$ at the discrete DCT-II grid points $x_j = (j - 1/2)/n_{T_2}$. Projecting g onto the DCT grid with n_{T_2} samples gives

$$\begin{aligned} c_k = (\Psi^T z)_k &= \sqrt{\frac{2}{n_{T_2}}} \sum_{j=1}^{n_{T_2}} \cos(\pi k x_j) g(\ell(x_j)) \\ &= \sqrt{2 n_{T_2}} \tilde{c}_k \left(1 + O(n_{T_2}^{-1})\right), \end{aligned} \tag{107}$$

$$\tilde{c}_k = \int_0^1 g(\ell(x)) \cos(\pi k x)\, dx.$$

The leading term follows from interpreting the sum as a midpoint Riemann approximation to the integral $\tilde{c}_k$. The coefficients $\tilde{c}_k$ encode the continuous field g, and hence the density $p(\ell) \propto e^{g(\ell)}$, independent of the grid size; the $\sqrt{2n_{T_2}}$ factor provides the amplitude scaling for the specific grid.

Ensuring a grid-independent representation for g is sufficient to ensure a regridding consistency for $\bar{x}$ due to the renormalization inherent in the softmax:

$$\begin{aligned} \sum_{i=1}^{n_{T_2}} e^{z_i} &= \sum_{i=1}^{n_{T_2}} e^{g(\ell(x_i))} \\ &= n_{T_2} \int_0^1 e^{g(\ell(x))}\, dx + O(1) \\ &= n_{T_2} Z \left(1 + O(n_{T_2}^{-1})\right), \end{aligned} \tag{108}$$

where $Z = \int_0^1 e^{g(\ell(x))}\, dx$. Writing the bin width explicitly as $\Delta\ell = L/n_{T_2}$ where $L = \log(T_2^{\max}/T_2^{\min})$, it follows that

$$\begin{aligned} \bar{x}_j = \frac{e^{z_j}}{\sum_i e^{z_i}} \approx \frac{e^{g(\ell(x_j))}}{n_{T_2} Z} = p(\ell_j)\, \Delta\ell, \\ \text{where} \quad p(\ell) = \frac{e^{g(\ell)}}{ZL}. \end{aligned} \tag{109}$$

and we see that $\bar{x}_j$ is naturally a consistent midpoint discretization $p(\ell_j)\,\Delta\ell$; a finer grid $n'_{T_2} > n_{T_2}$ results in coefficients c'_k larger by a factor $\sqrt{n'_{T_2}/n_{T_2}}$, representing the same g and thus the same $p \propto e^g$, and $\bar{x}'_j \approx p(\ell'_j)\,\Delta\ell'$ about a factor of $\Delta\ell'/\Delta\ell = n_{T_2}/n'_{T_2}$ smaller.

E.3 Scaling Consistency with the Sampling Interval

The discretization (109) holds for any continuous density $p(\ell)$, but the prior reproduces the same $p(\ell)$ across grids only if the cutoff frequency is chosen appropriately. Near the largest peak, the concentration estimate from equation (102) gives

$$\bar{x}_{\max} \approx \frac{F_c}{\alpha} \approx \left(\max_\ell p(\ell)\right) \Delta\ell \quad \Leftrightarrow \quad F_c \propto \Delta\ell. \tag{110}$$

Therefore F_c should be parameterized such that it scales with the $\log T_2$ bin width $\Delta\ell$, resulting in prior samples which are consistent discretizations of $p(\ell)$ at any resolution. For example, the peak mass is then approximately invariant to $\Delta\ell$:

$$M \approx m\bar{x}_{\max} \propto (\alpha/F_c)(\Delta\ell) \propto \alpha. \tag{111}$$

Thus the integrated peak mass is preserved under regridding. In this sense, prior samples drawn using different grid spacings will be consistent discretizations of an underlying density $p(\ell)$.

E.4 Scaling in Regions of Local Maxima with Multiple Peaks

Now assume that z contains two dominant clusters, with m_1 correlated neighbouring entries around z_1 and m_2 around $z_2 < z_1$. The log-sum-exp of z can then be approximated by

$$\mathrm{LSE}(z) \approx \log(m_1 e^{z_1} + m_2 e^{z_2}) \tag{112}$$

and then peaks of $\bar{x}$ are approximated as

$$\bar{x}_{\max,1} \approx \frac{e^{z_1}}{m_1 e^{z_1} + m_2 e^{z_2}} = \frac{1}{m_1 + m_2 e^{z_2 - z_1}}, \tag{113}$$

$$\bar{x}_{\max,2} \approx \frac{e^{z_2}}{m_1 e^{z_1} + m_2 e^{z_2}} = \frac{e^{z_2 - z_1}}{m_1 + m_2 e^{z_2 - z_1}}. \tag{114}$$

If $m_1 \approx m_2 \approx m = \alpha / F_c$, then these expressions reduce to

$$\bar{x}_{\max,1} \approx \frac{\sigma(z_1 - z_2)}{m}, \quad \bar{x}_{\max,2} \approx \frac{\sigma(z_2 - z_1)}{m}, \tag{115}$$

where $\sigma(z) = e^z/(1+e^z)$. Since both peaks scale as $1/m \propto \Delta\ell$ and contain $m \propto 1/\Delta\ell$ points, their masses $M_j \approx m\bar{x}_{\max,j}$ are approximately invariant to $\Delta\ell$.

Since $n_{\text{eff}} = n_{T_2}/m = n_{T_2} F_c/\alpha \propto n_{T_2}\Delta\ell$, the expected peak separation is

$$z_1 - z_2 = \mathcal{O}\left(\frac{\sigma_z}{\sqrt{2\log n_{\text{eff}}}}\right) = \mathcal{O}\left(\frac{\beta\sqrt{\Delta\ell}}{\sqrt{2\log n_{\text{eff}}}}\right). \tag{116}$$

If $1 \ll n_{\text{eff}} \ll n_{T_2}$ and σ_z is moderate, then $0 < \sigma(z_2 - z_1) < 1/2$ remains moderate and $\bar{x}_{\max,2} \propto 1/m$.

F Distributions on the Probability Simplex

F.1 The Probability Simplex

The *probability simplex* Δ_{n-1} – also known as the *standard simplex* or simply *the simplex* – is the set of positive n-dimensional vectors with unit sum:

$$\Delta_{n-1} = \{p \in \mathbb{R}^n \mid p > 0 \text{ and } \langle 1, p\rangle = 1\} \tag{117}$$

where 1 is the vector containing all ones. We treat Δ_{n-1} as embedded in $\mathbb{R}^n$, lying on the hyperplane $\mathcal{P}_{n-1}$:

$$\Delta_{n-1} \subset \mathcal{P}_{n-1} = \{p \in \mathbb{R}^n \mid \langle 1, p\rangle = 1\} \subset \mathbb{R}^n. \tag{118}$$

An alternative definition of Δ_{n-1} is in terms of the *regular polytope* $\mathcal{S}_{n-1} \subset \mathbb{R}^{n-1}$, the volume containing positive vectors with sum less than one:

$$\mathcal{S}_{n-1} = \left\{x \in \mathbb{R}^{n-1} \mid x > 0 \text{ and } \langle 1, x\rangle < 1\right\}. \tag{119}$$

In terms of $\mathcal{S}_{n-1}$, the probability simplex is given by

$$\Delta_{n-1} = \left\{ p = \begin{pmatrix} x \\ 1 - \langle 1, x\rangle \end{pmatrix} \in \mathbb{R}^n \middle| x \in \mathcal{S}_{n-1} \right\}. \tag{120}$$

As $\Delta_{n-1} \subset \mathcal{P}_{n-1}$, the *tangent space* $T_p\Delta_{n-1}$ for any $p \in \Delta_{n-1}$ is $\mathcal{T}_{n-1}$, the hyperplane of vectors orthogonal to the normal vector 1:

$$T_p\Delta_{n-1} = \mathcal{T}_{n-1} = \{q \in \mathbb{R}^n \mid \langle 1, q\rangle = 0\}. \tag{121}$$

F.2 The Aitchison Geometry

The interior of the simplex Δ_{n-1} can be given an $(n-1)$-dimensional vector space structure based on the *perturbation* and *power* operations [33, 34]:

$$p \oplus q = \mathcal{C}[p_1 q_1, \ldots, p_n q_n] \tag{122}$$

$$\alpha \otimes p = \mathcal{C}[p_1^\alpha, \ldots, p_n^\alpha] \tag{123}$$

where $p, q \in \Delta_{n-1}$, $\alpha \in \mathbb{R}$, and $\mathcal{C}[\cdot]$ denotes the closure operation that projects the given vector onto Δ_{n-1} by dividing elementwise by the sum of the vector. Subtraction is defined as $p \ominus q = p \oplus (-1 \otimes q)$. A Hilbert space structure can then be defined via the dot product [35]

$$\langle p, q \rangle_a = \frac{1}{n} \sum_{i<j} \log \frac{p_i}{p_j} \log \frac{q_i}{q_j} = \sum_{i=1}^n \log \frac{p_i}{\mathcal{G}(p)} \log \frac{q_i}{\mathcal{G}(q)} \tag{124}$$

$$= \sum_{i=1}^n \log p_i \log q_i - \frac{1}{n} \sum_{i=1}^n \log p_i \sum_{i=1}^n \log q_i \tag{125}$$

$$= \langle \log p, \left(\mathrm{I}_n - \frac{1}{n} 1 1^T \right) \log q \rangle \tag{126}$$

$$= \langle \Psi^T \log p, \Psi^T \log q \rangle \tag{127}$$

where $\mathcal{G}(p) = (\prod_i p_i)^{1/n}$ is the geometric mean of p, $\Psi \Psi^T = \mathrm{I}_n - \frac{1}{n} 1 1^T$ is the projection matrix onto the plane $\mathcal{T}_{n-1}$, and Ψ is any orthonormal basis for $\mathcal{T}_{n-1}$. The Aitchison norm and distance derive from the inner product in the usual way: $\|p\|_a = \sqrt{\langle p, p \rangle_a}$ and $d_a(p, q) = \|p \ominus q\|_a$. The resulting geometry is known as the *Aitchison geometry*.

The Aitchison geometry depends only on the *relative* values of the components of p and q, so all operations are scale invariant. Specifically, if $p, q \in \mathbb{R}^n_+$, $\alpha \in \mathbb{R}$, and $\beta, \gamma > 0$, then $(\beta p) \oplus (\gamma q) = p \oplus q$, $\alpha \otimes (\beta p) = \alpha \otimes p$, and $\langle \beta p, \gamma q \rangle_a = \langle p, q \rangle_a$.

Linear operators can be extended to the Aitchison geometry straightforwardly. Let $\{\hat{e}_j\}_{j=1}^{n-1}$ be an orthonormal basis for the simplex with respect to the Aitchison inner product. The linear operator $\mathcal{A}$ can be represented by the matrix A with entries $A_{ij} = \langle \hat{e}_i, \mathcal{A} \hat{e}_j \rangle_a$. Given a vector $p \in \Delta_{n-1}$ with coordinates $u_j = \langle \hat{e}_j, p \rangle_a$, the action of $\mathcal{A}$ on p is then

$$\mathcal{A} p = \bigoplus_{i=1}^{n-1} \left(\sum_{j=1}^{n-1} A_{ij} u_j \right) \otimes \hat{e}_i. \tag{128}$$

F.3 Distributions over Isometric Log-Ratio Coefficients

The *isometric log-ratio (ILR)* transformation and its inverse are given by [35]

$$u = \mathrm{ilr}(x; \beta) = \frac{1}{\beta} \Psi^T \log x \tag{129}$$

$$x = \mathrm{ilr}^{-1}(u; \beta) = \mathrm{softmax}(\beta \Psi u) \tag{130}$$

where $\beta > 0$, $u \in \mathbb{R}^{n-1}$ are the *ILR coefficients*, $x \in \Delta_{n-1}$ is a vector on the simplex, $\Psi \in \mathbb{R}^{n \times (n-1)}$ is an orthonormal basis for the plane $\langle 1, \cdot \rangle = 0$, and $\mathrm{softmax}(z)_i = e^{z_i} / \sum_j e^{z_j}$. Note that $u = 0 \Leftrightarrow x = \frac{1}{n} 1$, and $\|u\| \to \infty \Leftrightarrow x \to \partial \Delta_{n-1}$.

Unconstrained ILR coefficients are much easier to work with than constrained vectors on the simplex. To convert between distributions over u and x, we use the change-of-variables formula:

$$p_u(u) = p_x(x) \left| \frac{\partial \tilde{x}}{\partial u} \right| = p_x(x) \left(\beta^{n-1} \sqrt{n} \prod_{i=1}^n x_i \right) \tag{131}$$

where the differential change in volume $\left|\frac{\partial \tilde{x}}{\partial u}\right|$ is the Jacobian determinant of $\tilde{x}(u)$, where $\tilde{x}$ are any $n-1$ independent components of x; see the derivation in appendix G.1 which extends the result from [36] to general β, Ψ.

F.3.1 Geometry of the ILR Transformation

The ILR transformation with $\beta = 1$ is an isometry with respect to the Aitchison inner product (124), meaning $\langle x, y\rangle_a = \langle u, v\rangle$ where $u, v \in \mathbb{R}^{n-1}$ are the ILR representations of $x, y \in \Delta_{n-1}$. By bilinearity of the inner product, it is sufficient to check this for basis vectors $\hat{e}_j, \hat{e}_k \in \mathbb{R}^{n-1}$:

$$\begin{aligned}
\langle x, y\rangle_a &= \langle \text{ilr}^{-1}(\hat{e}_j; \beta = 1), \text{ilr}^{-1}(\hat{e}_k; \beta = 1)\rangle_a \\
&= \langle \text{softmax}\big(\psi^j\big), \text{softmax}\Big(\psi^k\Big)\rangle_a \\
&= \langle (e^{\psi_1^j}, \ldots, e^{\psi_n^j}), (e^{\psi_1^k}, \ldots, e^{\psi_n^k})\rangle_a \\
&= \langle \Psi^T \psi^j, \Psi^T \psi^k\rangle \\
&= \langle \hat{e}_j, \hat{e}_k\rangle
\end{aligned}$$ ■

where ψ^j is the jth column of Ψ. We use definition (127) of the inner product $\langle \cdot, \cdot\rangle_a$, as well as the scale-invariance property to drop the softmax denominators. While $\beta = 1$ provides an isometry, we find in the next section that $\beta = \pi/\sqrt{6}$ is more convenient for learning distributions. This does not meaningfully change the geometric interpretation; the ILR coefficients are merely scaled by $1/\beta$.

F.3.2 Canonical Distribution on ILR Coefficients

Let $p_x(x) \equiv 1/|\Delta_{n-1}| = \Gamma(n)$ be the uniform distribution $\mathcal{U}(\Delta_{n-1})$. Applying (131), the corresponding distribution over u is

$$p_u(u \mid \beta) = \Gamma(n) \left(\beta^{n-1}\sqrt{n} \prod_{i=1}^{n} x_i\right). \tag{132}$$

To sample from p_u, we sample the flow $x \sim \mathcal{U}(\Delta_{n-1}) \to u = \text{ilr}(x; \beta)$ where x can be generated by normalizing n i.i.d. samples $z_i \sim \text{Exp}(1)$, i.e. $x_i = z_i / \sum_j z_j$. Then, $\Psi^T \log x = \Psi^T \log z - (\log \sum_j z_j)\Psi^T 1 = \Psi^T \log z$ since $\Psi^T 1 = 0$, and we have that $u = \frac{1}{\beta}\Psi^T \log z \sim p_u$. The mean and covariance of $u \sim p_u$ can then be computed directly:

$$\begin{aligned}
\mathop{\mathbb{E}}_{u\sim p_u}[u] &= \mathop{\mathbb{E}}_{x\sim\mathcal{U}(\Delta_{n-1})}\left[\frac{1}{\beta}\Psi^T \log x\right] && (133)\\
&= \frac{1}{\beta}\Psi^T \mathop{\mathbb{E}}_{z\sim\text{Exp}(1)}[\log z] && (134)\\
&= \frac{1}{\beta}\Psi^T(-\gamma 1) && (135)\\
&= 0 && (136)\\
\mathop{\mathbb{E}}_{u\sim p_u}\left[uu^T\right] &= \mathop{\mathbb{E}}_{x\sim\mathcal{U}(\Delta_{n-1})}\left[\frac{1}{\beta^2}\Psi^T \log x \log x^T \Psi\right] && (137)\\
&= \frac{1}{\beta^2}\Psi^T \mathop{\mathbb{E}}_{z\sim\text{Exp}(1)}\left[\log z \log z^T\right]\Psi && (138)\\
&= \frac{1}{\beta^2}\Psi^T\left(\frac{\pi^2}{6}\text{I} + \gamma^2 11^T\right)\Psi && (139)
\end{aligned}$$

$$= \frac{\pi^2}{6\beta^2}\mathrm{I} \tag{140}$$

where we have used the facts that $z_i \sim \mathrm{Exp}(1)$ are i.i.d., $\mathbb{E}[\log z_i] = -\gamma$, $\mathbb{E}[(\log z_i)^2] = \gamma^2 + \pi^2/6$, $\Psi^T 1 = 0$, and $\Psi^T\Psi = \mathrm{I}$; $\gamma \approx 0.577$ is the Euler–Mascheroni constant. Thence, we define the *canonical distribution* on the ILR coefficients to be the distribution (132) with $\beta = \beta_0 \coloneqq \pi/\sqrt{6}$, denoted $p_u(u \mid \beta_0)$, such that $\mathbb{E}[u] = 0$ and $\mathbb{E}[uu^T] = \mathrm{I}$.

F.3.3 Marginal Properties of the Canonical ILR Distribution

The ILR coefficients $u \sim p_u(u \mid \beta_0)$ have zero mean and identity covariance, but the $n-1$ components u_j are not independent by the Darmois–Skitovich theorem, as they arise from a linear transformation $u = (-\frac{1}{\beta_0}\Psi^T)y$ of i.i.d. standard Gumbel variables $y_i = -\log z_i$. However, u_j can be made nearly identically distributed by choosing Ψ such that the cumulants $\kappa_m(u_j)$ match for as many orders m as possible:

$$\kappa_m(u_j) = \sum_{i=1}^{n} \left(-\frac{\Psi_{ij}}{\beta_0}\right)^m \kappa_m(y_i) = \frac{\kappa_m^{(G)}}{(-\beta_0)^m} c_j^{(m)} \tag{141}$$

where $c_j^{(m)} = \sum_{i=1}^{n} \Psi_{ij}^m$, and $\kappa_m^{(G)}$ is the mth cumulant of the standard Gumbel distribution. For any Ψ, $\Psi^T 1 = 0 \Rightarrow c_j^{(1)} \equiv 0$ and $\Psi^T\Psi = \mathrm{I} \Rightarrow c_j^{(2)} \equiv 1$, so cumulants of order $m = 1, 2$ are always identical. For $m > 1$, $\kappa_m^{(G)} = (m-1)!\zeta(m) > 0$ where ζ is the Riemann zeta function, so for $\kappa_m(u_j)$ to be the same for all j one must have

$$c_{j_1}^{(m)} = c_{j_2}^{(m)} \quad \forall j_1, j_2 \in \{1, \ldots, n-1\}. \tag{142}$$

A basis Ψ does not in general satisfy this condition for $m \geq 3$. The type-II DCT basis, however, satisfies (142) for all $m < n$ when n is prime, with $c_j^{(m)} \equiv 0$ for odd m. The marginals $p(u_j)$ are thus very close to being symmetric and identically distributed for moderately large prime n. We prove this by direct calculation.

The DCT basis elements are $\Psi_{ij} = \sqrt{\frac{2}{n}}\cos(j\theta_i)$ where $\theta_i = \frac{\pi(2i-1)}{2n}$, and so $c_j^{(m)} = \left(\frac{2}{n}\right)^{\frac{m}{2}} \sum_{i=1}^{n} \cos^m(j\theta_i)$. Using the cosine power-reduction formula, the sum is expanded to

$$\sum_{i=1}^{n} \cos^m(j\theta_i) = na_0 + \frac{1}{2^{m-1}} \sum_{k=0}^{\lfloor \frac{m-1}{2} \rfloor} \binom{m}{k} a(\ell_{jk}) \tag{143}$$

where $a_0 = \frac{1}{2^m}\binom{m}{\frac{m}{2}}$ if m is even and $a_0 = 0$ if m is odd, $\ell_{jk} = j(m-2k)$, and the cosine series $a(\ell)$ equals

$$a(\ell) = \sum_{i=1}^{n} \cos(\ell\theta_i) = \Re\left(\sum_{i=1}^{n} e^{\sqrt{-1}\ell\theta_i}\right) \tag{144}$$

$$= \begin{cases} n(-1)^q & \text{if } \ell = 2nq, q \in \mathbb{Z} \\ \Re\left(\sqrt{-1}\frac{1-(-1)^\ell}{2\sin(\frac{\ell\pi}{2n})}\right) = 0 & \text{otherwise.} \end{cases} \tag{145}$$

Consequently, if $\ell_{jk} = j(m-2k)$ is never a multiple of $2n$, then $a(\ell_{jk}) = 0$ and $c_j^{(m)} = \frac{n}{(2n)^{\frac{m}{2}}}\binom{m}{\frac{m}{2}}$ for even m and zero for odd m. If we take n to be prime and $m < n$, then n divides neither factor $j < n$ nor $m - 2k \leq m < n$ of ℓ_{jk}. Thus, the DCT basis guarantees that for prime n and $m < n$, the cumulants $\kappa_m(u_j)$ are equal for all j. ■

It is worth noting that a generic basis Ψ has $\frac{1}{2}(n-1)(n-2)$ degrees of freedom, but for prime n, the DCT basis satisfies the cumulant matching condition for $3 \leq m < n$, a total of $(n-2)(n-3)$ constraints. For $n > 5$, this exceeds the degrees of freedom. The highly structured DCT basis therefore satisfies – for large n – almost twice as many algebraic constraints as one naively expects to be feasible.

F.3.4 Power Distribution on Δ_{n-1}

Consider the distribution arising from scaling samples $u \sim p_u(u \mid \beta_0)$ from the canonical distribution by $\lambda > 0$:

$$p_{u_\lambda}(u_\lambda) = \frac{p_u(u \mid \beta_0)}{\lambda^{n-1}} = \Gamma(n) \left(\frac{\beta_0}{\lambda}\right)^{n-1} \sqrt{n} \prod_{i=1}^{n} x_i \tag{146}$$

where $x = \mathrm{ilr}^{-1}(u; \beta_0)$. Applying the canonical inverse $x_\lambda = \mathrm{ilr}^{-1}(u_\lambda; \beta_0)$ then gives the *power distribution on* Δ_{n-1}:

$$p_{x_\lambda}(x_\lambda) = \frac{p_{u_\lambda}(u_\lambda)}{\beta_0^{n-1}\sqrt{n}\prod_{i=1}^{n} x_{\lambda,i}} = \frac{\Gamma(n)}{\lambda^{n-1}} \prod_{i=1}^{n} \left(\frac{x_i}{x_{\lambda,i}}\right) \tag{147}$$

where $x_\lambda = \mathrm{ilr}^{-1}(\lambda u; \beta_0) = \mathrm{ilr}^{-1}(\lambda\, \mathrm{ilr}(x; \beta_0); \beta_0) = \lambda \otimes x$ is the power transformation (123) of $x \sim \mathcal{U}(\Delta_{n-1})$ by λ. Relative to $\mathcal{U}(\Delta_{n-1})$, small $\lambda < 1$ pulls p_{x_λ} toward the centroid $\frac{1}{n}1$, and large $\lambda > 1$ pushes p_{x_λ} out toward the simplex boundary.

F.3.5 Normal Distribution on Δ_{n-1}

Consider the isotropic normal distribution $p_u(u \mid \sigma) = \mathcal{N}(u \mid 0, \sigma^2 \mathrm{I})$. Applying the canonical inverse transform $x = \mathrm{ilr}^{-1}(u; \beta_0)$ gives the *normal distribution on* Δ_{n-1} [36]:

$$p(x \mid \sigma) = \frac{e^{-\|u\|^2/2\sigma^2}}{(\sqrt{2\pi}\sigma)^{n-1}} \left(\beta_0^{n-1}\sqrt{n}\prod_{i=1}^{n} x_i\right)^{-1} \tag{148}$$

Similar to the power distribution, small σ pulls $p(x \mid \sigma)$ toward the centroid $\frac{1}{n}1$, and large σ pushes $p(x \mid \sigma)$ out toward the simplex boundary.

G Custom Normalizing Flows

The orthogonal flow below, trained by Riemannian gradient descent [91], was explored during development but is not used in the final architecture, whose linear layers are the matrix-exponential flows (35).

G.1 Simplex Flow

Consider the *simplex flow* induced by the inverse ILR transformation, $u \in \mathbb{R}^{n-1} \sim p_u \to x = \mathrm{ilr}^{-1}(u; \beta) \in \Delta_{n-1}$. The inverse ILR transformation is a diffeomorphism, and therefore a density update can be computed to determine the induced distribution $p_x(x)$. To do so, let $\tilde{x} = Px$ be $n-1$ independent components of x where the projection matrix $P \in \mathbb{R}^{(n-1)\times n}$ drops the jth component of x. Then,

$$\begin{aligned} J \coloneqq \frac{\partial \tilde{x}}{\partial u} &= \frac{\partial(P\,\mathrm{softmax}(\beta\Psi u))}{\partial u} \\ &= P(\mathrm{diag}(x) - xx^T)(\beta\Psi). \end{aligned}$$

Now, we have that: $x^T x = \tilde{x}^T \tilde{x} + x_j^2$ by definition, $1^T x = 1$ implies $x_j = 1 - 1^T \tilde{x}$, and $\Psi\Psi^T = \mathrm{I} - \frac{1}{n} 1 1^T$. Denoting $X = \mathrm{diag}(x)$ and $\tilde{X} = \mathrm{diag}(\tilde{x})$, it then follows that

$$\begin{aligned}
JJ^T &= \beta^2 P(X - xx^T)(\mathrm{I} - \frac{1}{n} 1 1^T)(X - xx^T)P^T \\
&= \beta^2 P(X - xx^T)^2 P^T \\
&= \beta^2 PX(\mathrm{I} - 1x^T)(\mathrm{I} - x1^T)XP^T \\
&= \beta^2 PX(\mathrm{I} - x1^T - 1x^T + (x^T x)11^T)XP^T \\
&= \beta^2 \tilde{X}(\mathrm{I} - \tilde{x}1^T - 1\tilde{x}^T + (x^T x)11^T)\tilde{X} \\
&= \beta^2 \tilde{X}((\mathrm{I} - 1\tilde{x}^T)(\mathrm{I} - \tilde{x}1^T) + x_j^2 11^T)\tilde{X}.
\end{aligned}$$

To compute $|J| = \sqrt{\det(JJ^T)}$, it remains only to compute the determinant of the central term in brackets. Denote $A = \mathrm{I} - \tilde{x}1^T$. By the matrix determinant lemma, $\det(A) = 1 - 1^T\tilde{x} = x_j$, and similarly by the Sherman–Morrison formula, $A^{-1} = \mathrm{I} + \frac{1}{x_j}\tilde{x}1^T$. It then follows, again by the matrix determinant lemma, that:

$$\begin{aligned}
\det\big(A^T A + x_j^2 11^T\big) &= \det\big(A^T A\big)(1 + x_j^2 1^T (A^T A)^{-1} 1) \\
&= x_j^2 (1 + (x_j + 1^T\tilde{x})^2 1^T 1) \\
&= n x_j^2.
\end{aligned}$$

Finally, the volume correction term is given by

$$|J| = \sqrt{\det(JJ^T)} = \beta^{n-1}\sqrt{n}\prod_{i=1}^{n} x_i, \tag{149}$$

recovering the result from [36] when $\beta = 1$ ■. Note that the result does not depend on the specific basis Ψ.

G.2 Orthogonal Flow

An *orthogonal flow* layer applies a linear transformation with an orthogonal matrix Q [92]

$$x = Qu, \quad Q \in \mathcal{O}_n$$

where $\mathcal{O}_n$ is the manifold of orthogonal matrices:

$$\mathcal{O}_n = \left\{ Q \in \mathbb{R}^{n \times n} \mid Q^T Q = \mathrm{I} \right\}.$$

The tangent space at $Q \in \mathcal{O}_n$ is given by [93]:

$$\begin{aligned}
T_Q \mathcal{O}_n &= \left\{ X \in \mathbb{R}^{n \times n} \mid Q^T X + X^T Q = 0 \right\} \\
&= \left\{ X = QY \in \mathbb{R}^{n \times n} \mid Y^T = -Y \right\}
\end{aligned}$$

where the latter form expresses tangent vectors X in terms of their skew-symmetric *special orthogonal Lie algebra* $\mathfrak{so}(n)$ representer $Y = Q^T X$.

Orthogonal Q are convenient for flows since $|Q| = |\det(Q)| = 1$ implies the trivial density update

$$p_x(x) = p_u(u). \tag{150}$$

The matrix Q must stay orthogonal throughout training in order for the density update (150) to remain valid. This is often achieved by fixing Q to be a randomly initialized value drawn from the Haar (uniform) measure on $\mathcal{O}_n$, or alternatively by fixing $Q = P$ where P is a permutation matrix. Here, we experiment with allowing $Q \in \mathcal{O}_n$ to vary during training using *Riemannian gradient descent (RGD)*; see appendix J for a description of RGD.

G.2.1 Optimization on $\mathcal{O}_n$

The orthogonal manifold has a rich geometric structure which has been studied extensively in the optimization literature, particularly elegantly so by Edelman et al. [93]. Here, we list concrete definitions of Riemannian operations on $\mathcal{O}_n$ for use in RGD. Let $\mathcal{M} = \mathcal{O}_n \subset \mathbb{R}^{n\times n}$, $Q \in \mathcal{O}_n$, $A \in \mathbb{R}^{n\times n}$, $f : \mathbb{R}^{n\times n} \to \mathbb{R}$, $G = \nabla f(Q) \in \mathbb{R}^{n\times n}$, and $X = QY \in T_Q\mathcal{O}_n$ where $Y^T = -Y$. Then [93]

$$\operatorname{grad} f(Q) = \operatorname{proj}_{\mathcal{O}_n}(Q, G) = Q\frac{Q^TG - G^TQ}{2}$$

is the Riemannian gradient of f at Q,

$$\exp_Q(X) = Qe^Y \in \mathcal{O}_n$$

is the exponential map where e^Y is the *matrix exponential* of $Y = Q^TX$,

$$\log_Q(R) = Q\log(Q^TR) \in T_Q\mathcal{O}_n$$

is the logarithmic map from Q to $R \in \mathcal{O}_n$ where $\log(\cdot)$ is the *matrix logarithm*,

$$\operatorname{retr}_Q(X) = \operatorname{proj}_{\mathcal{O}_n}(Q + X) = UV^T \tag{151}$$

is the retraction of $Q + X = USV^T$ onto $\mathcal{O}_n$ via the singular value decomposition (SVD), and

$$\operatorname{par}_Q(Z; X) = Qe^{Y/2}We^{Y/2} \tag{152}$$

is the parallel transport of $Z = QW \in T_Q\mathcal{O}_n$ in the direction X. Note that $e^{Y/2} \approx \operatorname{retr}_{\mathrm{I}}(Y/2) = \operatorname{proj}_{\mathcal{O}_n}(\mathrm{I} + Y/2)$.

G.2.2 Accelerating the Retraction

The nearest orthogonal matrix $O^* = \operatorname{proj}_{\mathcal{O}_n}(A)$ to a matrix A (in 2- or Frobenius norm) has the closed-form solution [94]

$$O^* = \operatorname*{arg\,min}_{O\in\mathcal{O}_n} \|A - O\| = A(A^TA)^{-1/2} \tag{153}$$

$$= A(\mathrm{I} - \tfrac{1}{2}(A^TA - \mathrm{I}) + \cdots). \tag{154}$$

The series expansion of the matrix inverse square root is valid if the residual $E = A^TA - \mathrm{I}$ is small. In particular, if $\|E\|_2 < 1$ then the iteration

$$O_{i+1} = O_i(\mathrm{I} - \tfrac{1}{2}(O_i^TO_i - \mathrm{I})), \quad O_0 = A \tag{155}$$

converges quadratically to $O^* = A(A^TA)^{-1/2}$ [94], typically requiring only 1–4 iterations in 32-bit arithmetic. The SVD-based projection (151) can therefore be replaced by the iteration (155) when A is close to $\mathcal{O}_n$. This is common when the projected gradient step size η in (182) is small, as the residual norm for $A = Q - \eta X = Q(\mathrm{I} - \eta Y)$ is $\mathcal{O}(\eta^2)$:

$$\|E\| = \|(\mathrm{I} - \eta Y)^T(\mathrm{I} - \eta Y) - \mathrm{I}\| = \eta^2\|Y^TY\|.$$

Therefore, the retraction onto $\mathcal{O}_n$ uses the iteration (155) when $\|E\|_F^2 = \operatorname{tr}\big(E^TE\big) < 1$ (note: $\|E\|_2 \le \|E\|_F$), and resorts to the SVD-based projection (151) otherwise [95].

Additional mathematical material

H The Symmetric Dirichlet Distribution on Δ_{n-1}

Here we recall some basic properties of the symmetric Dirichlet distribution $\mathrm{Dir}(\alpha)$, defined over Δ_{n-1} by the probability density function

$$p(x \mid \alpha) = \frac{1}{\mathrm{B}_n(\alpha)} \prod_{i=1}^{n} x_i^{\alpha-1} \tag{156}$$

where $\alpha > 0$, $x \in \Delta_{n-1}$, and $\mathrm{B}_n(\alpha)$ is the multivariate *beta function* B with all arguments equal to α:

$$\begin{aligned} \mathrm{B}(\alpha_1, \ldots, \alpha_n) &= \frac{\prod_{i=1}^{n} \Gamma(\alpha_i)}{\Gamma\left(\sum_{i=1}^{n} \alpha_i\right)} \\ \Rightarrow \mathrm{B}_n(\alpha) &\coloneqq \mathrm{B}(\alpha, \ldots, \alpha) = \frac{\Gamma(\alpha)^n}{\Gamma(n\alpha)}. \end{aligned}$$

In using $\mathrm{Dir}(\alpha)$ as a prior on $x \in \Delta_{n-1}$, the concentration parameter $\alpha > 0$ has a regularizing effect: large α pushes inferred x toward the simplex centre – the uniform distribution $\frac{1}{n}1$ – and conversely, small α pushes x toward the simplex corners, approaching one-hot distributions as $\alpha \to 0^+$.

I The Symmetric Gaussian Restricted to Δ_{n-1}

Consider the symmetric Gaussian distribution defined over Δ_{n-1} by the probability density function

$$p(x \mid \sigma) = \frac{1}{\mathcal{Z}_n} e^{-\|x\|^2/2\sigma^2} \tag{157}$$

where $x \in \Delta_{n-1}$, $\sigma \in \mathbb{R}^+$, and the normalization constant is

$$\mathcal{Z}_n = \int_{\Delta_{n-1}} e^{-\|x\|^2/2\sigma^2}\, dx. \tag{158}$$

Similar to the usual Gaussian distribution, using this prior on $x \in \Delta_{n-1}$ acts as a regularizer: small σ pushes inferred x toward the simplex centroid $\frac{1}{n}1$, whereas $p(x \mid \sigma) \to \mathcal{U}(\Delta_{n-1})$ as $\sigma \to \infty$.

While $\mathcal{Z}_n$ does not admit a closed form, in appendix I.1 we derive analytic asymptotic expressions for $\log \mathcal{Z}_n$ in the small and large σ limits, and numerically compute minimax rational approximants for intermediate σ. In particular, we have the approximations:

$$\sigma \ll \sigma_0 : \begin{cases} \log \mathcal{Z}_n \approx -\frac{1}{2n\sigma^2} + \frac{n-1}{2} \log 2\pi\sigma^2 - \log \sqrt{n} \\ \frac{d \log \mathcal{Z}_n}{d \log \sigma} \approx \frac{1}{n\sigma^2} + n - 1. \end{cases} \tag{159}$$

$$\sigma \gg \sigma_0 : \begin{cases} \log \mathcal{Z}_n \approx -\log \Gamma(n) - \frac{a_1}{\sigma^2} + \frac{a_2}{\sigma^4} \\ \frac{d \log \mathcal{Z}_n}{d \log \sigma} \approx \frac{2a_1}{\sigma^2} - \frac{4a_2}{\sigma^4} \end{cases} \tag{160}$$

where $\sigma_0 = \sqrt{\frac{1}{n(n-1)}}$ is the distance from the simplex centroid to the nearest point on the simplex boundary, $a_1 = \frac{1}{n+1}$, and $a_2 = \frac{n-1}{2(n+1)^2(n+2)(n+3)}$.

I.1 Normalization Constant of the Symmetric Gaussian on Δ_{n-1}

Here, we derive approximations to the normalization constant $\mathcal{Z}_n$ of the symmetric Gaussian distribution $p(x \mid \sigma)$ introduced in appendix I. We first write $\mathcal{Z}_n$ as an expectation over the probability simplex:

$$\begin{aligned} \mathcal{Z}_n &= \int_{\Delta_{n-1}} e^{-\|x\|^2/2\sigma^2}\, dx \\ &= |\Delta_{n-1}| \cdot \mathbb{E}_{x\sim\mathrm{Dir}(1)}\left[e^{-\|x\|^2/2\sigma^2}\right] \end{aligned} \tag{161}$$

where $x \in \Delta_{n-1}$, $\sigma \in \mathbb{R}^+$, and $|\Delta_{n-1}| = 1/\Gamma(n)$ is the simplex volume. The latter interpretation of $\mathcal{Z}_n$ as an expectation is useful, as it allows us to use well-known properties of the Dirichlet distribution in order to integrate over Δ_{n-1}. The properties we will use are the mean and variance of $x \sim \mathrm{Dir}(\alpha)$

$$\mathbb{E}[x_i] = \frac{1}{n}, \quad \mathrm{Var}[x_i] = \frac{1 - 1/n}{n(1+n\alpha)} \tag{162}$$

and the following expressions involving higher moments of $x \sim \mathrm{Dir}(1)$:

$$\begin{aligned} \mathbb{E}\big[x_i^2\big] &= \frac{2}{n(n+1)} \\ \sum_{i=1}^{n} \mathbb{E}\big[x_i^2\big] &= \frac{2}{n+1} \\ \mathbb{E}\big[x_i^4\big] &= \frac{24}{n(n+1)(n+2)(n+3)} \\ \mathbb{E}\big[x_i^2 x_j^2\big] &= \frac{4}{n(n+1)(n+2)(n+3)} \quad (i \neq j). \\ \sum_{i,j=1}^{n} \mathbb{E}\big[x_i^2 x_j^2\big] &= \frac{4(n+5)}{(n+1)(n+2)(n+3)} \end{aligned} \tag{163}$$

We will derive expressions for $\log \mathcal{Z}_n$ in the limit of large and small σ, and find minimax rational approximants for intermediate σ. A convenient dividing value between large and small σ is the distance between the simplex centroid and the nearest point on the simplex boundary:

$$\sigma_0 = \left\| \frac{1}{n} 1_n - \frac{1}{n-1} \begin{pmatrix} 1_{n-1} \\ 0 \end{pmatrix} \right\| = \sqrt{\frac{1}{n(n-1)}}. \tag{164}$$

I.1.1 Wide Width Limit ($\sigma \gg \sigma_0$)

For large σ, since $\|x\| \leq 1$ is bounded, we can Taylor expand the integrand:

$$\begin{aligned} \frac{\mathcal{Z}_n}{|\Delta_{n-1}|} &= \mathbb{E}_{x\sim\mathrm{Dir}(1)}\left[e^{-\|x\|^2/2\sigma^2}\right] \\ &= \mathbb{E}\left[1 - \frac{\|x\|^2}{2\sigma^2} + \frac{1}{2}\left(\frac{\|x\|^2}{2\sigma^2}\right)^2 + \mathcal{O}\left(\frac{\|x\|^6}{\sigma^6}\right)\right] \\ &= 1 - \frac{1}{2\sigma^2}\sum_{i=1}^{n} \mathbb{E}\big[x_i^2\big] + \frac{1}{8\sigma^4}\sum_{i,j=1}^{n} \mathbb{E}\big[x_i^2 x_j^2\big] + \mathcal{O}\left(\frac{1}{\sigma^6}\right) \end{aligned}$$

$$= 1 - \frac{a_1}{\sigma^2} + \frac{a_2'}{\sigma^4} + \mathcal{O}\left(\frac{1}{\sigma^6}\right)$$

where the constants

$$a_1 = \frac{1}{n+1} \quad \text{and} \quad a_2' = \frac{n+5}{2(n+1)(n+2)(n+3)}$$

are computed using the moments of Dir(1) from (163). Note that as expected, $\mathcal{Z}_n \to |\Delta_{n-1}|$ and $p(x \mid \sigma) \to \mathcal{U}(\Delta_{n-1})$ as $\sigma \to \infty$. Now, taking the logarithm we have that

$$\begin{aligned}\log \frac{\mathcal{Z}_n}{|\Delta_{n-1}|} &= \log\left(1 - \frac{a_1}{\sigma^2} + \frac{a_2'}{\sigma^4} + \mathcal{O}\left(\frac{1}{\sigma^6}\right)\right) \\ &= -\frac{a_1}{\sigma^2} + \frac{a_2}{\sigma^4} + \mathcal{O}\left(\frac{1}{\sigma^6}\right)\end{aligned}$$

where Taylor expansion of the logarithm gives

$$a_2 = a_2' - \frac{1}{2}a_1^2 = \frac{n-1}{2(n+1)^2(n+2)(n+3)}.$$

Using the same approach, the derivative is given by

$$\begin{aligned}&\frac{1}{|\Delta_{n-1}|}\frac{d\mathcal{Z}_n}{d\log\sigma} = \mathbb{E}_{x\sim\mathrm{Dir}(1)}\left[\frac{\partial}{\partial\log\sigma}e^{-\|x\|^2/2\sigma^2}\right] \\ &= \mathbb{E}\left[\frac{\|x\|^2}{\sigma^2}e^{-\|x\|^2/2\sigma^2}\right] \\ &= \mathbb{E}\left[\frac{\|x\|^2}{\sigma^2} - \frac{\|x\|^4}{2\sigma^4} + \mathcal{O}\left(\frac{\|x\|^6}{\sigma^6}\right)\right] \\ &= \frac{1}{\sigma^2}\sum_{i=1}^{n}\mathbb{E}\big[x_i^2\big] - \frac{1}{2\sigma^4}\sum_{i,j=1}^{n}\mathbb{E}\big[x_i^2x_j^2\big] + \mathcal{O}\left(\frac{1}{\sigma^6}\right) \\ &= \frac{2a_1}{\sigma^2} - \frac{4a_2'}{\sigma^4} + \mathcal{O}\left(\frac{1}{\sigma^6}\right)\end{aligned}$$

where $\frac{\partial}{\partial\log\sigma} = \sigma\cdot\frac{\partial}{\partial\sigma}$, and a_1, a_2' are as previously defined. The derivative of $\log\mathcal{Z}_n$ is then

$$\begin{aligned}\frac{d\log\mathcal{Z}_n}{d\log\sigma} = \frac{1}{\mathcal{Z}_n}\frac{d\mathcal{Z}_n}{d\log\sigma} &= \frac{\frac{2a_1}{\sigma^2} - \frac{4a_2'}{\sigma^4} + \mathcal{O}\left(\frac{1}{\sigma^6}\right)}{1 - \frac{a_1}{\sigma^2} + \frac{a_2'}{\sigma^4} + \mathcal{O}\left(\frac{1}{\sigma^6}\right)} \\ &= \frac{2a_1}{\sigma^2} - \frac{4a_2}{\sigma^4} + \mathcal{O}\left(\frac{1}{\sigma^6}\right)\end{aligned}$$

where the final expression follows from Taylor expansion of the denominator, and again $a_2 = a_2' - \frac{1}{2}a_1^2$. Dropping higher-order terms, we arrive at the approximations for $\sigma \gg \sigma_0$:

$$\log\mathcal{Z}_n \approx -\log\Gamma(n) - \frac{a_1}{\sigma^2} + \frac{a_2}{\sigma^4} \tag{165}$$

$$\frac{d\log\mathcal{Z}_n}{d\log\sigma} \approx \frac{2a_1}{\sigma^2} - \frac{4a_2}{\sigma^4}. \tag{166}$$

I.1.2 Narrow Width Limit ($\sigma \ll \sigma_0$)

For small σ, the Gaussian integrand becomes sharply peaked about the simplex centroid $\mu = \frac{1}{n}1$, and therefore little error is incurred by extending the integral domain from Δ_{n-1} to $\mathcal{P}_{n-1}$. To integrate over $\mathcal{P}_{n-1}$, we split $\|x\|^2$ for $x \in \mathcal{P}_{n-1}$ into orthogonal components:

$$\begin{aligned}
\|x\|^2 &= \|\mu\|^2 + \|x-\mu\|^2 \\
&= \frac{1}{n} + \|\tilde{x}-\tilde{\mu}\|^2 + ((1-1^T\tilde{x}) - \frac{1}{n})^2 \\
&= \frac{1}{n} + \|\tilde{x}-\tilde{\mu}\|^2 + (1^T(\tilde{x}-\tilde{\mu}))^2 \\
&= \frac{1}{n} + (\tilde{x}-\tilde{\mu})^T(\mathrm{I}+11^T)(\tilde{x}-\tilde{\mu})
\end{aligned}$$

where $\tilde{x} = (x_1, \ldots, x_{n-1}) \in \mathbb{R}^{n-1}$, $x_n = 1 - 1^T\tilde{x}$, and $\tilde{\mu} = \frac{1}{n}1 \in \mathbb{R}^{n-1}$. Next, we let $\Sigma^{-1} = \frac{1}{\sigma^2}\left(\mathrm{I}+11^T\right) \in \mathbb{R}^{(n-1)\times(n-1)}$ and change variables to $\tilde{y} = \Sigma^{-1/2}(\tilde{x}-\tilde{\mu})$, $d\tilde{y} = |\Sigma|^{-1/2}d\tilde{x}$. From the matrix determinant lemma, $|\Sigma|^{1/2} = \sigma^{n-1}/\sqrt{n}$. Then, letting $f : \mathbb{R}^+ \to \mathbb{R}$ be of the form $f(t) = g(t)e^{-t/2}$ with g sub-exponential, we have:

$$\begin{aligned}
&\int_{\Delta_{n-1}} f\left(\frac{\|x\|^2}{\sigma^2}\right) dx \\
&= \int_{\mathcal{S}_{n-1}} f\left(\tfrac{1}{n\sigma^2} + (\tilde{x}-\tilde{\mu})^T\Sigma^{-1}(\tilde{x}-\tilde{\mu})\right) d\tilde{x} \\
&\approx |\Sigma|^{1/2} \int_{\mathbb{R}^{n-1}} f\left(\tfrac{1}{n\sigma^2} + \|\tilde{y}\|^2\right) d\tilde{y} \\
&= \sqrt{\frac{|2\pi\Sigma|}{e^{1/n\sigma^2}}} \int_{\mathbb{R}^{n-1}} g\left(\tfrac{1}{n\sigma^2} + \|\tilde{y}\|^2\right) \frac{e^{-\|\tilde{y}\|^2/2}}{\sqrt{2\pi}^{n-1}} d\tilde{y} \\
&= \sqrt{\frac{|2\pi\Sigma|}{e^{1/n\sigma^2}}} \mathbb{E}_{\tilde{y}\sim\mathcal{N}(0,\mathrm{I})}\left[g\left(\tfrac{1}{n\sigma^2} + \|\tilde{y}\|^2\right)\right]
\end{aligned}$$

where the integral over $\mathcal{S}_{n-1}$ is approximated as an integral over $\mathbb{R}^{n-1}$ (i.e. Δ_{n-1} is extended to $\mathcal{P}_{n-1}$). For $\mathcal{Z}_n$, taking $g(t) \equiv 1$ above gives

$$\mathcal{Z}_n \approx \hat{\mathcal{Z}}_n \coloneqq \sqrt{\frac{|2\pi\Sigma|}{e^{1/n\sigma^2}}} = \frac{\sigma^{n-1}}{\sqrt{n}} \frac{\sqrt{2\pi}^{n-1}}{e^{1/2n\sigma^2}}.$$

The same method applies to $\frac{d\log\mathcal{Z}_n}{d\log\sigma}$ with $g(t) = t$:

$$\begin{aligned}
\frac{d\log\mathcal{Z}_n}{d\log\sigma} &= \frac{1}{\mathcal{Z}_n}\int_{\Delta_{n-1}} \frac{\partial}{\partial\log\sigma} e^{-\|x\|^2/2\sigma^2}\, dx \\
&= \frac{1}{\mathcal{Z}_n}\int_{\Delta_{n-1}} \frac{\|x\|^2}{\sigma^2} e^{-\|x\|^2/2\sigma^2}\, dx \\
&\approx \frac{\hat{\mathcal{Z}}_n}{\mathcal{Z}_n}\mathbb{E}_{\tilde{y}\sim\mathcal{N}(0,\mathrm{I})}\left[\tfrac{1}{n\sigma^2} + \|\tilde{y}\|^2\right] \\
&\approx \frac{1}{n\sigma^2} + n - 1
\end{aligned}$$

where $\mathcal{Z}_n \approx \hat{\mathcal{Z}}_n$, and $\mathbb{E}\left[\|\tilde{y}\|^2\right]$ is the trace of the covariance matrix $\mathrm{tr}(\mathrm{I}) = n-1$. The final approximants for $\log\mathcal{Z}_n$ when $\sigma \ll \sigma_0$ are thus

$$\log\mathcal{Z}_n \approx -\frac{1}{2n\sigma^2} + \frac{n-1}{2}\log 2\pi\sigma^2 - \log\sqrt{n} \tag{167}$$

$$\frac{d\log \mathcal{Z}_n}{d\log \sigma} \approx \frac{1}{n\sigma^2} + n - 1. \tag{168}$$

To justify these approximations, we must bound the excess integral over $\mathcal{P}_{n-1} \setminus \Delta_{n-1}$ (or $\mathbb{R}^{n-1} \setminus \mathcal{S}_{n-1}$). By construction, $\|\tilde{y}\|^2 = \|x-\mu\|^2/\sigma^2$ for $x \in \mathcal{P}_{n-1}$, and therefore $\mathbb{R}^{n-1} \setminus \mathcal{S}_{n-1}$ lies outside the ball $\|\tilde{y}\|^2 \le \sigma_0^2/\sigma^2$. As the integrand is positive, the remainder is bounded by:

$$\begin{aligned}
|\mathcal{Z}_n - \hat{\mathcal{Z}}_n| &= \int_{\mathcal{P}_{n-1}\setminus\Delta_{n-1}} e^{-\|x\|^2/2\sigma^2}\,dx \\
&\le \hat{\mathcal{Z}}_n \int_{\|\tilde{y}\|^2 > \sigma_0^2/\sigma^2} \frac{e^{-\|\tilde{y}\|^2/2}}{\sqrt{2\pi}^{n-1}}\,d\tilde{y} \\
&= \hat{\mathcal{Z}}_n \int_{\sigma_0^2/\sigma^2}^{\infty} \frac{t^{\frac{n-1}{2}-1}e^{-t/2}}{2^{\frac{n-1}{2}}\Gamma(\frac{n-1}{2})}\,dt \\
&= \hat{\mathcal{Z}}_n P_{\chi^2_{n-1}}(t \ge \tfrac{\sigma_0^2}{\sigma^2}) \\
&\le \hat{\mathcal{Z}}_n e^{-c_1\sigma_0^2/\sigma^2}
\end{aligned}$$

where $t = \|\tilde{y}\|^2$, $P_{\chi^2_{n-1}}$ is the upper tail of the χ^2 distribution with $n-1$ degrees of freedom, and c_1 is a positive constant [96, 97]. The relative error of $\hat{\mathcal{Z}}_n$ (i.e. the absolute error of (167)) is thus $\mathcal{O}(e^{-c_1\sigma_0^2/\sigma^2})$ as $\sigma/\sigma_0 \to 0^+$. ■

Similarly for the derivative, we have

$$\begin{aligned}
&\left|\frac{d\mathcal{Z}_n}{d\log\sigma} - \frac{d\hat{\mathcal{Z}}_n}{d\log\sigma}\right| = \int_{\mathcal{P}_{n-1}\setminus\Delta_{n-1}} \frac{\|x\|^2}{\sigma^2} e^{-\|x\|^2/2\sigma^2}\,dx \\
&\le \hat{\mathcal{Z}}_n \int_{\|\tilde{y}\|^2 > \sigma_0^2/\sigma^2} \left(\tfrac{1}{n\sigma^2} + \|\tilde{y}\|^2\right) \frac{e^{-\|\tilde{y}\|^2/2}}{\sqrt{2\pi}^{n-1}}\,d\tilde{y} \\
&= \hat{\mathcal{Z}}_n \left(\frac{P_{\chi^2_{n-1}}(t \ge \frac{\sigma_0^2}{\sigma^2})}{n\sigma^2} + \int_{\sigma_0^2/\sigma^2}^{\infty} \frac{t^{\frac{n-1}{2}} e^{-t/2}}{2^{\frac{n-1}{2}}\Gamma(\frac{n-1}{2})}\,dt \right) \\
&= \hat{\mathcal{Z}}_n \left(\frac{P_{\chi^2_{n-1}}(t \ge \frac{\sigma_0^2}{\sigma^2})}{(\sigma^2/\sigma_0^2)/(n-1)} + \frac{P_{\chi^2_{n+1}}(t \ge \frac{\sigma_0^2}{\sigma^2})}{1/(n-1)} \right) \\
&\le \hat{\mathcal{Z}}_n (n-1) \left(\tfrac{\sigma_0^2}{\sigma^2} e^{-c_1\sigma_0^2/\sigma^2} + e^{-c_2\sigma_0^2/\sigma^2} \right) \\
&\le \hat{\mathcal{Z}}_n (n-1) e^{-c_3\sigma_0^2/\sigma^2}
\end{aligned}$$

where c_2, c_3 are positive constants. The absolute error for (168) is bounded by

$$\left|\frac{d\log\mathcal{Z}_n}{d\log\sigma} - \frac{d\log\hat{\mathcal{Z}}_n}{d\log\sigma}\right| \le \frac{1}{\mathcal{Z}_n}\left|\frac{d\mathcal{Z}_n}{d\log\sigma} - \frac{d\hat{\mathcal{Z}}_n}{d\log\sigma}\right| + \left|\frac{d\log\hat{\mathcal{Z}}_n}{d\log\sigma}\right| \frac{|\hat{\mathcal{Z}}_n - \mathcal{Z}_n|}{\mathcal{Z}_n}.$$

Since $\hat{\mathcal{Z}}_n/\mathcal{Z}_n \to 1$ and $\frac{d\log\hat{\mathcal{Z}}_n}{d\log\sigma} = \frac{1}{n\sigma^2} + n - 1$, this decreases as $\mathcal{O}(e^{-c\sigma_0^2/\sigma^2})$ for some $c > 0$ as $\sigma/\sigma_0 \to 0^+$. ■

I.1.3 Moderate Width ($\sigma \approx \sigma_0$)

In the regime where σ is neither small nor large, we approximate $\mathcal{Z}_n$ numerically. Since directly evaluating the high-dimensional integral (158) is infeasible, we first rewrite equation (158) into a tractable form. We use an identity for integrating products of univariate

functions on the simplex: let $f : \Delta_{n-1} \to \mathbb{R}$ be of the form $f(x) = \prod_{i=1}^{n} f_i(x_i)$, where $f_i : \mathbb{R} \to \mathbb{R}$ are univariate. Then, it can be shown that [98]:

$$\int_{\Delta_{n-1}} f(x)\,dx = \int_{\Delta_{n-1}} \prod_{i=1}^{n} f_i(x_i)\,dx \tag{169}$$

$$= (f_1 \otimes f_2 \otimes \cdots \otimes f_n)\,(1) \tag{170}$$

$$= \mathcal{L}^{-1}\left\{\prod_{i=1}^{n} \mathcal{L}\{f_i\}(s)\right\}(1) \tag{171}$$

where $\otimes$ denotes the convolution $(f \otimes g)(t) = \int_0^t f(\tau)g(t-\tau)\,d\tau$, $\mathcal{L}\{\cdot\}(s)$ denotes the Laplace transform, and $\mathcal{L}^{-1}\{\cdot\}(t)$ its inverse. We apply this identity to (158) by letting $f_i(t) = g(t) = e^{-t^2/2\sigma^2}$. The Laplace transform $G(s)$ of $g(t)$ is

$$G(s) \coloneqq \mathcal{L}\left\{e^{-t^2/2\sigma^2}\right\}(s) = \sqrt{\frac{\pi\sigma^2}{2}}\,\operatorname{erfcx}\left(\frac{\sigma s}{\sqrt{2}}\right) \tag{172}$$

where $\operatorname{erfcx}(t) = e^{t^2}\operatorname{erfc}(t)$ is the *scaled complementary error function*. It follows that

$$\mathcal{Z}_n = \mathcal{L}^{-1}\{G(s)^n\}(1). \tag{173}$$

While the inverse Laplace transform in (173) admits no closed form, there exist robust numerical algorithms to compute $\mathcal{L}^{-1}\{\cdot\}(t)|_{t=1}$. We use the method of de Hoog et al. [99] provided by the `invertlaplace` function from the open-source Python package `mpmath` [100], a library for arbitrary-precision floating-point arithmetic.

Similarly, $\frac{d\mathcal{Z}_n}{d\log\sigma}$ can be computed by first differentiating (173) under the integral sign. Noting that

$$\frac{\partial}{\partial\log\sigma}G(s) = (1+\sigma^2 s^2)\sqrt{\frac{\pi\sigma^2}{2}}\,\operatorname{erfcx}\left(\frac{\sigma s}{\sqrt{2}}\right) - \sigma^2 s \tag{174}$$

$$= (1+\sigma^2 s^2)G(s) - \sigma^2 s \tag{175}$$

it follows that

$$\frac{d\mathcal{Z}_n}{d\log\sigma} = \mathcal{L}^{-1}\left\{\frac{\partial}{\partial\log\sigma}G(s)^n\right\}(1) \tag{176}$$

$$= n\mathcal{Z}_n + n\sigma^2\mathcal{L}^{-1}\left\{s^2 G(s)^n - sG(s)^{n-1}\right\}(1) \tag{177}$$

and finally

$$\frac{d\log\mathcal{Z}_n}{d\log\sigma} = n + n\sigma^2\frac{\mathcal{L}^{-1}\left\{s^2 G(s)^n - sG(s)^{n-1}\right\}(1)}{\mathcal{L}^{-1}\{G(s)^n\}(1)}. \tag{178}$$

While (173) and (178) may in principle be evaluated for any σ, they remain too expensive to be computed during training, for example as part of a loss function. Instead, we fix a relative tolerance $\varepsilon > 0$ and (separately for both $\log\mathcal{Z}_n$ and $\frac{d\log\mathcal{Z}_n}{d\log\sigma}$) split the input domain into three regimes:

1. $\log\sigma < a$: find a such that (167) (or (168)) is ε-accurate;
2. $\log\sigma > b$: find b such that (165) (or (166)) is ε-accurate;
3. $a \le \log\sigma \le b$: find an ε-accurate minimax rational approximation to (173) (or (178)).

For the small and large σ regimes, the constants a and b are found using numerical root-finding methods: the largest value of a and smallest value of b are found such that the appropriate limiting forms for $\log \mathcal{Z}_n$ (or $\frac{d\log \mathcal{Z}_n}{d\log \sigma}$) agree with (173) (or (178)) with relative error ε. For the moderate σ regime, we first note from (158) that $\mathcal{Z}_n$ is a strictly increasing positive function of σ which asymptotically approaches $1/\Gamma(n)$ from below, and therefore $\log \mathcal{Z}_n < -\log \Gamma(n) \leq 0$ and $\frac{d\log \mathcal{Z}_n}{d\log \sigma} = \frac{\sigma}{\mathcal{Z}_n}\frac{d\mathcal{Z}_n}{d\sigma} > 0$. Then, we find minimax rational approximations

$$\left|\log\left(-\log \mathcal{Z}_n\right) - \frac{P}{Q}\right| \leq \varepsilon, \quad \left|\log\left(\frac{d\log \mathcal{Z}_n}{d\log \sigma}\right) - \frac{R}{S}\right| \leq \varepsilon$$

where $P, Q, R, S : \mathbb{R} \to \mathbb{R}$ are polynomial functions of $\log \sigma$. The additional logarithm serves to convert the minimax bound on the *absolute* error to a near-minimax bound on the *relative* error in the sense that, given $|\log f - \frac{P}{Q}| \leq \varepsilon$, one has:

$$\left|\frac{f - e^{\frac{P}{Q}}}{f}\right| = \left|1 - e^{\frac{P}{Q} - \log f}\right| \leq \left|\log f - \frac{P}{Q}\right| + \mathcal{O}(\varepsilon^2).$$

Minimax rational approximations are computed using the `minimax` function from the Matlab (R2022a) open-source package `chebfun` [101], a library for numerical computing with functions. We use $\varepsilon = 10^{-7}$, $\deg(P) = \deg(Q) = 6$, and $\deg(R) = \deg(S) = 8$.

J Optimization on Embedded Riemannian Manifolds

Consider $f : \mathbb{R}^n \to \mathbb{R}$ to be minimized under the constraint that the iterates $x_i \in \mathbb{R}^n$ lie on an embedded manifold $\mathcal{M} \subset \mathbb{R}^n$. Then, f can be locally minimized using *Riemannian gradient descent (RGD)*:

$$x_{i+1} = \exp_{x_i}(-\eta \operatorname{grad} f(x_i)) \tag{179}$$

where η is a step size, $\operatorname{grad} f : \mathcal{M} \to T_x\mathcal{M}$ is the *Riemannian gradient*, and $\exp_x : T_x\mathcal{M} \to \mathcal{M}$ is the *exponential map*. For $\mathcal{M} \subset \mathbb{R}^n$ with the induced Euclidean metric, one has that

$$\operatorname{grad} f(x) = \operatorname{proj}_{\mathcal{M}}(x, \nabla f(x)) \tag{180}$$

where $\nabla f(x) \in \mathbb{R}^n$, and $\operatorname{proj}_{\mathcal{M}}(x, \nabla f(x))$ is the projection of $\nabla f(x)$ onto $T_x\mathcal{M}$. The mapping $\exp_x(v)$ moves x along a geodesic with initial velocity v for unit time. Relatedly, the *parallel transport* $\operatorname{par}_{x\to y}(w)$ slides the tangent vector w along a geodesic from x to y with zero acceleration. If $y = \exp_x(v)$, we equivalently write $\operatorname{par}_x(w; v)$.

The exponential map is often approximated by the *retraction* of the first-order expansion $\exp_x(v) \approx x + v$ onto $\mathcal{M}$:

$$\exp_x(v) \approx \operatorname{retr}_x(v) = \operatorname{proj}_{\mathcal{M}}(x + v). \tag{181}$$

When using the retraction approximation, RGD is equivalent to *projected gradient descent* [102]:

$$x_{i+1} = \operatorname{proj}_{\mathcal{M}}(x_i - \eta \operatorname{proj}_{\mathcal{M}}(x_i, \nabla f(x_i))). \tag{182}$$

Adaptive methods such as Adam [103] are not directly usable on manifolds: coordinate systems can only be defined *locally* on a manifold, and therefore coordinatewise state such as momentum loses its meaning. The RAdam optimizer [104] addresses this subtlety as follows: 1) tangent vectors are parallel transported to the same coordinate system before being accumulated, 2) momentum normalization uses the coordinate-independent squared norm $\|\operatorname{grad} f(x)\|^2$ instead of the coordinatewise second moments $(\operatorname{grad} f(x))_i^2$, and 3) $x - \eta\Delta x$ is replaced by $\exp_x(-\eta\Delta x)$.

K Densities on Riemannian Manifolds with Application to the Standard Simplex

K.1 Density Transformations on Manifolds

We review how to transform probability densities under smooth, injective maps between Riemannian manifolds. For an intuitive derivation, see Appendix A in [29].

Let M and N be D-dimensional Riemannian manifolds with measures μ_M and μ_N induced by their metrics. For any measurable function $f\colon N \to \mathbb{R}$ and any diffeomorphism $T\colon M \to N$, the change-of-variables formula is

$$\int_U f(u)\, d\mu_N(u) = \int_X f\left(T(x)\right) |D_x T|\, d\mu_M(x) \tag{183}$$

where $X \subset M$, $U = T(X) \subset N$, and $|D_x T|$ is the Jacobian determinant computed in local coordinates. Let p_x and p_u be probability densities on M and N with respect to the Riemannian volume measures dx and du, where $u = T(x)$. Given p_x, by (183) the transformed density p_u is

$$p_u(u) = \frac{p_x(x)}{|D_x T|}. \tag{184}$$

Now suppose $M \subset \mathbb{R}^m$ and $N \subset \mathbb{R}^n$ are embedded with metrics induced by the ambient Euclidean inner product. If $E \in \mathbb{R}^{m \times D}$ is an orthonormal basis for the tangent space $T_x M$ and $J = \frac{\partial u}{\partial x} \in \mathbb{R}^{n \times m}$ is the Jacobian of T at x, then $|D_x T| = \sqrt{\det((JE)^T JE)}$ and the density update is

$$p_u(u) = \frac{p_x(x)}{\sqrt{\det(E^T J^T J E)}}. \tag{185}$$

K.2 The Standard Simplex Manifold Δ_{n-1}

Consider the standard $(n-1)$-simplex

$$\Delta^{n-1} = \Big\{ x \in \mathbb{R}^n : x_i > 0 \text{ and } \sum_{i=1}^{n} x_i = 1 \Big\}. \tag{186}$$

Δ^{n-1} is an $(n-1)$-dimensional embedded submanifold of $\mathbb{R}^n$ endowed with the metric induced by the Euclidean inner product. Its measure induced by the metric is denoted by dx.

K.2.1 Jacobian Determinant of the ILR Transform

We now derive the Jacobian determinant for the isometric log-ratio (ILR) transformation. For the ILR transform $\mathrm{ilr} : \Delta^{n-1} \to \mathbb{R}^{n-1}$ given by

$$\mathrm{ilr}(x) = \Psi^T \log(x) \tag{187}$$

the tangent space of the domain $T_x \Delta^{n-1}$ is spanned by the columns of Ψ. We thus take $E = \Psi$ to be a fixed orthonormal basis for $T_x \Delta^{n-1}$. The ambient Jacobian is

$$J = \frac{\partial\, \mathrm{ilr}(x)}{\partial x} = \Psi^T \operatorname{diag}\left(\frac{1}{x_1}, \ldots, \frac{1}{x_n} \right), \tag{188}$$

and so the effective Jacobian on $T_x\Delta^{n-1}$ is $JE = J\Psi$. The density update factor is then

$$\sqrt{\det(E^T J^T J E)} = |J\Psi| = \frac{1}{n\prod_{i=1}^n x_i}. \tag{189}$$

This can be derived in several steps. First, we note that the density update is independent of the choice of basis. If $\Psi_1, \Psi_2 \in \mathbb{R}^{n\times(n-1)}$ are orthonormal bases for the plane $\langle 1, \cdot\rangle = 0$, then $1^T\Psi_1 = 1^T\Psi_2 = 0$, $\Psi_1\Psi_1^T = \Psi_2\Psi_2^T = \mathrm{I}_n - \frac{1}{n}11^T$, and $\Psi_2 = \Psi_1 Q$ with $Q = \Psi_1^T\Psi_2$ orthogonal:

$$\begin{aligned}\Psi_1 Q &= (\Psi_1\Psi_1^T)\Psi_2 = (\mathrm{I}_n - \frac{1}{n}11^T)\Psi_2 = \Psi_2\\ Q^T Q &= \Psi_2^T(\Psi_1\Psi_1^T\Psi_2) = \Psi_2^T\Psi_2 = \mathrm{I}_{n-1}.\end{aligned}$$

Therefore we have

$$|J\Psi_2| = \left|\Psi_2^T \operatorname{diag}\left(\frac{1}{x}\right)\Psi_2\right| = \left|Q^T J\Psi_1 Q\right| = |J\Psi_1|.$$

Now, let $B \in \mathbb{R}^{n\times(n-1)}$ be the (non-orthogonal) basis with j^{th} column $e_j - e_n$, and let $\Psi = B(B^TB)^{-1/2}$ be the orthonormalization of B, noting that B^TB is positive definite such that the matrix inverse square root is unique, and

$$\begin{aligned}\Psi^T 1 &= (B^TB)^{-1/2}B^T 1 = 0,\\ \Psi^T\Psi &= (B^TB)^{-1/2}B^TB(B^TB)^{-1/2} = \mathrm{I}_{n-1}\\ \left|B^TB\right| &= \left|\begin{pmatrix}\mathrm{I}_{n-1}\\ -1^T\end{pmatrix}^T\begin{pmatrix}\mathrm{I}_{n-1}\\ -1^T\end{pmatrix}\right| = \left|\mathrm{I}_{n-1} + 11^T\right|\\ &= (1 + 1^T 1) = n.\end{aligned}$$

Then,

$$\begin{aligned}|J\Psi| &= \left|\Psi^T \operatorname{diag}\left(\frac{1}{x}\right)\Psi\right|\\ &= \left|B^T \operatorname{diag}\left(\frac{1}{x}\right)B\right| / \left|B^TB\right|\\ &= \left|B^T \operatorname{diag}\left(\frac{1}{x}\right)B\right| / n\end{aligned}$$

and we have

$$\begin{aligned}\left|B^T \operatorname{diag}\left(\frac{1}{x}\right)B\right| &= \left|\begin{pmatrix}\mathrm{I}_{n-1}\\ -1^T\end{pmatrix}^T \operatorname{diag}\left(\frac{1}{x}\right)\begin{pmatrix}\mathrm{I}_{n-1}\\ -1^T\end{pmatrix}\right|\\ &= \left|\begin{pmatrix}\mathrm{I}_{n-1}\\ -1^T\end{pmatrix}^T\begin{pmatrix}\operatorname{diag}(1/\tilde{x})\\ -(1/x_n)1^T\end{pmatrix}\right|\\ &= \left|\operatorname{diag}\left(\frac{1}{\tilde{x}}\right) + \frac{1}{x_n}11^T\right|\\ &= \left(\prod_i \frac{1}{\tilde{x}_i}\right)\left(1 + \frac{1}{x_n}1^T\operatorname{diag}(\tilde{x})1\right)\\ &= \left(\prod_i \frac{1}{\tilde{x}_i}\right)\left(1 + \frac{1 - x_n}{x_n}\right)\end{aligned}$$

$$= \prod_i \frac{1}{x_i}$$

$$\Rightarrow |J\Psi| = \frac{1}{n \prod_i x_i} \qquad \blacksquare$$

where we denoted $\tilde{x} = (x_1, \ldots, x_{n-1})$, and used the matrix determinant lemma $\det\big(A + uv^T\big) = \det(A)(1 + v^T A^{-1} u)$.

Let $p_x(x)$ denote the density on Δ^{n-1} (with respect to dx) and $p_u(u)$ denote the density in ILR coordinates. Then, by the change-of-variables formula,

$$\boxed{p_u(u) = p_x(x)\left(n \prod_{i=1}^{n} x_i\right).} \tag{190}$$

K.2.2 Jacobian Determinant of the Inverse ILR Transform

Alternatively, consider the inverse map $\text{ilr}^{-1}\colon \mathbb{R}^{n-1} \to \Delta^{n-1}$. Since the domain is $\mathbb{R}^{n-1}$, we choose the standard basis $E = \text{I}_{n-1}$. Letting $z = \Psi u$, the inverse ILR transform is

$$\text{ilr}^{-1}(u) = \text{softmax}(z) = \frac{\exp(z)}{\sum_{j=1}^{n} \exp(z_j)}. \tag{191}$$

The Jacobian of the softmax function is

$$\frac{\partial x}{\partial z} = \text{diag}(x) - xx^T \tag{192}$$

and thus the Jacobian of ilr^{-1} is

$$J = \frac{\partial x}{\partial z}\frac{\partial z}{\partial u} = \big(\text{diag}(x) - xx^T\big)\,\Psi. \tag{193}$$

The Jacobian determinant

$$\sqrt{\det(J^T J)} = n \prod_{i=1}^{n} x_i \tag{194}$$

can be derived using a similar method as for the forward transform. First, denote $X = \text{diag}(x) - xx^T$. Since $1^T X = 0^T$, the projection operator $\Psi\Psi^T$ does not change X:

$$\Psi\Psi^T X = \left(\text{I}_n - \frac{1}{n} 1 1^T\right) X = X.$$

This lets us factor the Jacobian determinant:

$$\begin{aligned}
\det\big(J^T J\big) &= \det\big(\Psi^T X^T X \Psi\big) \\
&= \det\big(\Psi^T X^T (\Psi\Psi^T X)\Psi\big) \\
&= \big|\Psi^T X \Psi\big|^2
\end{aligned}$$

By the same argument as in appendix K.2.1, the determinant $\big|\Psi^T X \Psi\big|$ is independent of the choice of basis; we choose $\Psi = B(B^T B)^{-1/2}$ as before, where the jth column of $B \in \mathbb{R}^{n \times (n-1)}$ is $e_j - e_n$. Then,

$$\begin{aligned}
\big|\Psi^T X \Psi\big| &= \big|B^T X B\big| / \big|B^T B\big| \\
&= \big|B^T X B\big| / n
\end{aligned}$$

$$\begin{aligned}\left|B^T X B\right| &= \left|\begin{pmatrix}\mathrm{I}_{n-1}\\ -1^T\end{pmatrix}^T \left(\operatorname{diag}(x) - xx^T\right)\begin{pmatrix}\mathrm{I}_{n-1}\\ -1^T\end{pmatrix}\right| \\ &= \left|\operatorname{diag}(\tilde{x}) + x_n 1 1^T - \tilde{y}\tilde{y}^T\right|\end{aligned}$$

where $\tilde{y} = \tilde{x} - x_n 1$. This can be computed straightforwardly, albeit tediously, with two applications of the matrix determinant lemma $\det\left(A + uv^T\right) = \det(A)(1 + v^T A^{-1} u)$ and one application of the Sherman–Morrison formula $(A + uv^T)^{-1} = A^{-1} - \frac{A^{-1}uv^T A^{-1}}{1+v^T A^{-1}u}$. Let $A_1 = \operatorname{diag}(\tilde{x})$, $A_2 = A_1 + x_n 1 1^T$, and $A_3 = A_2 - \tilde{y}\tilde{y}^T = B^T X B$. Then,

$$\begin{aligned}\det(A_3) &= \det(A_2)(1 - \tilde{y}^T A_2^{-1}\tilde{y}) \\ &= \det(A_1)(1 + x_n 1^T A_1^{-1} 1)(1 - \tilde{y}^T A_2^{-1}\tilde{y}) \\ &= \det(A_1)\Big((1 + x_n 1^T A_1^{-1} 1)(1 - \tilde{y}^T A_1^{-1}\tilde{y}) \\ &\qquad + x_n(\tilde{y}^T A_1^{-1} 1 1^T A_1^{-1}\tilde{y})\Big)\end{aligned}$$

Let $c = \sum_i \frac{1}{\tilde{x}_i}$ and recall that $\sum_i \tilde{x}_i = 1 - x_n$. Then, we have

$$\begin{aligned}1^T A_1^{-1} 1 &= \sum_i \frac{1}{\tilde{x}_i} = c \\ 1^T A_1^{-1}\tilde{y} &= \sum_i \left(1 - \frac{x_n}{\tilde{x}_i}\right) = n - 1 - x_n c \\ 1 - \tilde{y}^T A_1^{-1}\tilde{y} &= 1 - \sum_i \frac{(\tilde{x}_i - x_n)^2}{\tilde{x}_i} \\ &= 1 - \sum_i \left(\tilde{x}_i - 2x_n + \frac{x_n^2}{\tilde{x}_i}\right) \\ &= 1 - \big((1 - x_n) - 2(n-1)x_n + x_n^2 c\big) \\ &= x_n(2n - 1 - x_n c).\end{aligned}$$

Inserting $\det(A_1) = \prod_i \tilde{x}_i$ and $A_3 = B^T X B$ then gives

$$\begin{aligned}\frac{\left|B^T X B\right|}{\prod_i x_i} &= |(1 + x_n c)(2n - 1 - x_n c) \\ &\qquad + (n - 1 - x_n c)^2| \\ &= |2n - 1 + (n-1)^2 + (-1+1)\, x_n^2 c^2 \\ &\qquad + ((2n-1) - 1 - 2(n-1))\, x_n c| \\ &= n^2.\end{aligned}$$

Finally, the Jacobian determinant is

$$\begin{aligned}\sqrt{\det(J^T J)} &= \left|\Psi^T X \Psi\right| \\ &= \left|B^T X B\right|/n \\ &= n\prod_i x_i.\end{aligned}$$

■

Thus, by the change-of-variables formula, we recover the same density update:

$$\boxed{p_x(x) = \frac{p_u(u)}{n\prod_{i=1}^n x_i}.} \tag{195}$$

K.2.3 Changing to the Intrinsic Measure

Thus far, the density $p_x(x)$ is given with respect to the measure dx induced on the simplex by the Euclidean metric on $\mathbb{R}^n$. This is inconvenient to work with; for example, to compute integrals over Δ^{n-1}, it is more natural to work with the intrinsic (or coordinate) measure. We choose the coordinate chart

$$\varphi(x) = (x_1, x_2, \ldots, x_{n-1}) \tag{196}$$

for Δ^{n-1}, with inverse

$$\varphi^{-1}(\tilde{x}) = \Big(\tilde{x}_1, \tilde{x}_2, \ldots, \tilde{x}_{n-1}, 1 - \sum_{i=1}^{n-1} \tilde{x}_i\Big). \tag{197}$$

A straightforward computation relates the induced measure dx on Δ^{n-1} with the Lebesgue measure $d\tilde{x}$ on $\mathbb{R}^{n-1}$:

$$dx = \sqrt{\det\Big(J_{\varphi^{-1}}^T J_{\varphi^{-1}}\Big)}\, d\tilde{x} = \sqrt{n} d\tilde{x}, \tag{198}$$

$$\text{where } \; J_{\varphi^{-1}} = \begin{pmatrix} \mathrm{I}_{n-1} \\ -1^T \end{pmatrix}, \;\; J_{\varphi^{-1}}^T J_{\varphi^{-1}} = \mathrm{I}_{n-1} + 11^T.$$

Changing variables to intrinsic coordinates, the density $p_{\tilde{x}}(\tilde{x})$ with respect to $d\tilde{x}$ satisfies $p_x(x)dx = p_{\tilde{x}}(\tilde{x})d\tilde{x}$, and so

$$\boxed{p_{\tilde{x}}(\tilde{x}) = \sqrt{n} p_x(x) = \frac{p_u(u)}{\sqrt{n}\prod_{i=1}^n x_i}.} \tag{199}$$

This is analogous to converting a density $p_1(x, y, z)$ with respect to the area measure $d\Omega$ on the sphere $S^2 \subset \mathbb{R}^3$ into a density $p_2(\theta, \phi)$ with respect to the intrinsic measure $d\theta d\phi$ using the spherical coordinate chart $\varphi^{-1}(\theta, \phi) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$. These measures satisfy $d\Omega = \sin\theta d\theta d\phi$, resulting in $p_2(\theta, \phi) = p_1(\varphi^{-1}(\theta, \phi))\sin\theta$.

K.3 The Softmax Manifold $\Delta_{n-1} \times \mathbb{R}$

In this section, we consider the manifold $\Delta_{n-1} \times \mathbb{R} \subset \mathbb{R}^{n+1}$, the product manifold of the standard $(n-1)$-simplex and the real line. This manifold is endowed with the product metric, and its induced measure is denoted by dx.

K.3.1 Forward Transformation

Consider the injective extension $\varphi : \mathbb{R}^n \to \mathbb{R}^{n+1}$ of the *softmax* function

$$x = \varphi(u) = \begin{pmatrix} \bar{x} \\ \log s \end{pmatrix} = \begin{pmatrix} e^{\beta u}/\langle 1, e^{\beta u}\rangle \\ -\log\langle 1, e^{\beta u}\rangle \end{pmatrix} \tag{200}$$

where $\beta > 0$, $u \in \mathbb{R}^n$ are the *logits*, $\log s \in \mathbb{R}$ where $s^{-1} = \langle 1, e^{\beta u}\rangle$ is the normalization constant, and $\bar{x} \in \Delta_{n-1}$ is a probability vector embedded in $\mathbb{R}^n$.

We now compute the density update (185) for the transformation (200). Following the notation in appendix K.1, we let $\mathcal{M} = \mathbb{R}^n$ and $\mathcal{N} = \Delta_{n-1} \times \mathbb{R} \subset \mathbb{R}^{n+1}$, and we choose $E = \mathrm{I} \in \mathbb{R}^{n\times n}$ to be the canonical basis for the tangent space $T_u\mathbb{R}^n \cong \mathbb{R}^n$. What remains is to compute the differential volume correction $\sqrt{\det(J^T J)}$ for the transformation φ. We have that

$$\frac{\partial \bar{x}}{\partial u} = \beta\,\mathrm{diag}(\bar{x}) - \beta\bar{x}\bar{x}^T \tag{201}$$

$$\frac{\partial \log s}{\partial u} = -\beta \bar{x}^T \tag{202}$$

where $\operatorname{diag}(\bar{x})$ is the diagonal matrix with $\bar{x}$ on the diagonal. It follows that

$$\begin{aligned} J^T J &= \left(\frac{\partial \bar{x}}{\partial u}\right)^T \frac{\partial \bar{x}}{\partial u} + \left(\frac{\partial \log s}{\partial u}\right)^T \frac{\partial \log s}{\partial u} \\ &= \beta^2 \left((\operatorname{diag}(\bar{x}) - \bar{x}\bar{x}^T)^2 + \bar{x}\bar{x}^T \right) \\ &= \beta^2 \operatorname{diag}(\bar{x}) \left(XX^T + 11^T \right) \operatorname{diag}(\bar{x}) \end{aligned}$$

where $X = \mathrm{I} - 1\bar{x}^T$. Since $\det(\operatorname{diag}(\bar{x})) = \prod_i \bar{x}_i$, it remains only to compute the determinant of the matrix $XX^T + 11^T$. This can be accomplished using the identities

1. $\det\left(A + vw^T\right) = \det(A) + w^T \operatorname{adj}(A) v$, the matrix determinant lemma, and
2. $\operatorname{adj}\left(\mathrm{I} + vw^T\right) = -vw^T$, a special case of the Sherman–Morrison formula when $w^T v = -1$,

where $\operatorname{adj}(A)$ denotes the *adjugate matrix* of A. In our case, $\det(X) = 1 - 1^T \bar{x} = 0$ and $\operatorname{adj}(X) = \operatorname{adj}\left(\mathrm{I} - 1\bar{x}^T\right) = 1\bar{x}^T$, and therefore we have that

$$\begin{aligned} \det\left(XX^T + 11^T\right) &= \det\left(XX^T\right) + 1^T \operatorname{adj}\left(XX^T\right) 1 \\ &= 0 + 1^T (1\bar{x}^T)^T (1\bar{x}^T) 1 \\ &= (1^T 1)(\bar{x}^T 1)^2 \\ &= n. \end{aligned}$$

The volume correction term is therefore given by

$$\sqrt{\det(J^T J)} = \sqrt{n} \beta^n \prod_{i=1}^{n} \bar{x}_i. \tag{203}$$

and the resulting density update is

$$\boxed{p(x) = \frac{p(u)}{\sqrt{n} \beta^n \prod_{i=1}^{n} \bar{x}_i}.} \tag{204}$$

K.3.2 Inverse Transformation

Now, we derive the density transformation for the inverse softmax transformation:

$$u = \varphi^{-1}(x) = \frac{1}{\beta}(\log \bar{x} - \log s) \tag{205}$$

where $\varphi : \mathbb{R}^n \to \mathbb{R}^{n+1}$ is the injective extension of the softmax operation introduced in equation (200) in appendix K.3.1. As before, $\beta > 0$, $u \in \mathbb{R}^n$ are the logits, $\bar{x} \in \Delta_{n-1}$ is a probability vector embedded in $\mathbb{R}^n$, and s^{-1} is the normalization constant; subtraction by $\log s$ is performed elementwise.

Note that to compute the density update for (205), one could simply rearrange (204) to solve for $\log p(u)$ in terms of $\log p(x)$. However, computing the inverse density update directly provides both an illustration of the method on a nontrivial manifold and a means to verify the forward result.

Following the notation in appendix K.1, we let $\mathcal{M} = \Delta_{n-1} \times \mathbb{R} \subset \mathbb{R}^{n+1}$ and $\mathcal{N} = \mathbb{R}^n$. Since the tangent space of a product of manifolds is the product of the individual tangent spaces, we define the block orthonormal basis $E \in \mathbb{R}^{(n+1)\times n}$ as

$$E = \begin{bmatrix} \Psi & 0 \\ 0^T & 1 \end{bmatrix} \tag{206}$$

where $\Psi \in \mathbb{R}^{n\times(n-1)}$ is an orthonormal basis for the tangent space $T_{\bar{x}}\Delta_{n-1}$. The basis Ψ is fixed, and satisfies both:

1. $\Psi^T\Psi = \mathrm{I}$: the columns of Ψ are orthonormal, and
2. $1^T\Psi = 0^T$: the columns of Ψ are in $T_{\bar{x}}\Delta_{n-1}$.

Note that these two properties imply that $[\Psi\ \hat{1}]$ is an orthonormal basis for $\mathbb{R}^n$, where $\hat{1}$ is the unit vector with all entries $\frac{1}{\sqrt{n}}$, and it follows that $\mathrm{I} = [\Psi\ \hat{1}][\Psi\ \hat{1}]^T = \Psi\Psi^T + \frac{1}{n}11^T$.

The Jacobian $J \in \mathbb{R}^{n\times(n+1)}$ is given by

$$\frac{\partial u}{\partial \bar{x}} = \frac{1}{\beta}\operatorname{diag}\big(\tfrac{1}{\bar{x}}\big), \quad \frac{\partial u}{\partial \log s} = -\frac{1}{\beta}1 \tag{207}$$

$$\Rightarrow J = \frac{1}{\beta}\begin{bmatrix}\operatorname{diag}\big(\tfrac{1}{\bar{x}}\big) & -1\end{bmatrix} \tag{208}$$

where the reciprocal $\frac{1}{\bar{x}}$ is computed elementwise. Using $\det\big((JE)^TJE\big) = \det\big(JE(JE)^T\big)$, we compute the matrix products in a convenient order:

$$\begin{aligned} EE^T &= \begin{bmatrix}\Psi\Psi^T & 0\\ 0^T & 1\end{bmatrix} \\ \Rightarrow E(JE)^T &= \frac{1}{\beta}\begin{pmatrix}\Psi\Psi^T\operatorname{diag}\big(\tfrac{1}{\bar{x}}\big)\\ -1^T\end{pmatrix} \\ \Rightarrow JE(JE)^T &= \frac{1}{\beta^2}(\operatorname{diag}\big(\tfrac{1}{\bar{x}}\big)\Psi\Psi^T\operatorname{diag}\big(\tfrac{1}{\bar{x}}\big) + 11^T) \\ &= \frac{1}{\beta^2}\operatorname{diag}\big(\tfrac{1}{\bar{x}}\big)(\Psi\Psi^T + \bar{x}\bar{x}^T)\operatorname{diag}\big(\tfrac{1}{\bar{x}}\big). \end{aligned} \tag{209}$$

The determinant of $\Psi\Psi^T + \bar{x}\bar{x}^T$ is computed using the matrix determinant lemma and the Sherman–Morrison formula. First, since Ψ is not full rank we have $\det\big(\Psi\Psi^T\big) = 0$. Then, using $\Psi\Psi^T = \mathrm{I} - \frac{1}{n}11^T$, by Sherman–Morrison we have that $\operatorname{adj}\big(\Psi\Psi^T\big) = \operatorname{adj}\big(\mathrm{I} - \frac{1}{n}11^T\big) = \frac{1}{n}11^T$. The matrix determinant lemma finally gives

$$\begin{aligned} \det\big(\Psi\Psi^T + \bar{x}\bar{x}^T\big) &= \det\big(\Psi\Psi^T\big) + \bar{x}^T\operatorname{adj}\big(\Psi\Psi^T\big)\bar{x} \\ &= 0 + \bar{x}^T(\tfrac{1}{n}11^T)\bar{x} \\ &= \tfrac{1}{n}(\bar{x}^T1)^2 \\ &= \tfrac{1}{n}. \end{aligned} \tag{210}$$

Mirroring (203) and (204), the resulting volume correction is

$$\sqrt{\det((JE)^TJE)} = \frac{1}{\sqrt{n}\beta^n}\prod_{i=1}^{n}\frac{1}{\bar{x}_i} \tag{211}$$

and the corresponding density update is

$$\boxed{\Rightarrow p(u) = p(x)\left(\sqrt{n}\beta^n\prod_{i=1}^{n}\bar{x}_i\right)} \tag{212}$$

as expected from the forward transformation ■.

K.3.3 Changing to the Intrinsic Measure on $\Delta^{n-1} \times \mathbb{R}$

Thus far, the density p_x is given with respect to the measure dx induced on the product manifold $\Delta^{n-1} \times \mathbb{R}$ by the Euclidean metric on $\mathbb{R}^{n+1}$. As discussed in appendix K.2.3, in practice it is more convenient to use the intrinsic (or coordinate) measure. A natural coordinate chart for $\Delta^{n-1} \times \mathbb{R}$ is given by

$$\varphi(x) = (\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_{n-1}, \log s), \tag{213}$$

with inverse

$$\varphi^{-1}(\tilde{x}) = \Big(\tilde{x}_1, \tilde{x}_2, \ldots, \tilde{x}_{n-1}, 1 - \sum_{i=1}^{n-1} \tilde{x}_i, \log s\Big) \tag{214}$$

where $x = (\bar{x}, \log s)$. The induced measure dx and the Lebesgue measure $d\tilde{x}$ on $\mathbb{R}^{n-1} \times \mathbb{R} \cong \mathbb{R}^n$ are related by

$$dx = \sqrt{\det\Big(J_{\varphi^{-1}}^T J_{\varphi^{-1}}\Big)}\, d\tilde{x} = \sqrt{n} d\tilde{x} \tag{215}$$

$$\text{where} \quad J_{\varphi^{-1}} = \begin{bmatrix} \mathrm{I}_{n-1} & 0 \\ -1^T & 0 \\ 0^T & 1 \end{bmatrix},$$

$$J_{\varphi^{-1}}^T J_{\varphi^{-1}} = \begin{bmatrix} \mathrm{I}_{n-1} + 11^T & 0 \\ 0 & 1 \end{bmatrix}.$$

Changing variables to intrinsic coordinates gives the density $p_{\tilde{x}}$

$$\boxed{p_{\tilde{x}}(\tilde{x}) = \sqrt{n} p(x) = \frac{p(u)}{\beta^n \prod_{i=1}^n \bar{x}_i}.} \tag{216}$$

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